

The Female Genital Tract

Lora Hedrick Ellenson and Drucilla J. Roberts

CHAPTER CONTENTS

Infections, 896

- Infections of the Lower Genital Tract, 897
 - Herpes Simplex Virus*, 897
 - Other Lower Female Genital Tract Infections*, 897
- Infections Involving the Lower and Upper Genital Tract, 898
 - Pelvic Inflammatory Disease*, 898

Vulva, 899

Bartholin Cyst, 899

Nonneoplastic Epithelial Disorders, 899

- Lichen Sclerosus, 900
- Lichen Simplex Chronicus, 900

Benign Exophytic Lesions, 900

- Condyloma Acuminatum, 900

Squamous Neoplastic Lesions, 900

- Vulvar Intraepithelial Neoplasia and Vulvar Carcinoma, 900

Glandular Neoplastic Lesions, 901

- Papillary Hidradenoma, 902
- Extramammary Paget Disease, 902

Vagina, 903

Developmental Anomalies, 903

Premalignant and Malignant Neoplasms of the Vagina, 904

- Squamous Intraepithelial Lesions and Squamous Cell Carcinoma, 904
- Embryonal Rhabdomyosarcoma, 904

Cervix, 904

Inflammations, 904

- Acute and Chronic Cervicitis, 904

Endocervical Polyps, 905

Premalignant and Malignant Neoplasms of the Cervix, 905

- Cervical Squamous Intraepithelial Lesions, 906
- Cervical Carcinoma, 907
- Cervical Cancer Screening and Prevention, 909

Uterine Corpus, 910

Endometrial Histology in the Menstrual Cycle, 910

Functional Endometrial Disorders (Dysfunctional Uterine Bleeding), 910

- Anovulatory Cycle, 910

Inflammatory Disorders, 911

- Acute Endometritis, 911
- Chronic Endometritis, 911

Endometriosis and Adenomyosis, 912

Endometrial Polyps, 914

Endometrial Hyperplasia, 915

Malignant Tumors of the Endometrium, 915

- Carcinoma of the Endometrium, 915
- Endometrioid Endometrial Carcinoma, 916

Serous Endometrial Carcinoma, 918

Carcinosarcoma, 919

Tumors of Endometrial Stroma, 920

- Adenosarcoma, 920
- Stromal Tumors, 920

Tumors of the Myometrium, 920

- Leiomyoma, 920
- Leiomyosarcoma, 921

Fallopian Tubes, 922

- Inflammations, 922
- Tumors and Cysts, 922

Ovaries, 922

Nonneoplastic and Functional Cysts, 923

- Follicle and Luteal Cysts, 923
- Polycystic Ovaries and Stromal Hyperthecosis, 923

Ovarian Tumors, 923

- Classification, 923
- Epithelial Tumors, 924
 - Serous Tumors*, 924
 - Mucinous Tumors*, 926
 - Endometrioid Ovarian Tumors*, 927
 - Clear Cell Carcinoma*, 928
 - Transitional Cell Tumors (Brenner Tumors)*, 928
 - Clinical Course, Detection, and Prevention of Ovarian Epithelial Tumors*, 928

Germ Cell Tumors, 929

- Teratoma*, 929
- Dysgerminoma*, 930
- Yolk Sac Tumors*, 930
- Choriocarcinoma*, 931
- Other Germ Cell Tumors*, 931

Sex Cord–Stromal Tumors, 931

- Granulosa Cell Tumors*, 931
- Fibromas and Thecomas*, 932
- Sertoli–Leydig Cell Tumors*, 932
- Other Sex Cord–Stromal Tumors*, 933
- Metastatic Tumors*, 933

Gestational and Placental Disorders, 933

Disorders of First and Early Second Trimester Gestations, 934

- Spontaneous Abortion, 934
- Ectopic Pregnancy, 935

Disorders of Mid to Late Second Trimester Gestations, 936

Disorders of the Third Trimester, Peripartum, and

Puerperium, 936

- Twin Placentas, 936
- Maternal Vascular Malperfusion, 936

Abnormalities of Placental Implantation, 937

Placental Infections, 937

Preeclampsia and Eclampsia, 937

Gestational Trophoblastic Disease, 939

Hydatidiform Mole, 940

Complete Mole, 940

Partial Mole, 940

Invasive Mole, 940

Choriocarcinoma, 940

Tumors of Extravascular Trophoblasts, 941

A brief review of the development and anatomy of the female genital tract is fundamental to understanding the diseases that affect this complex organ system. Normal development of the female genital tract proceeds through a series of tightly choreographed events involving the primordial germ cells, the müllerian (paramesonephric) ducts, the wolffian (mesonephric) ducts, and the urogenital sinus (Fig. 22.1).

- *Germ cells* arise in the wall of the yolk sac by the fourth week of gestation. By the fifth or sixth week, they migrate into the urogenital ridge and induce proliferation of the mesodermal epithelium, which gives rise to the epithelium and stroma of the ovary.
- The lateral *müllerian ducts* form at about the sixth week of development through invagination and fusion of the coelomic lining epithelium. The ducts progressively grow caudally into the pelvis, where they swing medially to fuse with the urogenital sinus at the müllerian tubercle (see Fig. 22.1A). Further caudal growth brings these fused ducts into contact with the urogenital sinus. The unfused upper portions of the müllerian ducts mature into the fallopian tubes, while the fused lower portion develops into the uterus, cervix, and upper vagina.
- The *urogenital sinus* develops when the cloaca is subdivided by the urorectal septum; it eventually forms the lower part of the vagina and the vestibule of the external genitalia (see Fig. 22.1B).
- The *mesonephric ducts* normally regress in the female, but remnants may persist into adult life as epithelial inclusions adjacent to the

ovaries, tubes, and uterus. In the cervix and vagina, these rests may be cystic and are termed *Gartner duct cysts*.

The epithelial lining of the female genital tract as well as the ovarian surface share a common origin from coelomic epithelium (mesothelium), which may explain why morphologically similar benign and malignant lesions arise in various sites within the female genital tract and the adjacent peritoneal surfaces.

Diseases of the female genital tract are extremely common and include complications of pregnancy, infections, tumors, and hormonally induced abnormalities. The following discussion presents the pathology of the major diseases that result in clinical problems. Additional details can be found in textbooks of gynecologic pathology and clinical obstetrics and gynecology. We will discuss the pathologic conditions peculiar to each segment of the female genital tract separately, but before doing so we will briefly review infections and pelvic inflammatory disease because they can affect many of the various anatomic structures concomitantly.

INFECTIONS

A large variety of organisms can infect the female genital tract. Some infections with microorganisms such as *Candida*, *Trichomonas*, and *Gardnerella* are very common and may cause significant discomfort but

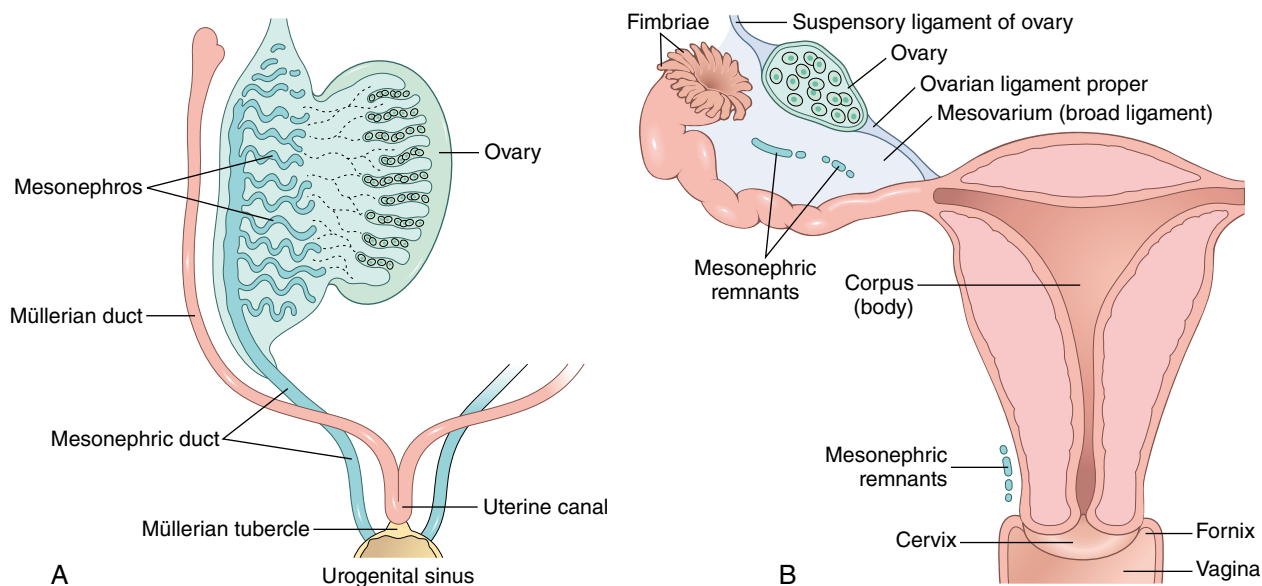


Fig. 22.1 Embryology and anatomy of the female genital tract. (A) Early in development, the mesonephric (blue) and müllerian (red) ducts merge at the urogenital sinus to form the müllerian tubercle. (B) By birth, the müllerian ducts have fused to form the fallopian tubes, uterus, and endocervix (red), merging with the vaginal squamous mucosa. The mesonephric ducts regress but may be found as a remnant in the ovary, adnexa, and cervix (Gartner duct cysts). (Modified from Langman J. *Medical Embryology*. Williams and Wilkins; 1981.)

are without serious sequelae. Others, such as *Neisseria gonorrhoeae* and *Chlamydia* infections, are major causes of infertility, while infections with organisms such as *Group B Streptococcus* infections are implicated in preterm deliveries, stillbirths, and neonatal infections. Viruses, especially herpes simplex viruses (HSVs) and human papillomaviruses (HPVs), also account for considerable morbidity; HSVs cause painful genital ulcerations, whereas HPVs are involved in the pathogenesis of cervical, vaginal, and vulvar cancers (described later).

Many of these infections are sexually transmitted, including trichomoniasis, gonorrhea, *Mycoplasma*, *Chlamydia*, HSV, and HPV, as well as less common infections such as syphilis, chancroid, granuloma inguinale, and lymphogranuloma venereum. Most of these conditions are considered in Chapter 8; HPV is also discussed in Chapter 7 due to its important role as a transforming virus. Here we touch only on aspects relevant to the female genital tract, including pathogens confined to the lower genital tract (vulva, vagina, and cervix) and those that involve the entire genital tract and are implicated in pelvic inflammatory disease.

Infections of the Lower Genital Tract

Herpes Simplex Virus

Genital HSV infection is common and may involve the vulva, vagina, or cervix. HSVs are DNA viruses that include two serotypes, HSV-1 and HSV-2. HSV-1 typically results in perioral infection, whereas HSV-2 usually involves genital mucosa and skin; however, depending on sexual practices, HSV-1 may be detected in the genital region and HSV-2 may cause oral infections as well (Chapter 8). By 40 years of age, approximately 30% of females are seropositive for antibodies against HSV-2.

About one-third of newly infected individuals are symptomatic. Lesions typically develop 3 to 7 days after transmission and are often associated with systemic symptoms such as fever, malaise, and tender inguinal lymph nodes. The earliest lesions usually consist of red papules that progress to vesicles and then to painful coalescent ulcers. The lesions are easily visible on vulvar skin and mucosa, while cervical and vaginal lesions present with purulent discharge and pelvic pain. Lesions around the urethra may cause painful urination and urine retention. The vesicles and ulcers contain numerous viral particles, accounting for the high transmission rate during active infection. The mucosal and cutaneous lesions heal spontaneously in 1 to 3 weeks, but during the acute infection the virus migrates to the regional lumbosacral nerve ganglia and establishes a latent infection. Because of viral latency, HSV infections persist indefinitely, and any decrease in immune function due to stress, trauma, concurrent viral infection, or hormonal changes can trigger reactivation of the virus and recurrence of the skin and mucosal lesions. As expected, recurrences are much more common in immunosuppressed individuals.

MORPHOLOGY

By the time an HSV lesion is biopsied, it typically has ulcerated. The epithelium is desquamated, and marked acute inflammation is present in the ulcer bed. Smears of the inflammatory exudate from active lesions show characteristic cytopathic changes consisting of **multinucleated squamous cells containing eosinophilic to basophilic viral inclusions with a “ground-glass” appearance (Fig. 22.2).**

Transmission of HSV takes place mainly during the active phase but occasionally may occur during the latent phase due to subclinical virus shedding. Condoms and antiviral therapies reduce the risk of transmission, but do not prevent it completely. As with other sexually

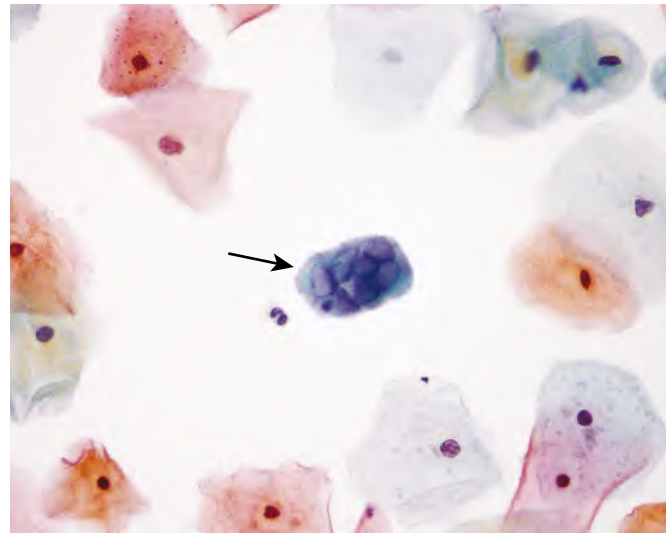


Fig. 22.2 Herpes simplex virus (HSV) infection (cervical smear). The cell in the center shows HSV cytopathic effect. Infected cells become multinucleated and contain intranuclear viral inclusions with a characteristic “ground-glass” appearance (arrow).

transmitted diseases, the virus is more readily transmitted to females than to males. Previous infection with HSV-1 seems to reduce susceptibility to HSV-2 infection. The gravest consequence of HSV infection is transmission to the neonate during birth. This risk is highest if the infection is active during delivery and particularly if it is a primary (initial) infection in the mother. Cesarean delivery is warranted in such cases to prevent transmission to the neonate. Another important consequence of active genital HSV infection is that it increases the risk of HIV-1 acquisition and transmission.

The diagnosis is based on typical clinical findings and HSV detection. The purulent exudate is aspirated from the lesions and inoculated into a tissue culture. The viral cytopathic effect can be seen after 48 to 72 hours, and the virus can then be serotyped. In addition, some laboratories offer more sensitive polymerase chain reaction (PCR) tests, enzyme-linked immunosorbent assays (ELISA), and direct immunofluorescent antibody tests for detection of HSV in the lesional secretions. Individuals with primary, acute HSV infection do not have serum anti-HSV antibodies. Detection of anti-HSV antibodies in the serum is indicative of recurrent/latent infection.

There is no effective treatment for latent HSV; however, antiviral agents shorten the length of the initial and recurrent symptomatic phases. The ultimate solution is an effective vaccine, a tantalizing goal yet to be realized.

Other Lower Female Genital Tract Infections

As mentioned earlier, a variety of other viruses, fungi, and bacteria can also cause symptomatic infections of the lower genital tract. Those that are most common include:

- *Molluscum contagiosum* is a cutaneous or mucosal lesion caused by poxvirus (Fig. 22.3A). There are four types of molluscum contagiosum viruses (MCVs), MCV-1 to MCV-4, with MCV-1 being the most prevalent and MCV-2 being most often sexually transmitted. The infections are common in young children between 2 and 12 years of age, in whom they are transmitted through direct contact or shared articles (e.g., towels). Molluscum may affect any area of the skin but is most common on the trunk, arms, and legs. In adults, molluscum infections are typically sexually transmitted and affect the genitals, lower abdomen, buttocks, and inner thighs.

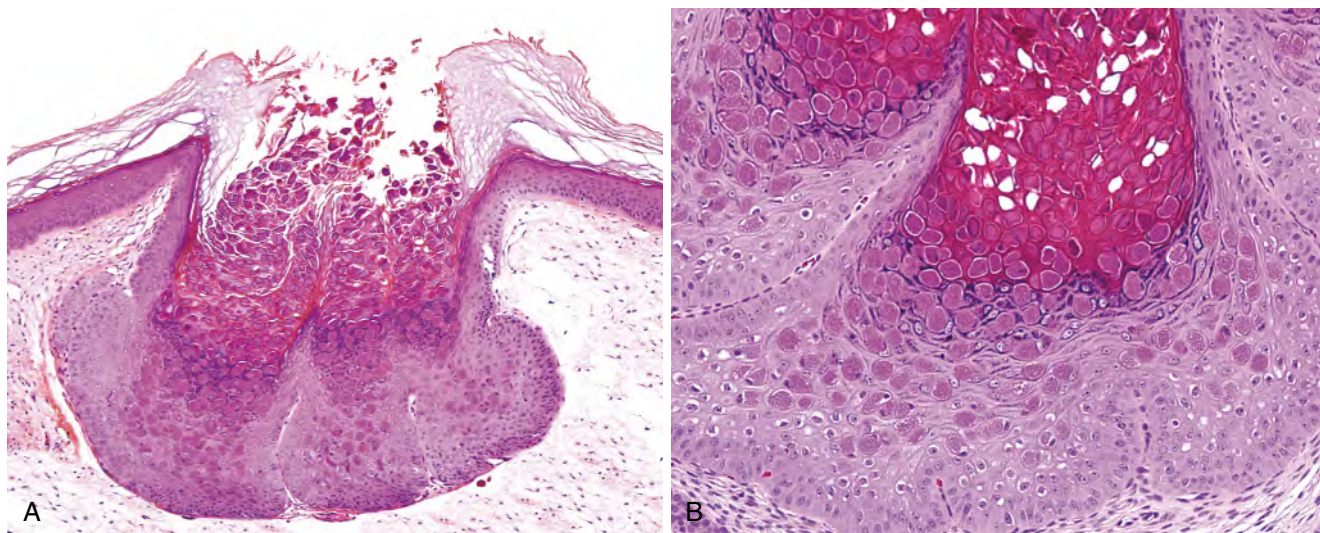


Fig. 22.3 Molluscum contagiosum infection. (A) Low-power appearance of a dome-shaped papule with dimpled center. (B) High-power magnification reveals intracytoplasmic viral inclusions.

The average incubation period is 6 weeks. Diagnosis is based on the characteristic clinical appearance of pearly, dome-shaped papules with a dimpled center. The papules measure 1 to 5 mm in diameter, and their central waxy core contains cells with *cytoplasmic viral inclusions* (see Fig. 22.3B).

- **Fungal infections**, especially those caused by yeast (*Candida*), are extremely common; in fact, yeast are part of many women's normal vaginal microflora, and the development of symptomatic candidiasis is typically a result of a disturbance in the patient's vaginal microbial ecosystem. Diabetes, antibiotics, pregnancy, and conditions that compromise neutrophil or Th17 T-cell function increase the risk of symptomatic candidal infection, which manifests itself by marked vulvovaginal pruritus, erythema, swelling, and curdlike vaginal discharge. Severe infection may result in mucosal ulcerations. The diagnosis is made by finding the pseudospores or filamentous fungal hyphae in wet KOH mounts of the discharge or on Papanicolaou (Pap) smear. Even though sexual transmission of yeast infection has been documented, candidiasis is not considered a sexually transmitted disease.
- *Trichomonas vaginalis* is a large, flagellated ovoid protozoan that is usually transmitted by sexual contact. Infected patients may be asymptomatic or may complain of yellow, frothy vaginal discharge, vulvovaginal discomfort, dysuria (painful urination), and dyspareunia (painful intercourse). The vaginal and cervical mucosa typically has a fiery-red appearance, with marked dilation of cervical mucosal vessels resulting in characteristic colposcopic appearance of "strawberry cervix."
- *Gardnerella vaginalis* is a gram-negative coccobacillus that is implicated as the main cause of *bacterial vaginosis* (vaginosis). Patients typically present with thin, green-gray, malodorous (fishy) vaginal discharge. Pap smears reveal superficial and intermediate squamous cells covered with a shaggy coat of coccobacilli. Bacterial cultures in such cases reveal *G. vaginalis* and other bacteria, including anaerobic peptostreptococci and aerobic α -hemolytic streptococci. In pregnant patients, bacterial vaginosis has been implicated in premature labor.
- *Ureaplasma urealyticum* and *Mycoplasma hominis* species account for some cases of vaginitis and cervicitis and have been implicated in chorioamnionitis and premature delivery in pregnant patients.
- *Chlamydia trachomatis* infections mainly take the form of cervicitis. However, in some patients the infection may ascend to the

uterus and fallopian tubes, resulting in endometritis and salpingitis; thus, chlamydia is one of the causes of pelvic inflammatory disease.

Infections Involving the Lower and Upper Genital Tract Pelvic Inflammatory Disease

Pelvic inflammatory disease (PID) is an infection that begins in the vulva or vagina and spreads upward to involve most of the structures in the female genital system, resulting in pelvic pain, adnexal tenderness, fever, and vaginal discharge. *Neisseria gonorrhoeae* continues to be a common cause of PID, the most serious complication of gonorrhea in females. *Chlamydia* infection is another well-recognized cause of PID. Infections after spontaneous or induced abortions and normal or abnormal deliveries (called *puerperal infections*) are also important causes of PID. In these situations, the infections are typically polymicrobial and may be caused by staphylococci, streptococci, coliforms, and *Clostridium perfringens*.

With gonococcus, inflammatory changes start to appear approximately 2 to 7 days after inoculation. The initial infection most commonly involves the endocervical mucosa, but it may also begin in the Bartholin gland and other vestibular, or periurethral, glands. From these sites, the organisms may spread upward to involve the fallopian tubes and tubo-ovarian region. The nongonococcal bacterial infections that follow induce abortion, dilation and curettage of the uterus, and other surgical procedures are thought to spread upward from the uterus through the lymphatics or venous channels rather than on the mucosal surfaces. Therefore, these infections tend to produce more inflammation within the deeper layers of the organs than gonococcal infections.

MORPHOLOGY

Gonococcal infection is characterized by marked acute inflammation of involved mucosal surfaces. Smears of the inflammatory exudate disclose phagocytosed gram-negative diplococci within neutrophils; however, definitive diagnosis requires culture or detection of gonococcal RNA or DNA. If infection spreads, the endometrium is usually spared (for unclear reasons), but within the fallopian tubes, an **acute suppurative salpingitis** ensues (Fig. 22.4A). The tubal mucosa becomes congested and diffusely infiltrated by neutrophils,

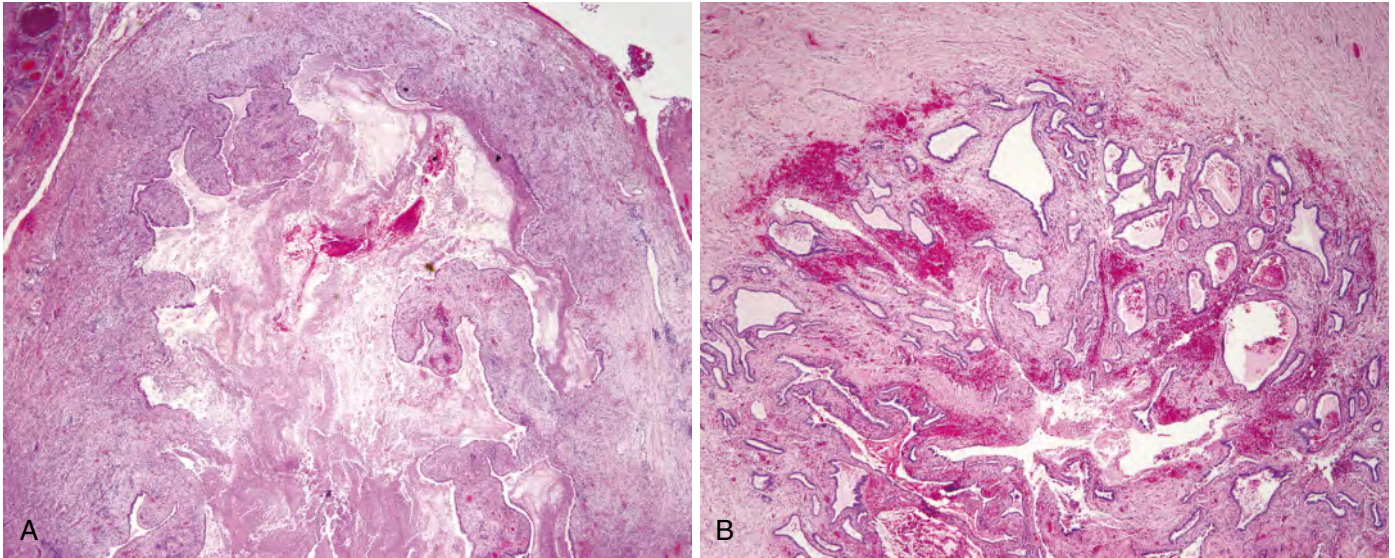


Fig. 22.4 Salpingitis. (A) Acute salpingitis; note the dilated tube lumen and edematous tubal plicae expanded by inflammatory cell infiltrates. Pus fills the center of the fallopian tube. (B) Chronic salpingitis showing scarring and fusion of the plicae with formation of glandlike spaces. Such scarring may cause infertility or ectopic tubal pregnancy.

plasma cells, and lymphocytes, resulting in epithelial injury and sloughing of the plicae. The tubal lumen fills with purulent exudate that may leak out of the fimbriated end. The infection may then spread to the ovary to create a **salpingo-oophoritis**. Collections of pus may accumulate within the ovary and tube (**tubo-ovarian abscess**) or tubal lumen (**pyosalpinx**) (see Fig. 22.4A). With time, the infecting organisms may disappear, but the tubal plicae, denuded of epithelium, adhere to one another and slowly fuse in a reparative, scarring process that forms glandlike spaces and blind pouches, referred to as **chronic salpingitis** (Fig. 22.4B). The scarring of the tubal lumen and fimbriae may prevent the uptake and passage of oocytes, leading to infertility or ectopic pregnancy. **Hydrosalpinx (dilation of the fallopian tube)** may also develop due to fusion of the fimbriae and subsequent accumulation of the tubal secretions resulting in tubal distention.

As compared to gonococcal infections, PID caused by staphylococci, streptococci, and the other puerperal invaders tend to show less involvement of the mucosa and fallopian tube lumen, and more inflammation within the deeper tissue layers. These infections often spread throughout the wall to involve the serosa and the broad ligaments, pelvic structures, and peritoneum. Bacteremia is a more frequent complication of streptococcal or staphylococcal PID than of gonococcal infections.

The acute complications of PID include peritonitis and bacteremia, which in turn may result in endocarditis, meningitis, and suppurative arthritis. The chronic sequelae of PID include infertility and tubal obstruction, ectopic pregnancy, pelvic pain, and intestinal obstruction due to adhesions between the bowel and pelvic organs. In the early stages, gonococcal infections are usually controlled with antibiotics, although antimicrobial-resistant isolates are increasing. Infections that become walled off in tubo-ovarian abscesses are difficult to eradicate with antibiotics, and it sometimes becomes necessary to remove the organs surgically. Postabortion and postpartum PIDs may also be amenable to treatment with antibiotics but are far more difficult to control because of the broad spectrum of pathogens that may be involved.

VULVA

Diseases of the vulva in the aggregate constitute only a small fraction of gynecologic practice. Many inflammatory diseases that affect skin elsewhere on the body also occur on the vulva, such as psoriasis, eczema, and allergic dermatitis. Because it is constantly exposed to secretions and moisture, the vulva is more prone to superficial infections than skin elsewhere on the body. Nonspecific vulvitis is particularly likely to occur in the setting of immunosuppression. Most skin cysts (epidermal inclusion cysts) and skin tumors such as squamous cell carcinoma, basal cell carcinoma, and melanoma can also occur in the vulva. Here we discuss vulvar disorders that are relatively specific and common, including Bartholin cyst, nonneoplastic epithelial disorders, benign exophytic lesions, and tumors of the vulva.

BARTHOLIN CYST

Infection of the Bartholin gland produces acute inflammation (adenitis) and may result in an abscess. Bartholin duct cysts are relatively common, occur at all ages, and result from obstruction of the duct by an inflammatory process. These cysts are usually lined by transitional or squamous epithelium. They may become large, up to 3 to 5 cm in diameter, and produce pain and local discomfort. Bartholin duct cysts are either excised or opened permanently (marsupialization).

NONNEOPLASTIC EPITHELIAL DISORDERS

Leukoplakia is a descriptive clinical term for opaque, white, plaque-like epithelial thickening that may produce pruritus and scaling. Leukoplakia (literally, *white plaques*) may be caused by a variety of benign, premalignant, or malignant disorders, including:

- Inflammatory dermatoses (e.g., psoriasis, chronic dermatitis)
- Lichen sclerosus and squamous cell hyperplasia
- Neoplasias, such as vulvar intraepithelial neoplasia (VIN), Paget disease, and invasive carcinoma

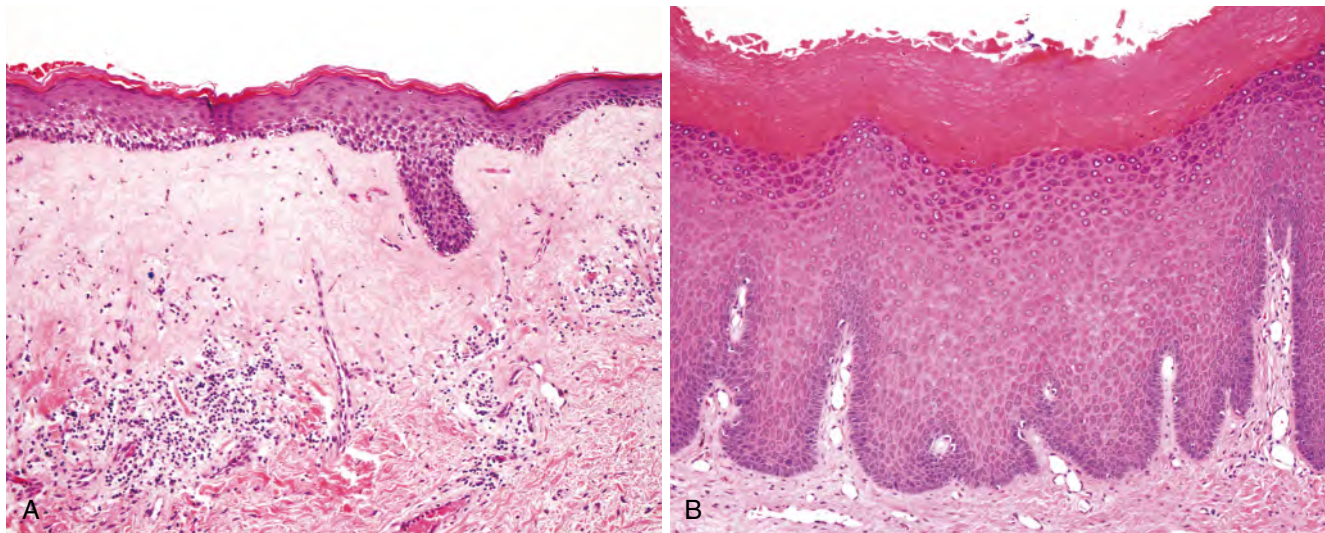


Fig. 22.5 Nonneoplastic epithelial vulvar disorders. (A) Lichen sclerosus. There is marked thinning of the epidermis, sclerosis of the superficial dermis, and chronic inflammatory cells in deeper dermis. (B) Lichen simplex chronicus, displaying thickened epidermis and hyperkeratosis.

Inflammatory dermatoses associated with leukoplakia are described in Chapter 25, while neoplastic disorders are discussed later in this chapter. Here the major nonneoplastic causes of leukoplakia—lichen sclerosus and lichen simplex chronicus—are briefly discussed.

Lichen Sclerosus

Lichen sclerosus presents as smooth, white plaques or macules that in time may enlarge and coalesce, producing a surface that resembles porcelain or parchment (see Chapter 25). When the entire vulva is affected, the labia become atrophic and adhered, and the vaginal orifice constricts. Histologically, the lesion is characterized by marked thinning of the epidermis (Fig. 22.5A), degeneration of the basal epithelial cells, excessive keratinization (hyperkeratosis), sclerotic changes of the superficial dermis, and a bandlike lymphocytic infiltrate in the underlying dermis. The disease occurs in all age groups but is most common in postmenopausal females. It may also be encountered elsewhere on the skin. Its pathogenesis is uncertain, but the presence of activated T cells in the subepithelial inflammatory infiltrate and the increased frequency of autoimmune disorders in affected females suggest an autoimmune reaction is involved. Although lichen sclerosus is not itself a premalignant lesion, females with lichen sclerosus have a slightly increased chance of developing HPV-independent squamous cell carcinoma of the vulva.

Lichen Simplex Chronicus

Previously called *hyperplastic dystrophy* or *squamous cell hyperplasia*, lichen simplex chronicus is a nonspecific condition resulting from rubbing or scratching of the skin to relieve pruritus. Clinically it presents as leukoplakia, and histologic examination reveals thickening of the epidermis (acanthosis) and hyperkeratosis (Fig. 22.5B). Lymphocytic infiltration of the dermis is sometimes present. The hyperplastic epithelium may show mitotic activity but lacks cellular atypia. While lichen simplex chronicus is not considered premalignant, it is sometimes present at the margins of HPV-independent vulvar cancers.

BENIGN EXOPHYTIC LESIONS

Benign raised (exophytic) or wartlike lesions of the vulva may be caused by infection or may be reactive conditions of unknown

etiology. *Condyloma acuminatum*, a papillomavirus-induced lesion, also called a *genital wart*, and syphilitic *condyloma latum* (described in Chapter 21) are consequences of sexually transmitted infections. Vulvar *fibroepithelial polyps*, or skin tags, are similar to skin tags occurring elsewhere on the skin. Vulvar *squamous papillomas* are benign exophytic proliferations covered by nonkeratinized squamous epithelium that develop on vulvar surfaces; they may be single or numerous (vulvar papillomatosis). The etiology of fibroepithelial polyps and squamous papillomas is unknown.

Condyloma Acuminatum

Condylomata acuminata are benign genital warts caused by low-risk HPV, mainly types 6 and 11. They may be solitary, but are more frequently multifocal, and they may involve vulvar, perineal, and perianal regions as well as the vagina and, less commonly, the cervix. The lesions are identical to those found on the penis and around the anus in males (Chapter 21). On histologic examination, they consist of papillary, exophytic, treelike cores of stroma covered by thickened squamous epithelium (Fig. 22.6A). The surface epithelium shows characteristic viral cytopathic changes referred to as *koilocytic atypia* (Fig. 22.6B), which manifest as nuclear enlargement, hyperchromasia, and a cytoplasmic perinuclear halo (see the Cervix section later in this chapter). Condylomata acuminata are not precancerous lesions. HPV vaccines (described later) provide excellent protection against infection by low-risk HPV and genital warts.

SQUAMOUS NEOPLASTIC LESIONS

Vulvar Intraepithelial Neoplasia and Vulvar Carcinoma

Carcinoma of the vulva is an uncommon malignant neoplasm (approximately one-eighth as frequent as cervical cancer) representing about 3% of genital cancers in the female; approximately two-thirds occur in females older than 60 years. Squamous cell carcinoma is the most common histologic type of vulvar cancer. In terms of etiology, pathogenesis, and histologic features, vulvar squamous cell carcinoma is divided into two groups:

- HPV-associated squamous cell carcinomas are related to infection with high-risk HPVs, most commonly HPV-16. These are less

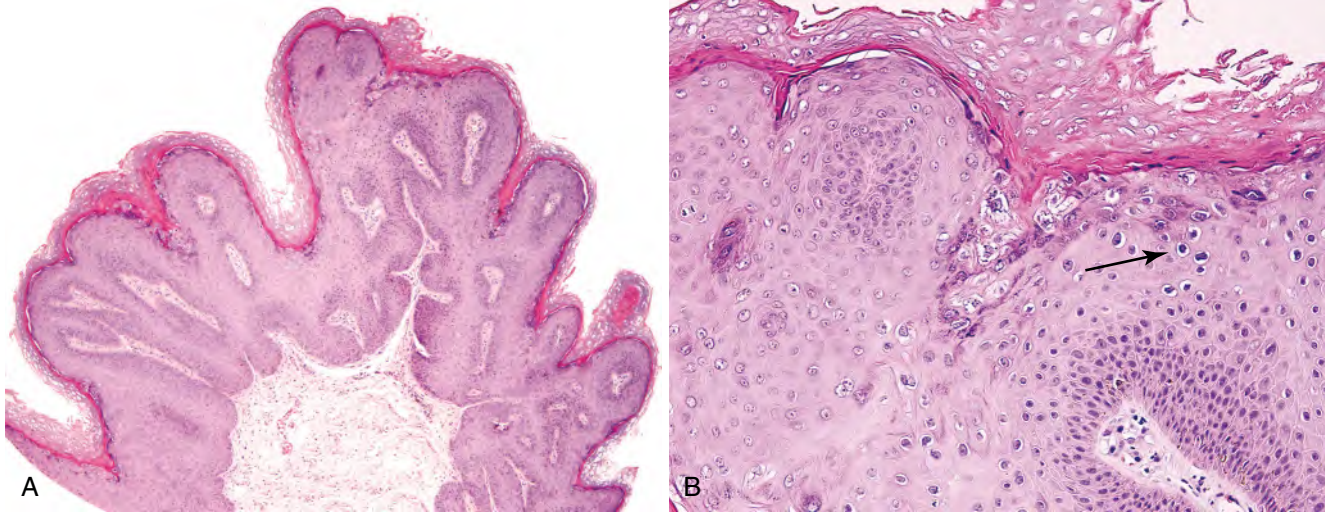


Fig. 22.6 Condyloma acuminatum. (A) Low-power view showing exophytic, papillary architecture. (B) High-power view reveals human papillomavirus cytopathic effect (koilocytic atypia) characterized by atypical, enlarged, hyperchromatic nuclei with perinuclear halos (arrow).

common (30% of cases) and occur in younger females (average 60 years of age).

- HPV-independent squamous cell carcinomas are more common (70% of cases) and occur predominantly in older females (average 75 years of age).

HPV-associated carcinoma develops from in situ precursor lesions called squamous intraepithelial lesions, also known as *vulvar intraepithelial neoplasia* (VIN). This form of VIN occurs mainly in reproductive age females and the risk factors are the same as those associated with cervical squamous intraepithelial lesions (e.g., young age at first intercourse, multiple sexual partners, male partner with multiple sexual partners), as both are related to HPV infection. VIN is frequently multicentric, and 10% to 30% of patients with VIN also have vaginal or cervical HPV-related lesions. Spontaneous regression of classic VIN has been reported, usually in younger females. The risk of progression to invasive carcinoma is higher in females who are older than 45 years of age or who are immunosuppressed.

HPV-independent squamous cell carcinoma occurs most often in individuals with long-standing lichen sclerosus or lichen simplex chronicus and is not related to HPV. It arises from a precursor lesion referred to as *differentiated vulvar intraepithelial neoplasia* (dVIN). It is postulated that chronic epithelial irritation in lichen sclerosus or lichen simplex chronicus may contribute to a gradual evolution to the malignant phenotype, presumably through acquisition of driver mutations in oncogenes and tumor suppressor genes. Somatic mutations in *TP53* are commonly found in dVIN and HPV-independent vulvar squamous cell carcinoma. More recently other morphologic forms of HPV-independent vulvar intraepithelial neoplasia and associated squamous carcinomas have been described. Studies have shown that these lesions lack somatic mutations in *TP53* and instead harbor mutations in other tumor suppressor genes and oncogenes (e.g., *PIK3CA*, *NOTCH1*, and *HRAS*). Ongoing molecular and morphologic studies will likely result in future changes in the nomenclature and classification of these HPV-independent lesions.

MORPHOLOGY

HPV-associated VIN may present as a discrete and white (hyperkeratotic) or slightly raised and pigmented lesion on the vulva. Microscopically, it is

characterized by epidermal thickening, nuclear atypia, increased mitoses, and lack of cellular maturation (Fig. 22.7A), features analogous to those seen in cervical squamous intraepithelial lesions (see the Cervix section later in this chapter). Invasive carcinomas that arise from classic VIN may be exophytic or indurated with central ulceration. On histologic examination, the tumors can have a basaloid appearance (Fig. 22.7B) and consist of nests and cords of small, tightly packed cells that lack maturation and resemble the basal layer of the normal epithelium. The tumor may have foci of central necrosis. Alternatively, it can have a warty appearance due to an exophytic, papillary architecture and prominent koilocytic atypia.

HPV-independent differentiated VIN is characterized by marked atypia of the basal layer of the squamous epithelium and normal-appearing differentiation of the more superficial layers (Fig. 22.8A). Invasive keratinizing squamous cell carcinomas that arise in differentiated VIN contain nests and tongues of malignant squamous epithelium with prominent central keratin pearls (Fig. 22.8B).

The risk of cancer development in VIN depends on duration and extent of disease, and the immune status of the patient. Invasive carcinomas associated with lichen sclerosus, squamous cell hyperplasia, and differentiated VIN may develop in an insidious fashion and may be misinterpreted as dermatitis or leukoplakia for long periods and behave more aggressively than HPV-associated squamous cell carcinoma. Once invasive cancer develops, the risk of metastatic spread is determined by the size of tumor, the depth of invasion, and whether there is lymphatic invasion. The initial spread is to inguinal, pelvic, iliac, and periaortic lymph nodes. Ultimately, hematogenous dissemination to the lungs, liver, and other internal organs may occur. Patients with lesions less than 2 cm in diameter have a 90% 5-year survival rate after treatment with vulvectomy and lymphadenectomy; however, larger lesions with lymph node involvement have a poor prognosis.

GLANDULAR NEOPLASTIC LESIONS

Like the breast, the vulva contains modified apocrine sweat glands. Presumably because of these “breastlike” features, the vulva may be involved by tumors with counterparts in the breast, the most common of which are papillary hidradenoma and extramammary Paget disease.

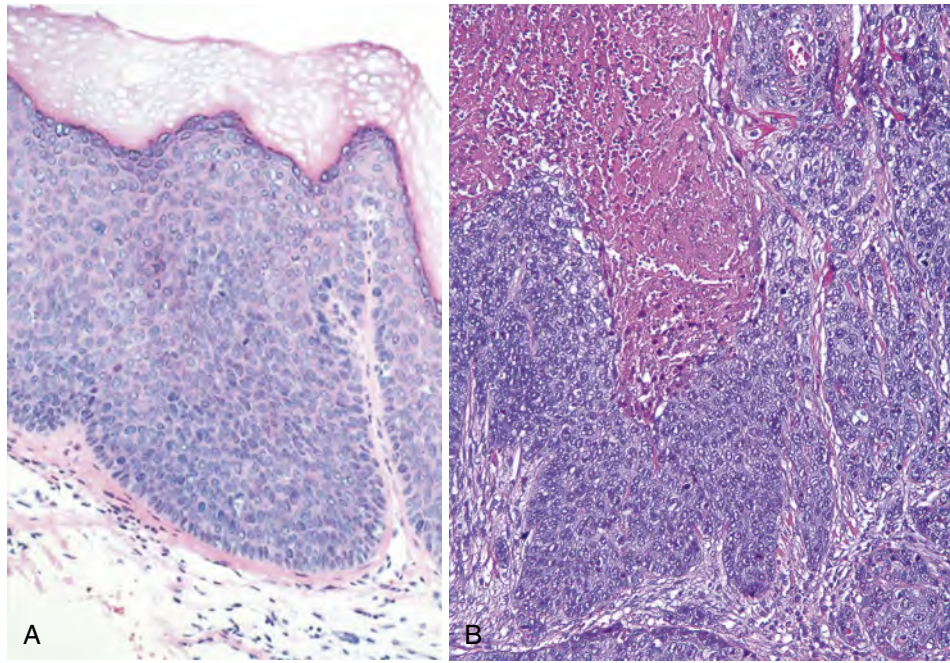


Fig. 22.7 Human papillomavirus (HPV)-related vulvar preneoplastic and malignant lesions. (A) Classic vulvar intraepithelial neoplasia (HPV positive), showing small, immature basaloid cells encompassing the full thickness of the epithelium. No invasion is present. (B) Basaloid vulvar carcinoma (HPV positive), composed of invasive small, immature (basaloid) cells. There is a focus of necrosis (*red area*).

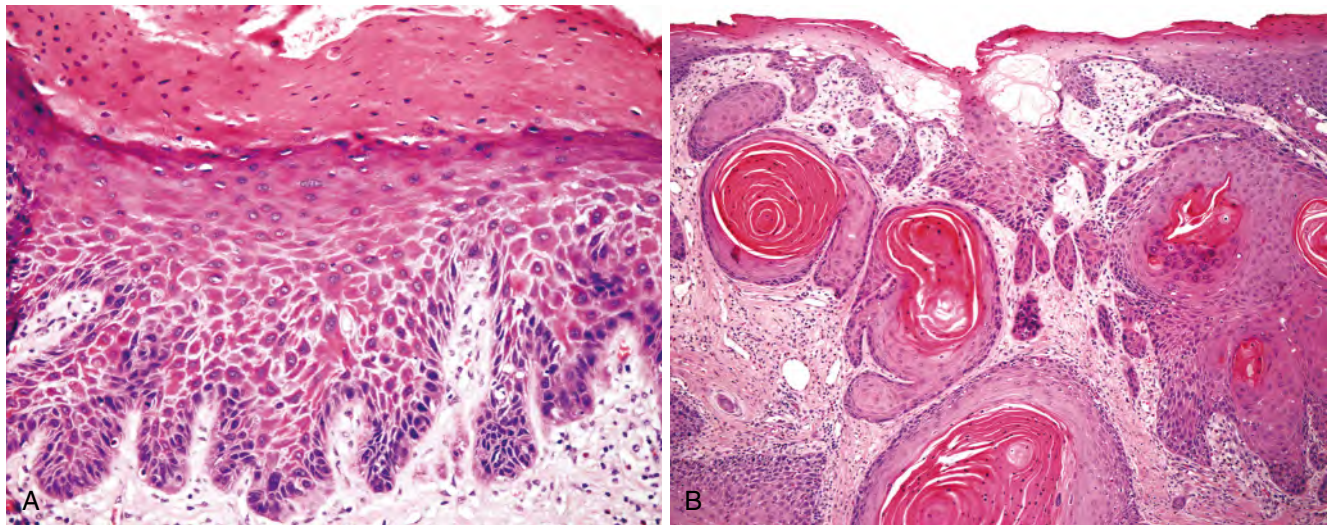


Fig. 22.8 Non-human papillomavirus (HPV) vulvar preneoplastic and malignant lesions. (A) Differentiated vulvar intraepithelial neoplasia (HPV negative), showing maturation of the superficial layers, hyperkeratosis, and atypia of basal epithelial cells. No invasion is present. (B) Well-differentiated, keratinizing squamous cell carcinoma of the vulva (HPV negative) composed of invasive tumor nests with central keratin pearls.

Papillary Hidradenoma

Papillary hidradenoma presents as a sharply circumscribed nodule, most commonly on the labia majora or interlabial folds, and may be confused clinically with carcinoma because of its tendency to ulcerate. Its histologic appearance is identical to that of intraductal papilloma of the breast and consists of papillary projections covered by two cell layers, an upper layer of columnar secretory cells and a deeper layer of flattened myoepithelial cells. These myoepithelial elements are characteristic of sweat glands and sweat gland tumors (Fig. 22.9).

Extramammary Paget Disease

This curious and rare lesion of the vulva is similar in its manifestations to Paget disease of the breast (Chapter 23). In the vulva, it presents as a pruritic, red, crusted, maplike area, usually on the labia majora.

MORPHOLOGY

Paget disease is a distinctive intraepithelial proliferation of malignant cells. **Paget cells** are larger than surrounding keratinocytes and are seen singly or

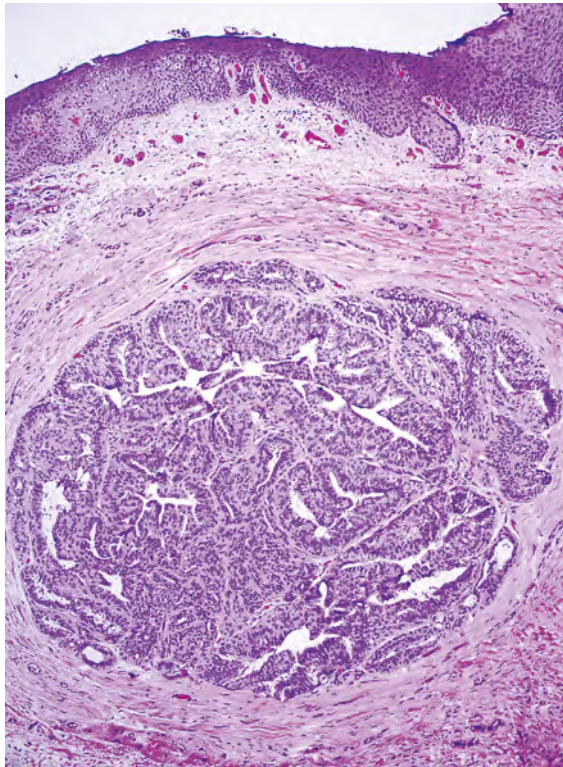


Fig. 22.9 Papillary hidradenoma of the vulva, a well-circumscribed tumor composed of benign papillary projections covered with columnar secretory epithelium and underlying myoepithelial cells.

in small clusters within the epidermis (Fig. 22.10A). The cells have pale cytoplasm containing mucopolysaccharide that stains with periodic acid–Schiff (PAS), Alcian blue, or mucicarmine stains. In addition, unlike squamous epithelium, the cells express cytokeratin 7 (Fig. 22.10B). Ultrastructurally, Paget cells display apocrine, eccrine, and keratinocyte differentiation and presumably arise from multipotent cells found within the mammary gland–like ducts of the vulvar skin.

In contrast to Paget disease of the nipple, in which 100% of patients have an underlying ductal breast carcinoma, vulvar Paget disease is typically not associated with underlying cancer and is confined to the vulvar epidermis. Treatment consists of wide local excision, which is necessary because Paget cells spread laterally within the epidermis and may be present beyond the confines of the grossly visible lesion. Intraepidermal Paget disease may persist for many years or even decades without invasion or metastases. In the rare instances when invasion develops, the prognosis is poor.

KEY CONCEPTS

Vulvar Squamous Cell Carcinoma

- Approximately 30% of vulvar cancers are caused by infection with high-risk HPVs, principally HPV-16. These cancers develop from an in situ lesion termed squamous intraepithelial lesion (also referred to as vulvar intraepithelial neoplasia).
- Most vulvar cancers (70%) are not related to HPV and develop in a background of lichen sclerosis or lichen simplex chronicus, most commonly from a premalignant lesion called differentiated vulvar intraepithelial neoplasia (dVIN).

VAGINA

The vagina is a portion of the female genital tract that is remarkably resistant to diseases that affect nearby structures. For example, in the adult, inflammation that begins in the vulva and perivulvar structures often spreads to the cervix without significant involvement of the vagina. Primary lesions of the vagina are rare, the most serious being vaginal squamous cell carcinoma; they are discussed only briefly here.

DEVELOPMENTAL ANOMALIES

Septate, or double, vagina is an uncommon anomaly that arises from a failure of müllerian duct fusion and is accompanied by a double uterus (*uterus didelphys*). These and other anomalies of the external genitalia may be the manifestations of genetic syndromes, in utero exposure to diethylstilbestrol (DES, used to prevent threatened abortions in the

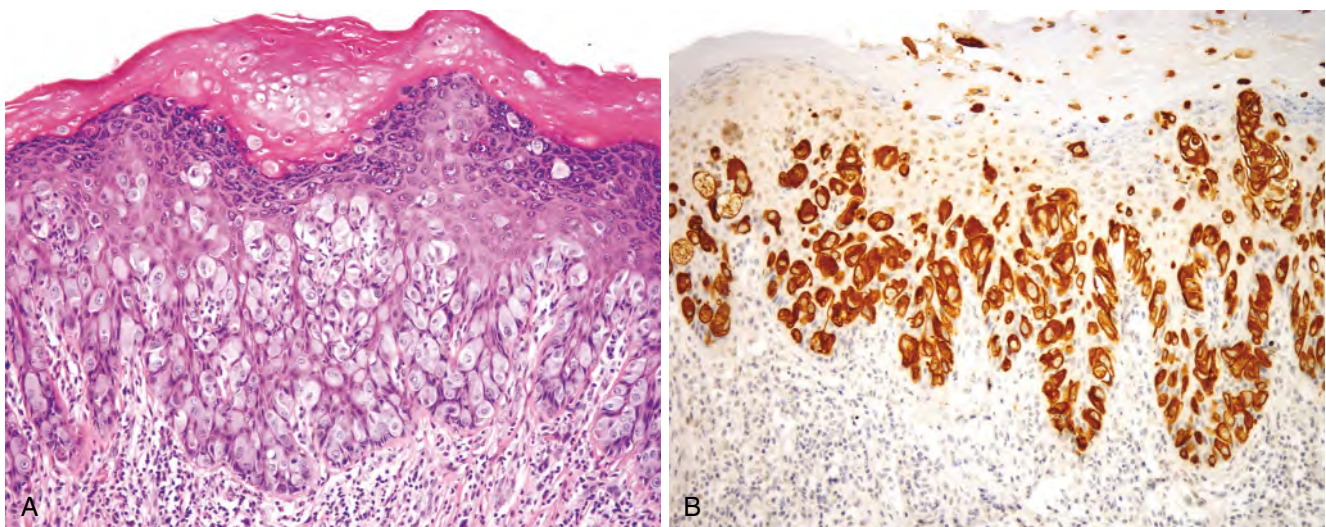


Fig. 22.10 Paget disease of the vulva. (A) The epidermis is infiltrated by large cells with pale-pink cytoplasm that are spreading along the basal portion of the squamous epithelium. There is inflammation in the underlying dermis. (B) Immunostaining for cytokeratin 7 highlights the intraepidermal Paget cells.

1940s to 1960s), or other unknown factors that perturb reciprocal epithelial-stromal signaling during fetal development.

During embryonal development, the vagina is initially covered by columnar, endocervical type epithelium. This is normally replaced by squamous epithelium advancing upward from the urogenital sinus. Small patches of residual glandular epithelium may persist into adult life; such areas are recognized as *vaginal adenosis*, which on examination presents as red, granular areas that stand out from the surrounding normal pale-pink vaginal mucosa. Microscopically, adenosis consists of columnar mucinous epithelium indistinguishable from endocervix. Adenosis is found in only a small percentage of females but has been reported in 35% to 90% of females exposed to DES in utero. Rare cases of clear cell carcinoma arising in DES-related adenosis were recorded in teenagers and young females in the 1970s and 1980s, resulting in discontinuation of DES treatment.

Gartner duct cysts are relatively common lesions found along the lateral walls of the vagina that are derived from wolffian (mesonephric) duct rests. They consist of 1- to 2-cm fluid-filled submucosal cysts. Other cysts, including mucus cysts, which occur in the proximal vagina, are derived from müllerian epithelium. Another müllerian-derived lesion, endometriosis (described later), may occur in the vagina and clinically simulate a neoplasm.

PREMALIGNANT AND MALIGNANT NEOPLASMS OF THE VAGINA

Most benign tumors of the vagina occur in reproductive age females and include stromal tumors (stromal polyps), leiomyomas, and hemangiomas. The most common malignant tumor to involve the vagina is carcinoma spreading from the cervix, followed by primary squamous cell carcinoma of the vagina. Infants may develop a unique, rare malignancy—rhabdomyosarcoma.

Squamous Intraepithelial Lesions and Squamous Cell Carcinoma

Virtually all primary carcinomas of the vagina are squamous cell carcinomas associated with high-risk HPV infection. Vaginal carcinoma is extremely uncommon (about 0.6 per 100,000 females yearly) and accounts for about 1% of malignant neoplasms in the female genital tract. The greatest risk factor is a previous carcinoma of the cervix or vulva; 1% to 2% of females with an invasive cervical carcinoma eventually develop a vaginal squamous cell carcinoma. Squamous cell carcinoma of the vagina arises from a premalignant lesion, squamous intraepithelial lesion (also referred to as *vaginal intraepithelial neoplasia*), analogous to cervical squamous intraepithelial lesion (described later). Most often the invasive tumor affects the upper vagina, particularly the posterior wall at the junction with the ectocervix. The lesions in the lower two-thirds of the vagina metastasize to the inguinal nodes, whereas lesions in the upper vagina tend to spread to regional iliac nodes.

Embryonal Rhabdomyosarcoma

This uncommon vaginal tumor composed of malignant embryonal rhabdomyoblasts is most frequently found in infants and children younger than 5 years of age. These tumors tend to grow as polypoid, rounded, bulky masses that have the appearance and consistency of grapelike clusters (in the past, this was referred to as *sarcoma botryoides* for this grapelike growth pattern) (Fig. 22.11). The tumor cells are small and have oval nuclei, with small protrusions of cytoplasm from one end, resembling a tennis racket. Rarely, striations (indicative of muscle differentiation) can be seen within the cytoplasm. Beneath the vaginal epithelium, the tumor cells are crowded in a so-called cambium layer, but in the deep regions they lie within a loose edematous fibromyxomatous



Fig. 22.11 Sarcoma botryoides (embryonal rhabdomyosarcoma) of the vagina appearing as a polypoid mass protruding from the vagina. (Courtesy Dr. Michael Donovan, Children's Hospital, Boston.)

stroma that may contain many inflammatory cells. Such lesions can be mistaken for benign inflammatory polyps. The tumors tend to invade locally and cause death by penetration into the peritoneal cavity or by obstruction of the urinary tract. Conservative surgery coupled with chemotherapy offer the best hope, particularly in cases diagnosed sufficiently early.

CERVIX

Anatomically the cervix consists of the external vaginal portio (ectocervix) and the endocervical canal. The ectocervix is visible on vaginal examination and is covered by a mature squamous epithelium that is continuous with the vaginal wall. The squamous epithelium converges centrally at a small opening termed the *external os* that leads to the endocervical canal. The endocervix is lined by columnar, mucus-secreting epithelium. The point where the squamous and the columnar epithelium meet is referred to as the *squamocolumnar junction* (Fig. 22.12). The position of the junction is variable and changes with age and hormonal influence, but in general the junction moves upward into the endocervical canal with time. The replacement of the glandular epithelium by advancing squamous epithelium is a process called squamous metaplasia. The area of the cervix where the columnar epithelium coexists with the squamous epithelium is termed the “*transformation zone*.” The unique epithelial environment of the cervix renders it highly susceptible to infection with HPV, the main cause of cervical cancer. Immature squamous cells in the transformation zone are most susceptible to HPV infection, and as a result most cervical precursor lesions and cervical cancers develop in the transformation zone.

INFLAMMATIONS

Acute and Chronic Cervicitis

At the onset of menarche, the production of estrogens by the ovary stimulates maturation of the cervical and vaginal squamous mucosa and formation of intracellular glycogen vacuoles in the squamous cells. As these cells are shed, the glycogen provides a substrate for various endogenous vaginal aerobes and anaerobes, but particularly lactobacilli, which

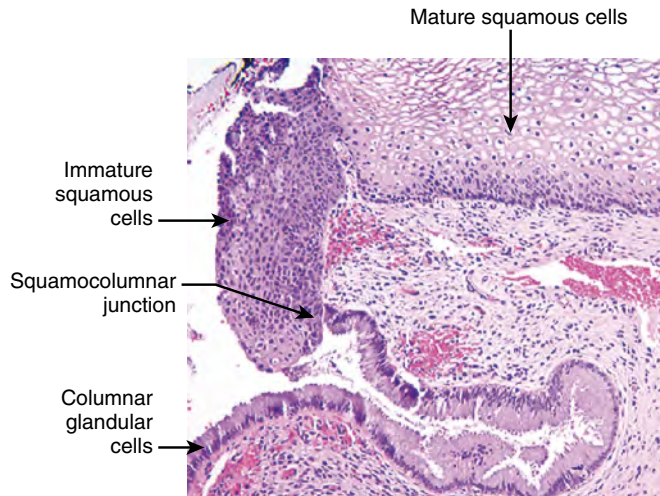


Fig. 22.12 Cervical squamocolumnar junction showing a transition from mature, glycogenized squamous epithelium to columnar endocervical glandular epithelium. The superficial, mature squamous epithelial cells are not susceptible to human papillomavirus (HPV) infection. The HPV-susceptible cells include immature basal squamous cells and endocervical glandular cells.

are the dominant microbial species in the normal vagina. Lactobacilli produce lactic acid, which maintains the vaginal pH below 4.5, suppressing the growth of other saprophytic and pathogenic organisms. In addition, at low pH, lactobacilli produce bacteriotoxic hydrogen peroxide (H_2O_2). If the pH becomes alkaline due to bleeding, sexual intercourse, or vaginal douching, H_2O_2 production by lactobacilli decreases. Antibiotic therapy that suppresses lactobacilli can also cause the pH to rise. In each of these settings, the altered vaginal environment promotes the overgrowth of other microorganisms, which may result in cervicitis or vaginitis. Some degree of cervical inflammation may be found in virtually all females and is usually of little clinical consequence. However, infections by gonococci, chlamydiae, mycoplasmas, and HSV may produce significant acute or chronic cervicitis and are important to identify due to their association with upper genital tract disease, complications during pregnancy, and sexual transmission. Marked cervical inflammation produces reparative and reactive changes of the epithelium and shedding of atypical-appearing squamous cells, and therefore may cause an abnormal Pap test result.

ENDOCERVICAL POLYPS

Endocervical polyps are common benign exophytic growths that arise within the endocervical canal. They vary from small, sessile “bumps” to large polypoid masses that may protrude through the cervical os. Histologically, they are composed of a fibrous stroma covered by mucus-secreting endocervical glands, often accompanied by inflammation (Fig. 22.13). Their main significance is that they may be the source of irregular vaginal “spotting” or bleeding that arouses suspicion of a more serious lesion. Simple curettage or surgical excision is curative.

PREMALIGNANT AND MALIGNANT NEOPLASMS OF THE CERVIX

Worldwide, cervical carcinoma is the fourth most common cancer in females, with an estimated 604,127 new cases in 2020, of which more than one-half will prove fatal. In the United States, it was estimated in 2023 that 13,960 females would be diagnosed with cervical

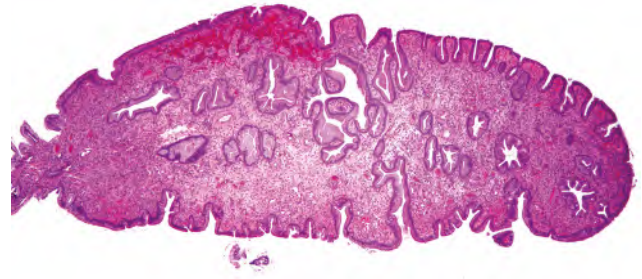


Fig. 22.13 Endocervical polyp composed of a dense fibrous stroma covered with endocervical columnar epithelium.

cancer and that 4310 females would die of the disease. Fifty years ago, carcinoma of the cervix was the leading cause of cancer death in females in the United States, but the death rate has declined by 75%. No form of cancer better documents the remarkable benefits of effective prevention, screening, early diagnosis, and curative therapy than does cancer of the cervix. Most credit for these dramatic gains belongs to the effectiveness of the Pap test in detecting cervical precursor lesions, some of which progress to cancer if untreated; in addition, the Pap test can also detect low-stage, highly curable cancers. The accessibility of the cervix to Pap testing and visual examination (colposcopy) as well as the slow progression of precursor lesions to invasive carcinoma (typically over the course of decades) provides ample time for screening, detection, and preventive treatment. Notably, recent studies show that HPV vaccination has led to an 81% drop in HPV infections and a 40% reduction in squamous intraepithelial lesions caused by the HPV types that are most commonly associated with cancer.

Pathogenesis

High-risk HPVs are by far the most important factor in the development of cervical cancer. HPVs are DNA viruses that are grouped into those of high and low oncogenic risk based on their genotypes. There are 15 high-risk HPVs that are currently known, but HPV-16 alone accounts for almost 60% of cervical cancer cases, and HPV-18 accounts for another 10% of cases; other HPV types contribute to less than 5% of cases individually. High-risk HPVs are also implicated in squamous cell carcinomas arising at many other sites, including the vagina, vulva, penis, anus, tonsil, and other oropharyngeal locations. As noted earlier, low-risk HPVs are the cause of sexually transmitted vulvar, perineal, and perianal warts (condyloma acuminatum).

Genital HPV infections are extremely common; most of them are asymptomatic, do not cause any tissue changes, and therefore are not detected on Pap test. The prevalence of HPV in cervical smears in females with normal Pap test results peaks between 20 and 24 years of age, around the onset of sexual activity, and subsequently declines as protective immunity is established. Most HPV infections are transient; overall, 50% of HPV infections are cleared within 8 months, and 90% of infections are cleared within 2 years. The duration of the infection is related to HPV type, as infections with high-risk HPVs take longer to clear on average than infections with low-risk HPVs (13 months vs. 8 months, respectively). Persistent infection increases the risk of the development of cervical precursor lesions and subsequent carcinoma.

Productive, persistent HPV infection requires viral entry into immature basal epithelial cells. Surfaces covered with mature, intact squamous epithelium, such as the ectocervix, vagina, vulva, penis, and oropharynx, are normally resistant to HPV infection. Sites in the female

genital tract that are susceptible to infection include areas of squamous epithelial trauma and repair, where the virus may access basal cells, and the immature metaplastic squamous cells that are present at the squamocolumnar junction of the cervix (see Fig. 22.12). The cervix, with its relatively large areas of immature squamous metaplastic epithelium, is particularly vulnerable to HPV infection. Other sites in the body that are vulnerable to HPV infection include the squamocolumnar junction of the anus and the squamous cells of oropharyngeal tonsillar crypts, both relatively common sites of HPV-associated cancers in individuals who practice anal or oral sex, respectively.

The ability of HPV to act as a carcinogen depends on the viral E6 and E7 proteins, which interfere with the activity of the key tumor suppressor proteins p53 and RB, respectively. Although HPV infects immature squamous cells, viral replication occurs in maturing squamous cells. Normally, these more mature cells are arrested in the G₁ phase of the cell cycle, but they continue to actively progress through the cell cycle when infected with HPV, which uses the host cell DNA synthesis machinery to replicate its own genome. As you will recall from Chapter 7, viral E7 protein promotes the proteosomal degradation of RB, and also binds and inhibits p21 and p27, two important cyclin-dependent kinase inhibitors. Removal of these controls enhances cell cycle progression, and impairs the ability of cells to repair DNA damage. The DNA repair defect in infected cells is exacerbated by E6 proteins encoded by high-risk HPV subtypes, which bind p53 and promote its degradation by the proteasome. In addition, E6 up-regulates the expression of telomerase, which leads to cellular immortalization. The net effect is increased proliferation of cells that are prone to acquire additional mutations that may lead to cancer development. By contrast to high-risk HPVs, the E7 proteins of low-risk HPVs bind RB with lower affinity, while the E6 proteins of low-risk HPVs fail to bind p53 altogether and instead appear to dysregulate growth and survival by interfering with the Notch signaling pathway.

Another factor that contributes to malignant transformation by HPV is the physical state of the virus. The viral DNA is integrated into the host cell genome in most cancers. This configuration increases the expression of *E6* and *E7* genes, and may also dysregulate oncogenes near the sites of viral insertion, such as *MYC*. Alternatively, viral DNA

is extrachromosomal (episomal) in precursor lesions associated with high-risk HPVs and in condylomata associated with low-risk HPVs.

Even though HPV is firmly established as the major cause of cervical cancer, it is worth remembering that although a high percentage of young females are infected with one or more HPV types during their reproductive years, only a few develop cancer. Thus, other factors, such as exposure to co-carcinogens and host immune status, influence whether an HPV infection regresses or persists and eventually leads to cancer.

Cervical Squamous Intraepithelial Lesions

The classification of cervical precursor lesions has evolved over time, and the terms from the various classification systems are used interchangeably. Hence a brief review of the terminology is warranted. The oldest classification system grouped lesions as having mild dysplasia on one end and severe dysplasia/carcinoma in situ on the other. This was followed by the *cervical intraepithelial neoplasia* (CIN) classification, with mild dysplasia termed CIN 1, moderate dysplasia termed CIN 2, and severe dysplasia termed CIN 3. Because the decision regarding patient management is two-tiered (observation vs. surgical treatment), the three-tier classification system has been recently simplified to a two-tiered system, with CIN 1 renamed low-grade squamous intraepithelial lesion (LSIL) and CIN 2 and CIN 3 combined into one category referred to as high-grade squamous intraepithelial lesion (HSIL) (Fig. 22.14).

Most LSIL cases regress spontaneously, and only a small percentage progress to HSIL. LSIL represents a productive HPV infection in which there is a high level of viral replication, but only mild alterations in the growth of host cells. For these reasons, LSIL is not treated like a premalignant lesion. LSIL is approximately 10 times more common than HSIL.

By contrast to LSIL, HSIL has a high risk for progression to carcinoma. In HSIL, there is a progressive deregulation of the cell cycle by HPV, which results in increased cellular proliferation, decreased or arrested epithelial maturation, and a lower rate of viral replication as compared with LSIL. Derangement of the cell cycle in HSIL may become irreversible and lead to a fully transformed malignant phenotype.

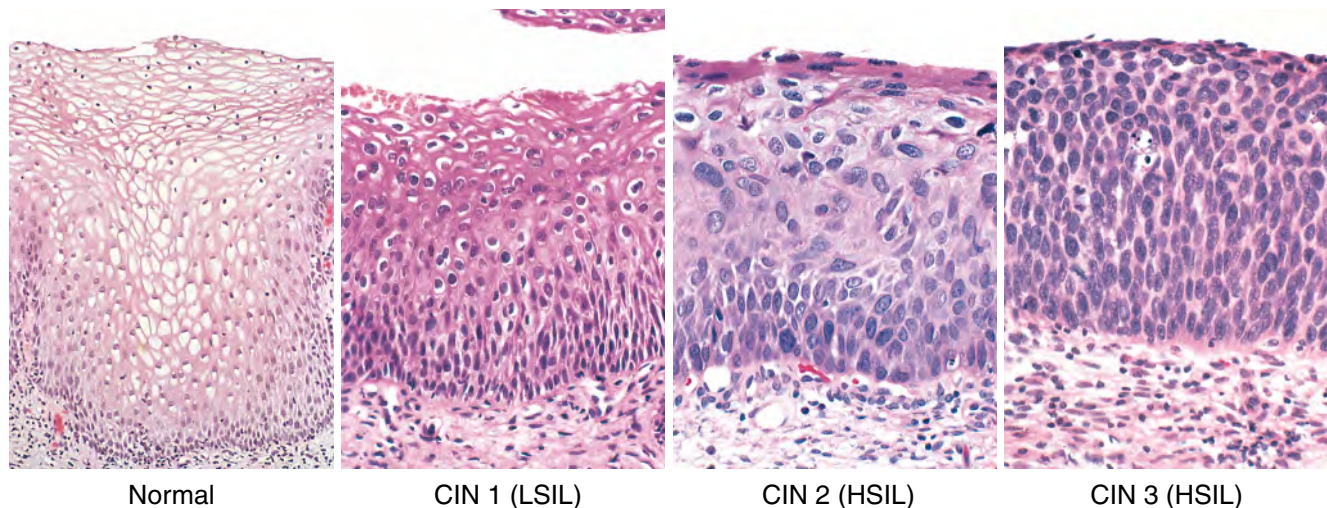


Fig. 22.14 Spectrum of cervical intraepithelial neoplasia with normal squamous epithelium for comparison. Cervical intraepithelial neoplasia 1 (CIN 1)/low-grade squamous intraepithelial lesion (LSIL) with koilocytic atypia; CIN 2/high-grade squamous intraepithelial lesion (HSIL) with progressive atypia and expansion of the immature basal cells above the lower third of the epithelial thickness; CIN 3/HSIL with diffuse atypia, loss of maturation, and expansion of immature basal cells to the epithelial surface.

MORPHOLOGY

The diagnosis of squamous intraepithelial lesion is based on identification of nuclear atypia characterized by nuclear enlargement, hyperchromasia (dark staining), coarse chromatin granules, and variation in nuclear size and shape (see Fig. 22.14). The nuclear changes are often accompanied by cytoplasmic “halos.” At an ultrastructural level, these “halos” consist of perinuclear vacuoles, a cytopathic change created in part by the HPV-encoded protein E5, which localizes to the membranes of the endoplasmic reticulum. Nuclear alterations associated with perinuclear halos are termed **kilocytic atypia**. The grading of a squamous intraepithelial lesion into low or high grade is based on expansion of the immature cell layer from its normal, basal location. If the immature squamous cells are confined to the lower one-third of the epithelium, the lesion is called LSIL; if they expand to the upper two-thirds of the epithelial thickness, it is called HSIL.

The histologic features of LSIL correlate with HPV replication and changes in host cell growth and gene expression (Fig. 22.15). The highest viral loads (assessed by in situ hybridization for HPV DNA; Fig. 22.15B) are found in maturing keratinocytes in the upper half of the epithelium. HPV E6 and E7 proteins prevent cell cycle arrest. As a result, cells in the upper portion of the epithelium express markers of actively dividing cells, such as Ki-67 (Fig. 22.15C), that are normally confined to the basal layer of the epithelium. Disturbed growth regulation also leads to overexpression of p16, a cyclin-dependent kinase inhibitor (Fig. 22.15D). Immunohistochemical expression of Ki-67 and p16 are both highly correlated with HPV infection and are useful for confirmation of the diagnosis in equivocal cases of SIL.

More than 80% of LSILs and 100% of HSILs are associated with high-risk HPVs, with HPV-16 being the most common HPV type in both lesions. Table 22.1 shows rates of regression and progression of SILs within a 2-year follow-up. Although the majority of HSILs develop from LSILs, approximately 20% of cases of HSIL develop de novo, independent of any preexisting LSIL. The rates of progression

TABLE 22.1 Natural History of Squamous Intraepithelial Lesions With Approximate 2-Year Follow-up

Lesion	Regress	Persist	Progress
LSIL	60%	30%	10% to HSIL
HSIL	30%	60%	10% to carcinoma ^a

^aProgression within 2 to 10 years.

HSIL, High-grade squamous intraepithelial lesion; LSIL, low-grade squamous intraepithelial lesion.

vary and although HPV type—especially HPV 16—is associated with increased risk, it is difficult to predict the outcome in an individual patient. These findings underscore that the risk of precursor lesions and cancer is conferred only in part by HPV type. Progression to invasive carcinoma, when it occurs, on average takes place over several decades.

Cervical Carcinoma

The average age of patients with invasive cervical carcinoma is between 45 and 50 years. Squamous cell carcinoma is the most common histologic subtype, accounting for approximately 80% of cases. The second most common tumor type is adenocarcinoma, which constitutes about 15% of cervical cancer cases, most of which develop from a precursor lesion called *adenocarcinoma in situ*. Adenosquamous and neuroendocrine carcinomas are rare cervical tumors that account for the remaining 5% of cases. All the aforementioned tumor types are caused by high-risk HPVs. The progression time from in situ to invasive adenosquamous and neuroendocrine carcinomas is shorter than in squamous cell carcinoma, and patients with these tumors often present with advanced disease and have a less favorable prognosis. There is a rare, but aggressive, form of HPV-independent cervical adenocarcinoma, that often exhibits gastric-type differentiation, has frequent mutations in *STK11* and *TP53*, and may be seen in association with Peutz-Jeghers syndrome (germline *STK11* mutations).

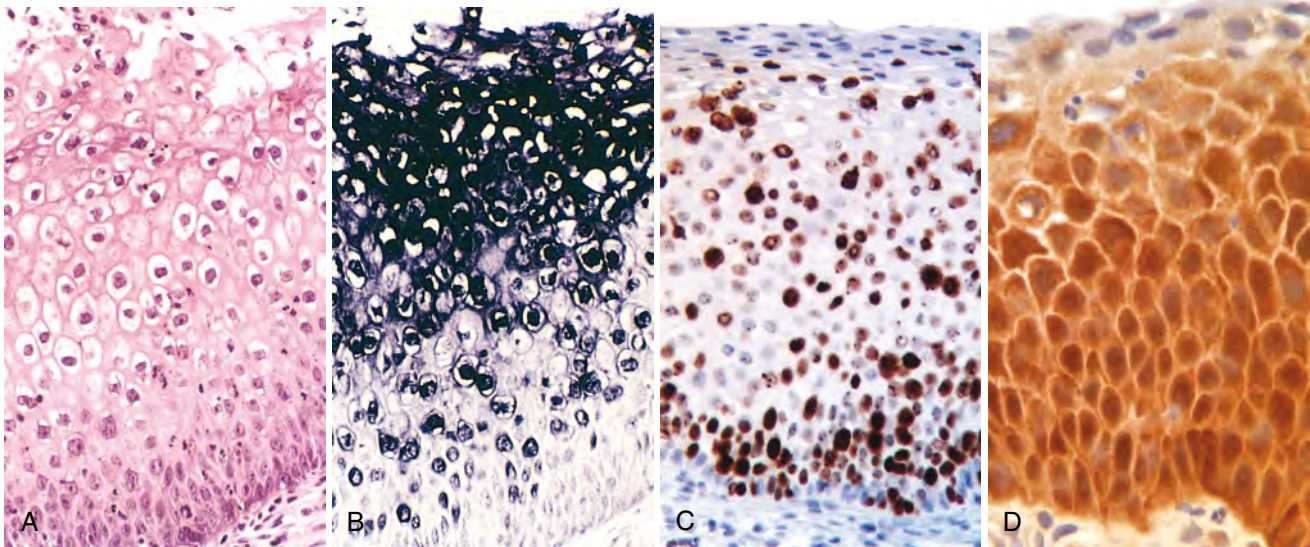


Fig. 22.15 Pathologic features of cervical HPV infection. (A) Low-grade squamous intraepithelial lesion. Hematoxylin and eosin staining shows koilocytic change, seen as nuclear enlargement and perinuclear “halos.” (B) In situ hybridization test for human papillomavirus deoxyribonucleic acid (HPV DNA). The dark granular staining denotes HPV DNA, which is most abundant in the superficial koilocytes. (C) Widespread positivity for the proliferation marker Ki-67 (seen as brown nuclear staining), illustrates expansion of the proliferating cells from the normal basal location to the superficial layers of the epithelium. (D) Up-regulation of the cyclin-dependent kinase inhibitor p16 (brown staining) characterizes high-risk HPV infections.

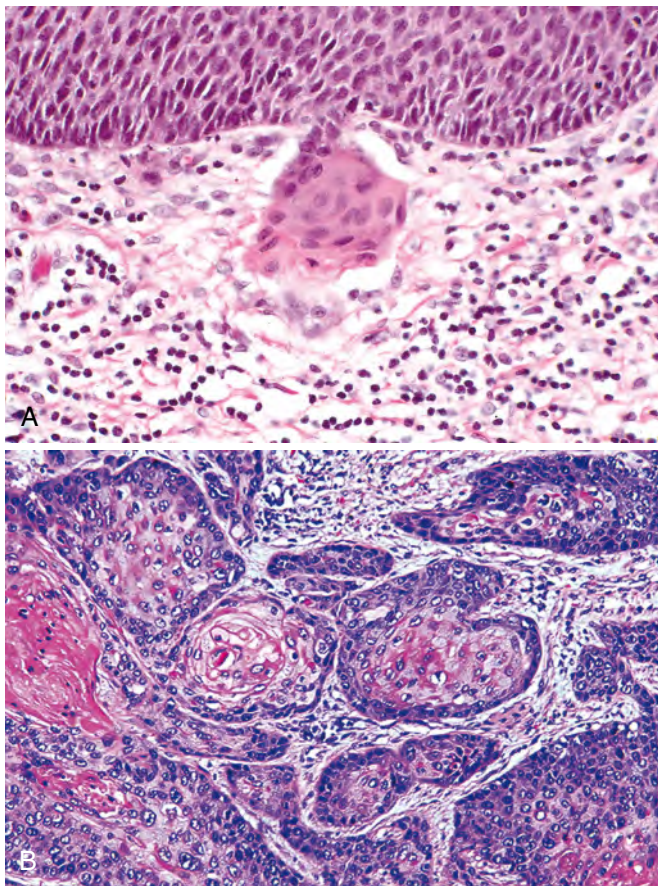


Fig. 22.16 Squamous cell carcinoma of the cervix. (A) Early invasion in squamous cell carcinoma showing an invasive nest breaking through the basement membrane of a high-grade squamous intraepithelial lesion. (B) Invasive squamous cell carcinoma.

MORPHOLOGY

Invasive cervical carcinoma may manifest as fungating (exophytic) or infiltrative masses. **Squamous cell carcinoma** is composed of nests and tongues of malignant squamous epithelium, either keratinizing or nonkeratinizing, which invade the underlying cervical stroma (Fig. 22.16). **Adenocarcinoma** is characterized by proliferation of glandular epithelium composed of malignant endocervical cells with large, hyperchromatic nuclei and relatively mucin-depleted cytoplasm, resulting in a dark appearance of the glands, as compared with the normal endocervical epithelium (Fig. 22.17). Adenosquamous carcinoma is composed of intermixed malignant glandular and squamous epithelium. Neuroendocrine cervical carcinoma, like those occurring elsewhere in the body may have a small (small cell neuroendocrine carcinoma) or large (large cell neuroendocrine carcinoma) cell morphology, but differs from those in most other anatomic locations in being positive for high-risk HPVs.

Advanced cervical carcinoma spreads by direct extension to contiguous tissues, including paracervical soft tissue, urinary bladder, ureters (resulting in hydronephrosis), rectum, and vagina. Lymphovascular invasion results in local and distant lymph nodes metastases. Distant metastases may also be found in the liver, lungs, bone marrow, and other organs.

Cervical cancer is staged as:

Stage 0—Carcinoma in situ (CIN 3, HSIL)

Stage I—Carcinoma confined to the cervix

Ia—Preclinical carcinoma, that is, diagnosed only by microscopy

Ia1—Stromal invasion of 3 mm or less depth (so-called superficially invasive squamous cell carcinoma)

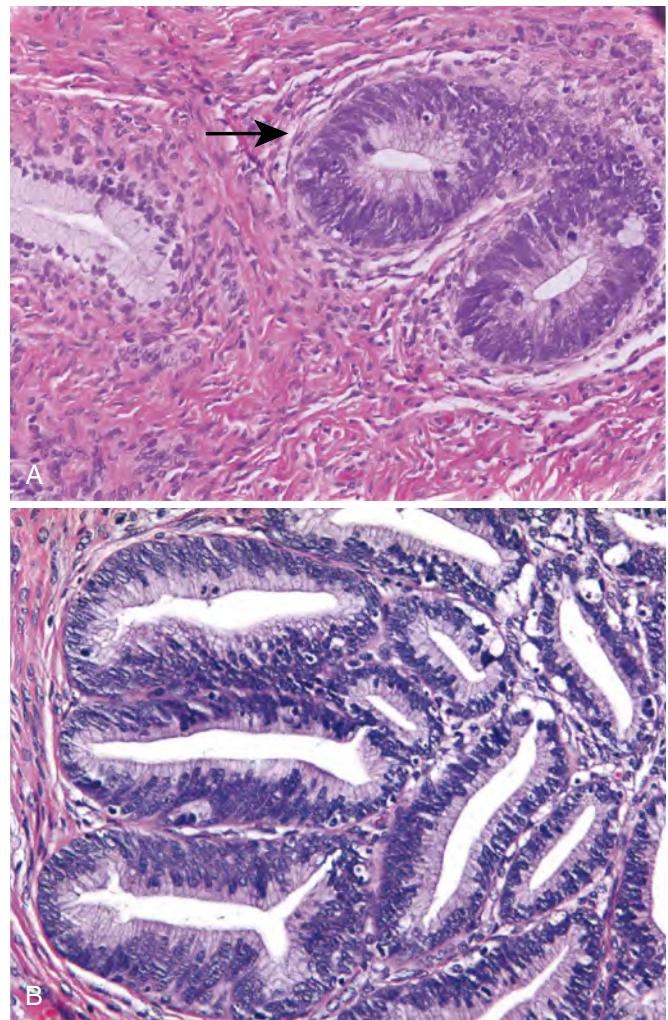


Fig. 22.17 Adenocarcinoma of the cervix. (A) Adenocarcinoma in situ (arrow) showing dark glands with atypical, enlarged nuclei adjacent to a normal, pale endocervical gland. (B) Invasive adenocarcinoma.

Ia2—Maximum depth of invasion of stroma deeper than 3 mm and no deeper than 5 mm taken from base of epithelium

Ib—Histologically invasive carcinoma confined to the cervix and greater than stage Ia2

Stage II—Carcinoma extends beyond the cervix but not to the pelvic wall. Carcinoma involves the vagina but not the lower third.

Stage III—Carcinoma involves the lower third of the vagina, extends to the pelvic wall, causes hydronephrosis, and/or involves pelvic and/or paraaortic lymph nodes.

Stage IV—Carcinoma has extended beyond the true pelvis or has involved the mucosa of the bladder or rectum. This stage also includes cancers with metastatic dissemination.

Clinical Features

More than one-half of invasive cervical cancers are detected in females who did not participate in regular screening. While superficially invasive squamous cell carcinomas may be treated by cervical cone excision alone, most invasive cancers are managed by hysterectomy with lymph node dissection and, for advanced lesions, radiation and chemotherapy. The prognosis for invasive carcinomas depends on the stage of the

cancer at diagnosis and to some degree on histologic subtype, with neuroendocrine carcinoma and gastric-type HPV-independent adenocarcinoma having very poor prognoses. With current treatments, the 5-year survival rate is 100% for superficially invasive squamous cell carcinomas and less than 20% for tumors extending beyond the pelvis. Most patients with advanced cervical cancer die of the consequences of local tumor invasion (e.g., ureteral obstruction, pyelonephritis, and uremia) rather than distant metastases.

Cervical Cancer Screening and Prevention

As is well known, cytologic cancer screening has significantly reduced mortality from cervical cancer. In countries where such screening is not widely practiced, cervical cancer continues to exact a high toll. The reason that cytologic screening is so effective in preventing cervical cancer is that most cancers arise from precursor lesions over the course of years. These lesions shed abnormal cells that can be detected on cytologic examination. Using a spatula or brush, the transformation zone of the cervix is circumferentially scraped and the cells are smeared or spun down onto a slide. Following fixation and staining with the Papanicolaou method, the smears are screened microscopically by eye or (increasingly) with automated image analysis systems. The cellular changes seen on the Pap test, illustrating the spectrum from LSIL to HSIL, are shown in Fig. 22.18.

Testing for the presence of HPV DNA in the cervical scrape is a molecular method of cervical cancer screening. HPV testing has a higher sensitivity but lower specificity, as compared to the Pap test. HPV DNA testing may be added to cervical cytology for screening in females 30 years of age or older. HPV testing of females younger than 30 years of age is not recommended because of the high incidence of infection, and thus the particularly low specificity of HPV test results in this age group.

Cervical cancer screening and preventive measures are carried out in a stepwise fashion. Recommendations for the frequency of Pap screening vary, but in general the first smear should be at 21 years of age or within 3 years of onset of sexual activity, and thereafter every 3 years. After 30 years of age, females who have had normal cytology results and are negative for HPV may be screened every 5 years by either HPV testing alone or in combination with a Pap test. Recommendations are evolving and vary in different parts of the world. Females who have a normal cytology result but test positive for high-risk HPV DNA should have cervical cytology repeated every 6 to 12 months.

When the result of a screening test is abnormal, a colposcopic examination of the cervix and vagina is performed to identify the lesion. The mucosa is examined with a magnifying glass following application of acetic acid, which highlights abnormal epithelium as white spots (*aceto-white areas*). Abnormal appearing areas are biopsied. Females with biopsy-confirmed LSIL can be followed in a conservative fashion.

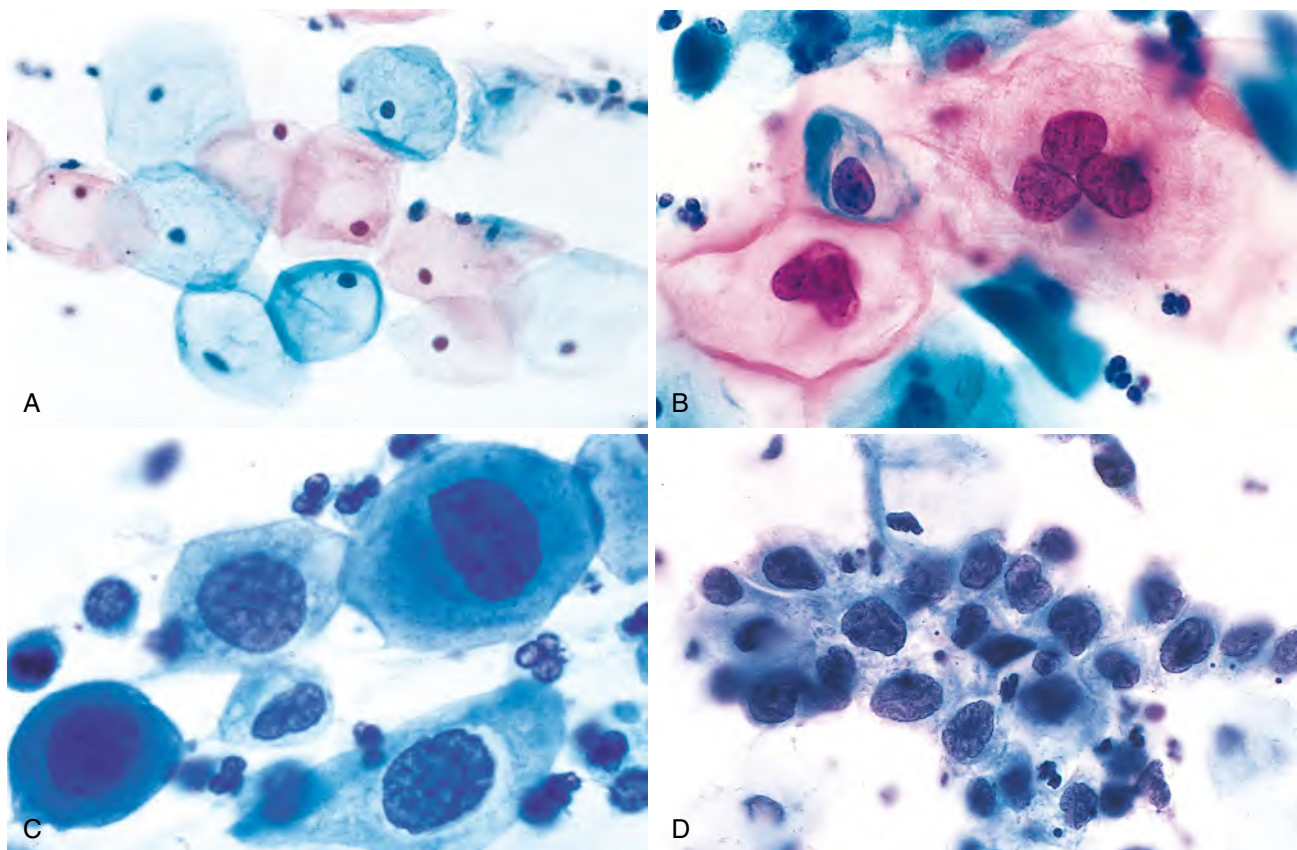


Fig. 22.18 The cytology of cervical intraepithelial neoplasia as seen on the Papanicolaou smear. The cytoplasmic staining of desquamated cells may be either red or blue. (A) Normal exfoliated superficial squamous cells. (B) Low-grade squamous intraepithelial lesion—koilocytes. (C) High-grade squamous intraepithelial lesion (HSIL). (D) HSIL. Note the reduction in cytoplasm and the increase in the nucleus-to-cytoplasm ratio, which occurs as the grade of the lesion increases. This reflects the progressive loss of cellular differentiation on the surface of the lesions from which these cells are exfoliated. (Courtesy Dr. Edmund S. Cibas, Brigham and Women's Hospital, Boston.)

Some gynecologists will perform local ablation (e.g., cryotherapy) of LSIL, particularly if there is concern about the reliability of patient follow-up. HSIL is treated with cervical conization (superficial excision).

An additional important aspect of cervical cancer prevention is vaccination against low- and high-risk HPV types, which is now recommended for all children and adults aged 9 to 26 years and selected adults between 27 to 45 years of age. Currently only one HPV vaccine is available in the United States that covers 9 HPV types (6, 11, 16, 18, 31, 33, 45, 52 and 58). Vaccination is now recommended for both boys and girls due to the role that males play in the spread of HPV to females and the toll that HPV-related anal and oropharyngeal cancers take in males. The vaccines offer protection for at least 6 years; longer follow-up studies are still pending. Because the HPV vaccines do not protect against all high-risk HPV types, current guidelines recommend that cervical cancer screening be continued.

KEY CONCEPTS

Cervical SIL and Squamous Carcinoma

- Cervical LSIL is a productive HPV infection that usually regresses spontaneously but occasionally progresses to HSIL.
- HSIL is characterized by progressive deregulation of the cell cycle and increasing cellular atypia. HSIL may progress to invasive carcinoma.
- Almost all cervical precursor lesions and cervical carcinomas are caused by high-risk HPV types, most commonly HPV-16.

UTERINE CORPUS

The uterus has two major components: the myometrium and the endometrium. The myometrium is composed of tightly interwoven bundles of smooth muscle that form the wall of the uterus. The internal cavity of the uterus is lined by the endometrium, which is composed of glands embedded in a cellular stroma. The uterus is affected by a variety of disorders, the most common of which result from endocrine imbalances, complications of pregnancy, and neoplastic proliferation.

ENDOMETRIAL HISTOLOGY IN THE MENSTRUAL CYCLE

The endometrium undergoes dynamic physiologic and morphologic changes during the menstrual cycle in response to sex steroid hormones coordinately produced in the ovary. The ovary is influenced by hormones produced by the pituitary gland in response to signals from the hypothalamus. Together, hypothalamic, pituitary, and ovarian factors and their interactions regulate maturation of ovarian follicles, ovulation, and menstruation.

The histologic appearance of the endometrium may be used to assess hormonal status, document ovulation, and determine causes of endometrial bleeding and infertility (Fig. 22.19). Progression through a normal menstrual cycle is correlated with the following histologic features:

- The cycle commences with *menses*, during which the superficial portion of the endometrium, referred to as the functionalis, is shed.
- The *proliferative phase* is marked by rapid growth of glands and stroma arising from the deeper portion of the endometrium (basalis). During the proliferative phase, the glands are straight, tubular structures lined by regular, tall, pseudostratified columnar cells. Mitotic figures are numerous, and there is no evidence of mucus secretion or vacuolation. The endometrial stroma is composed of spindle cells with scant cytoplasm that are also actively proliferating (see Fig. 22.19A).
- At *ovulation*, endometrial proliferation ceases, and differentiation commences in response to the effects of progesterone made by the corpus luteum in the ovary.

- *Postovulation* is initially marked by the appearance of *secretory vacuoles* beneath the nuclei in the glandular epithelium (see Fig. 22.19B). Secretory activity is most prominent during the third week of the menstrual cycle, when the basal vacuoles progressively move to the apical surface. By the fourth week, the glands are tortuous, producing a serrated appearance. This serrated or “sawtooth” appearance is accentuated by secretory exhaustion and shrinkage of the glands.
- *Stromal changes in the late secretory phase*, due predominantly to progesterone, are the most significant features. Prominent spiral arterioles appear, which is accompanied by an increase in ground substance and edema between the stromal cells, followed by stromal cell hypertrophy, increased cytoplasmic eosinophilia (*predecidual change*), and a resurgence of stromal mitoses (see Fig. 22.19C). Predecidual changes spread throughout the functionalis layer and are accompanied by a sparse infiltrate of neutrophils and lymphocytes, which in this context are considered normal.
- With the dissolution of the corpus luteum and the subsequent drop in progesterone levels, the functionalis layer degenerates and bleeding into the stroma occurs, followed by stromal breakdown and onset of the next menstrual cycle (see Fig. 22.19D).

The action of ovarian hormones on the endometrium primarily occurs through nuclear hormone receptors. During the proliferative phase, estrogen drives the proliferation of both glands and stroma, sometimes by promoting “cross-talk” between these two cell types. For example, much of the effect of estrogen on glandular proliferation occurs via stromal cells, which in response to estrogen, produce growth factors (e.g., insulin-like growth factor-1 and epidermal growth factor) that bind receptors expressed on the epithelial cells. During the secretory phase, progesterone down-regulates the expression of estrogen receptor in both the glands and the stroma, and as a result endometrial proliferation is suppressed. Progesterone also promotes the differentiation of the glands and causes functional changes in the stromal cells. Endometrial stem cells have been identified that likely have a central role in the regeneration of the endometrium after menses. They may also contribute to the development of ectopic endometrial tissue and endometrial cancer.

FUNCTIONAL ENDOMETRIAL DISORDERS (DYSFUNCTIONAL UTERINE BLEEDING)

Uterine bleeding that lacks an underlying structural abnormality is called dysfunctional uterine bleeding. It most commonly stems from hormonal disturbances (Table 22.2). Uterine bleeding can also be caused by well-defined pathologic conditions, such as chronic endometritis, endometrial polyps (Fig. 22.20A), submucosal leiomyomas (Fig. 22.20B), or endometrial neoplasms. As discussed earlier, the normal cyclical proliferation, differentiation, and shedding of the endometrium requires that all the involved pituitary and ovarian hormones be released at the proper time at physiological levels. Any disturbance of this finely tuned system may result in dysfunctional uterine bleeding, the most common causes of which are discussed in the following sections.

Anovulatory Cycle

The most frequent cause of dysfunctional bleeding is anovulation (failure to ovulate). Anovulatory cycles result from hormonal imbalances and are most common at menarche and in the perimenopausal period. Less commonly, anovulation is the result of:

- *Endocrine disorders*, such as thyroid disease, adrenal disease, or pituitary tumors

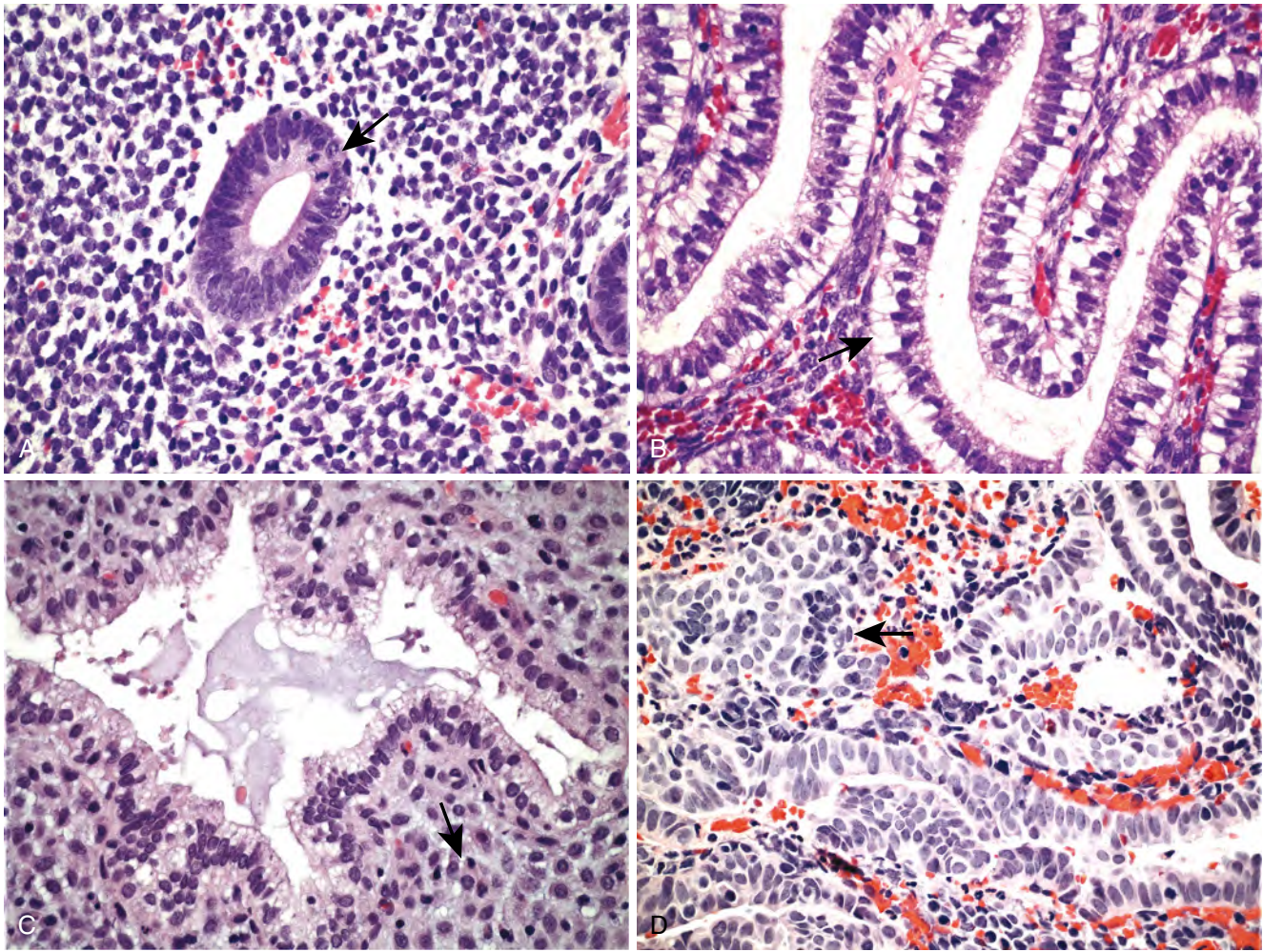


Fig. 22.19 Histology of the menstrual cycle. (A) Proliferative phase with mitoses (arrow). (B) Early secretory phase with subnuclear vacuoles (arrow). (C) Late secretory exhaustion and predecidual changes (arrow). (D) Menstrual endometrium with stromal breakdown (arrow) (see text).

- *Ovarian lesions*, such as a functioning ovarian tumor (granulosa cell tumors) or polycystic ovaries (see later in this chapter)
- *Generalized metabolic disturbances*, such as occur with obesity, malnutrition, or other chronic systemic diseases

Failure of ovulation results in excessive endometrial stimulation by estrogens that is unopposed by progesterone. Under these circumstances, the endometrial glands undergo mild architectural changes, including cystic dilation, that usually resolve due to a subsequent ovulatory cycle. However, repeated anovulation may result in bleeding that, in certain clinical situations, may prompt an endometrial biopsy. In this setting, biopsies reveal stromal condensation and eosinophilic epithelial metaplasia, features similar to those seen in menstrual endometrium. However, unlike menstrual endometrium, progesterone-dependent morphologic features (e.g., glandular secretory changes and stromal predecidualization) are absent because the source of progesterone, the corpus luteum, does not develop without ovulation. Most commonly, the endometrium is composed of pseudostratified glands and contains scattered mitotic figures (Fig. 22.20C). More severe consequences of repeated anovulation are discussed later in this chapter.

INFLAMMATORY DISORDERS

The endometrium and myometrium are uncommon sites of infections, primarily because the endocervix forms a barrier to ascending infection. Thus, although chronic inflammation in the cervix is common and usually insignificant, it is of concern in the endometrium.

Acute Endometritis

Acute endometritis is uncommon and limited to bacterial infections that arise after delivery or miscarriage. Retained products of conception are the usual predisposing factors; the causative agents include group A hemolytic streptococci, staphylococci, and other bacteria. The inflammatory response is chiefly limited to the stroma and is nonspecific. Removal of the retained gestational products by curettage and antibiotic therapy promptly clears these infections.

Chronic Endometritis

Chronic endometritis occurs in association with the following disorders:

- Chronic pelvic inflammatory disease
- Retained gestational tissue, postpartum or postabortion
- Intrauterine contraceptive devices

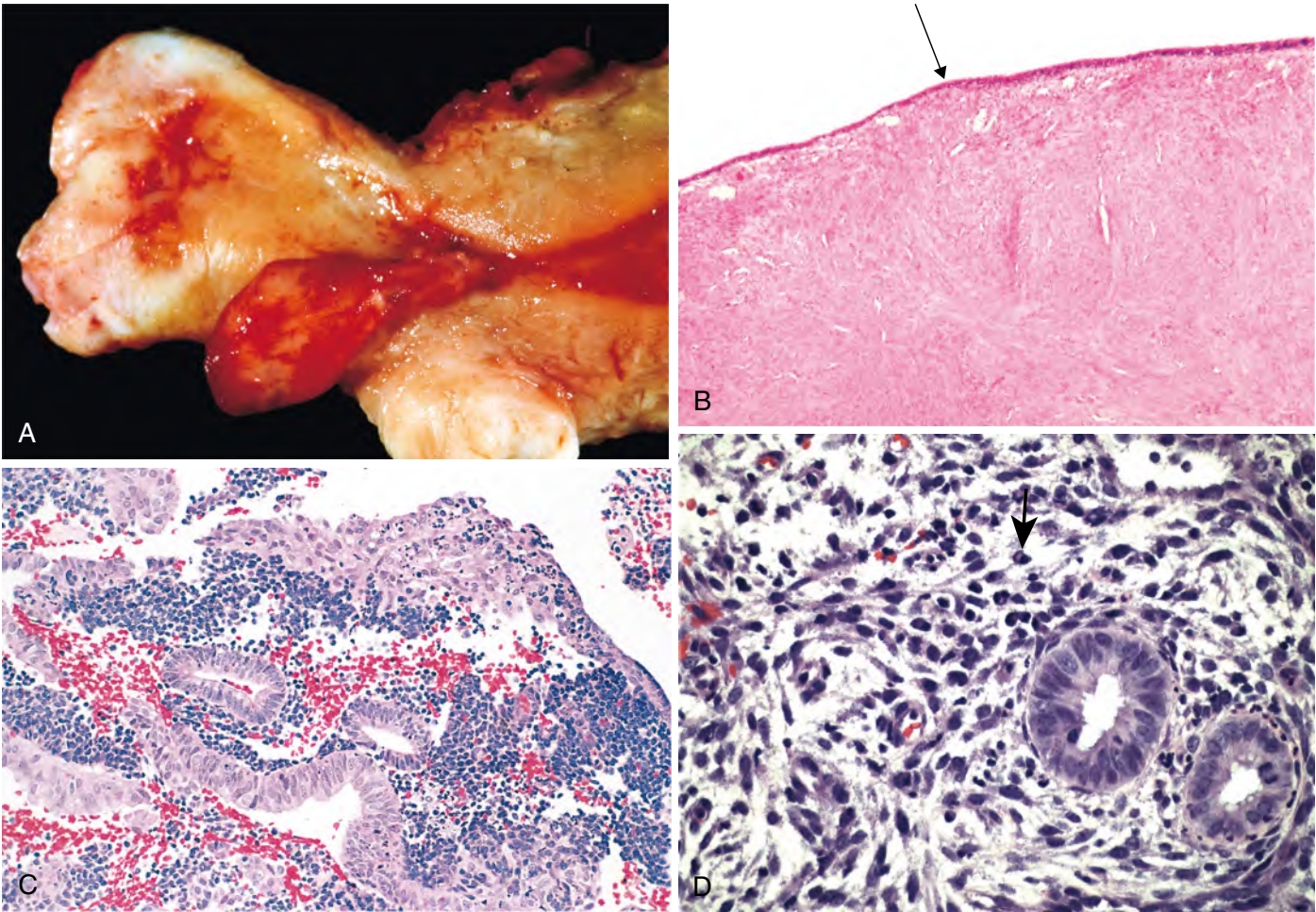


Fig. 22.20 Common causes of abnormal uterine bleeding. (A) Endometrial polyp. (B) Submucosal leiomyoma with attenuation of the endometrial lining (arrow). (C) The most common cause is dysfunctional uterine bleeding, seen here as anovulatory endometrium with stromal breakdown. Note the proliferative glands. (D) Chronic endometritis with plasma cells (arrow).

TABLE 22.2 Causes of Abnormal Uterine Bleeding by Age Group

Age Group	Causes
Prepuberty	Precocious puberty (hypothalamic, pituitary, or ovarian origin)
Adolescence	Anovulatory cycle, coagulation disorders
Reproductive age	Complications of pregnancy (abortion, trophoblastic disease, ectopic pregnancy) Anatomic lesions (leiomyoma, adenomyosis, polyps, endometrial hyperplasia, carcinoma) Dysfunctional uterine bleeding Anovulatory cycle Ovulatory dysfunctional bleeding (e.g., inadequate luteal phase)
Perimenopausal	Dysfunctional uterine bleeding Anovulatory cycle Anatomic lesions (carcinoma, hyperplasia, polyps)
Postmenopausal	Endometrial atrophy Anatomic lesions (carcinoma, hyperplasia, polyps)

- Tuberculosis, either from miliary spread or, more often, from drainage of tuberculous salpingitis. Endometrial tuberculosis is rare in high-resource countries.

The diagnosis of chronic endometritis rests on the identification of plasma cells in the stroma (see Fig. 22.20D), which are not seen in normal endometrium. In about 15% of cases, no cause is apparent. Some females with so-called “nonspecific” chronic endometritis have gynecologic symptoms such as abnormal bleeding, pain, discharge, and infertility. *Chlamydia* may be involved and is commonly associated with both acute (e.g., neutrophils) and chronic (e.g., lymphocytes, plasma cells) inflammatory infiltrates. The responsible organisms may not be detected by culture. If infection is suspected on clinical grounds, antibiotic therapy is indicated, even in the face of negative cultures, as it may prevent other sequelae (e.g., salpingitis).

ENDOMETRIOSIS AND ADENOMYOSIS

Endometriosis is defined by the presence of “ectopic” endometrial tissue at a site outside of the uterus. The abnormal tissue most commonly includes both endometrial glands and stroma, but may consist

only of stroma in some cases. It occurs in the following sites, in descending order of frequency: (1) ovaries, (2) uterine ligaments, (3) rectovaginal septum, (4) cul de sac, (5) pelvic peritoneum, (6) serosa of the large and small bowel and appendix, (7) mucosa of the cervix, vagina, and fallopian tubes, and (8) laparotomy scars.

Endometriosis can have significant clinical consequences; it often causes infertility, dysmenorrhea (painful menstruation), pelvic pain, and other problems. The disorder is principally a disease of females in active reproductive life, most often in the third and fourth decades, and affects approximately 10% of females. There are three types of endometriosis: superficial peritoneal endometriosis, ovarian endometriosis, and deep infiltrating endometriosis (Fig. 22.21). The superficial and ovarian forms of endometriosis are occasionally associated with the development of malignancy, whereas it is extremely rare for the deep infiltrating form to undergo malignant transformation.

Pathogenesis

The origin and pathogenesis of endometriosis remains elusive. Proposed origins of fall into two main categories: (1) from the uterine endometrium and (2) from cells outside the uterus that have the capacity to give rise to endometrial tissue. The leading pathogenic theories are:

- *The regurgitation theory* proposes that endometrial tissue implants at ectopic sites via retrograde flow of menstrual endometrium.
- *The benign metastasis theory* holds that endometrial tissue from the uterus can “spread” to distant sites (e.g., bone, lung, and brain) via blood vessels and lymphatic channels.
- *The metaplastic theory* suggests that endometrium arises directly from coelomic epithelium (mesothelium of pelvis or abdomen), from which the müllerian ducts and ultimately the endometrium itself originate during embryonic development. In addition, mesonephric remnants may undergo endometrial differentiation and give rise to ectopic endometrial tissue.
- *The extrauterine stem/progenitor cell theory* proposes that stem/progenitor cells from the bone marrow differentiate into endometrial tissue.

Retrograde menstruation through the fallopian tubes is common, and the regurgitation theory provides a plausible explanation for the origin of endometrial tissue in the peritoneal cavity, which constitutes most cases. However, it cannot explain endometriosis in females who are amenorrheic because of a variety of underlying etiologies (e.g.,

gonadal dysgenesis); endometriosis in the urogenital tract of males treated with high-dose estrogens for prostate cancer; and endometriosis in distant sites like the brain, lung, and bone. In addition, the relatively low incidence of endometriosis, despite the common occurrence of retrograde menstruation (up to 90% of females), suggests that additional factors are involved in the pathogenesis of the disorder.

Molecular analyses have provided additional insights. The endometriotic implants show certain differences when compared to the endometria of females without endometriosis (Fig. 22.22). These include:

- *Release of proinflammatory and angiogenic factors*, including prostaglandin E_2 , interleukin (IL)-1, tumor necrosis factor, IL-6 and IL-8, nerve growth factor (NGF), vascular endothelial growth factor (VEGF), monocyte chemoattractant protein-1 (MCP-1/CCL-2), matrix metalloproteinases, and tissue inhibitors of metalloproteinases. Some of these factors are released by associated macrophages recruited to endometriotic implants by proinflammatory factors. Like metastatic tumors, the ability of endometriotic implants to survive and grow is dependent on angiogenesis, which is mediated by typical proangiogenic factors such as VEGF. Similarly, the ability to implant requires remodeling of the extracellular matrix, which is carried out by factors such as matrix metalloproteinases. Altogether, these factors are proposed to play a role in both the establishment and maintenance of endometriotic implants. These factors may also contribute to avoidance of immune clearance.
- *Aromatase production by endometriotic stromal cells*, which is absent in normal endometrial stroma. Estrogen enhances the survival and persistence of endometriotic tissue, and inhibitors of aromatase are beneficial in the treatment of endometriosis. A link between inflammation and estrogen production is made plausible by the ability of prostaglandin E_2 to stimulate local synthesis of estrogen. It also has been shown that retinoic acid produced in the stromal cells promotes epithelial cell survival via paracrine signaling. In addition, epigenetic alterations (DNA methylation) have been described that lead to increased responsiveness to estrogen and decreased responsiveness to progesterone, alterations that promote endometrial proliferation and survival. These abnormalities are present not only in ectopic endometriotic tissue, but also, to a lesser degree, in the uterine endometrium of patients with endometriosis, suggesting that there is a fundamental defect in the endometrium.
- *Mutations in tumor suppressor genes and oncogenes*, such as *KRAS*, *PIK3CA*, *PPP2R1A*, and *ARID1A*, have been identified in

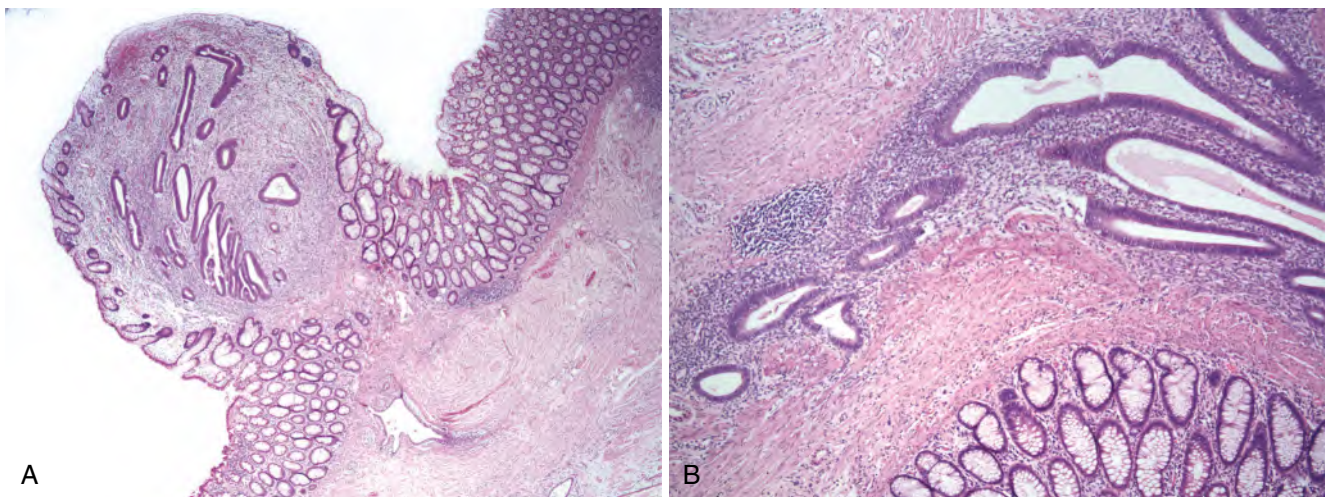


Fig. 22.21 Endometriosis. (A) Endometriosis involving the mucosa of the colon. (B) Higher magnification reveals endometrial glands and stroma adjacent to normal colonic mucosa.

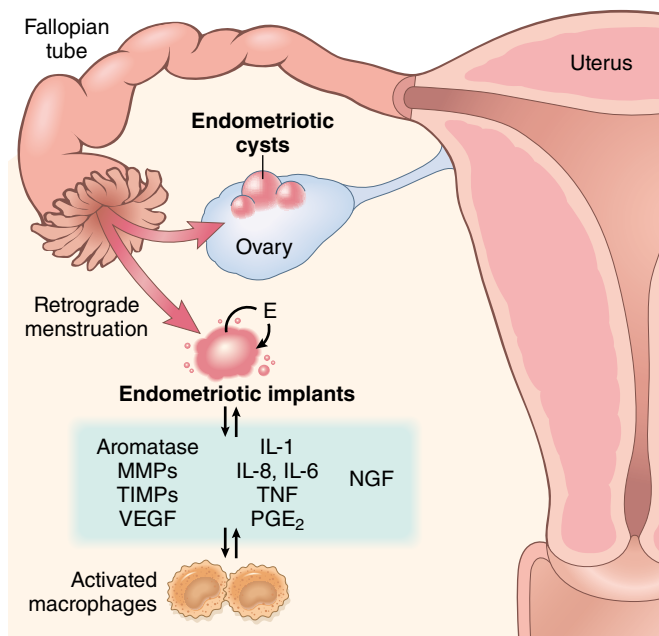


Fig. 22.22 Pathogenesis of endometriosis. Depicted is the interplay between factors expressed in endometriotic implants and activated macrophages that are hypothesized to play a role in the establishment and maintenance of endometriotic implants. *E*, Estrogen; *IL*, interleukin; *MMPs*, matrix metalloproteinase; *NGF*, nerve growth factor; *PGE₂*, prostaglandin *E₂*; *TIMPs*, tissue inhibitors of metalloproteinases; *TNF*, tumor necrosis factor; *VEGF*, vascular endothelial growth factor.

the epithelial cells of deeply infiltrating endometriosis. The significance of these mutations is unknown, but it is possible that they contribute to the locally aggressive behavior of deeply infiltrating endometriosis. An association between endometriosis and ovarian cancer of the endometrioid and clear cell types (discussed later) has been noted in several epidemiologic studies, with an approximate three-fold increase in endometriosis. These studies suggest a common origin of abnormal endometriotic tissue and ovarian cancers in some cases. Consistent with this idea, in some instances it has been shown that carcinomas harbor the same mutations as associated endometriosis in the same patient, strongly supporting a pathogenic role for these driver mutations in the development of both the premalignant and malignant lesions.

MORPHOLOGY

Endometriotic lesions bleed periodically in response to both extrinsic cyclic (ovarian) and intrinsic hormonal stimulation. This bleeding produces nodules with a red-blue to yellow-brown appearance on or just beneath the mucosal and/or serosal surfaces at sites of involvement. When lesions are widespread, organizing hemorrhage causes extensive fibrous adhesions between tubes, ovaries, and other structures and obliterates the pouch of Douglas. The ovaries may become markedly distorted by large cystic masses (3 to 5 cm in diameter) filled with brown fluid resulting from previous hemorrhage; these are referred to clinically as **chocolate cysts** or **endometriomas**. Deeply infiltrating endometriosis can invade tissues and cause fibrosis and adhesions.

The histologic diagnosis of endometriosis is usually straightforward but may be difficult in long-standing cases in which the endometrial tissue is obscured by secondary fibrosis. The diagnosis is readily made when both endometrial

glands and stroma are present (see Fig. 22.21B), with or without the presence of hemosiderin. In rare cases only stroma is identified. If only glands are present, other diagnoses with different clinical ramifications, such as endosalpinx, must be considered.

Atypical endometriosis, the likely precursor to endometriosis-related ovarian carcinoma, has two morphologic appearances. One shows cytologic atypia of the epithelium lining the endometriotic cyst, without major architectural changes. The second is marked by glandular crowding due to excessive epithelial proliferation, often associated with cytologic atypia, producing an appearance that resembles complex atypical endometrial hyperplasia (discussed later).

Clinical Features

Clinical signs and symptoms usually include severe dysmenorrhea, dyspareunia (pain with intercourse), and pelvic pain due to intrapelvic bleeding and periuterine adhesions. Menstrual irregularities are common, and infertility is the presenting complaint in 30% to 40% of females. In addition, although uncommon, malignancies may develop within endometriotic cysts, suggesting that these lesions contain “at-risk” epithelium.

A related disorder, **adenomyosis**, is defined as the presence of endometrial tissue within the uterine wall (myometrium). Adenomyosis remains in continuity with the endometrium, presumably signifying down-growth of endometrial tissue into and between the smooth muscle fascicles of the myometrium. Adenomyosis occurs in up to 20% of uteri. On microscopic examination, irregular nests of endometrial stroma, with or without glands, are arranged within the myometrium. Like endometriosis, the clinical symptoms of adenomyosis include menometrorrhagia (irregular and heavy menses), colicky dysmenorrhea, dyspareunia, and pelvic pain, particularly during the premenstrual period. It can coexist with endometriosis.

KEY CONCEPTS

Endometriosis and Adenomyosis

- Endometriosis is defined as endometrial glands and stroma outside of the uterus. The “ectopic” endometrial tissue may bleed cyclically.
- Most common sites of endometriosis are within the abdominal cavity, but occasionally it is found at distant sites.
- Several theories (regurgitation, metaplasia, metastasis, and stem cell origin) are proposed to explain the distribution of endometriosis.
- It commonly results in dysmenorrhea, pelvic pain, and infertility.
- Peritoneal and ovarian endometriosis may be a precursor to carcinoma (endometrioid and clear cell carcinoma).
- Adenomyosis, the presence of endometrial tissue within myometrium, can present with dysmenorrhea and pelvic pain.

ENDOMETRIAL POLYPS

Endometrial polyps are exophytic masses of variable size that project into the endometrial cavity. They may be single or multiple and are usually sessile and relatively small (Fig. 22.20C), measuring 0.5 to 3 cm in diameter, but are occasionally large and pedunculated. Polyps may be asymptomatic or may cause abnormal bleeding and infertility.

Cytogenetic and molecular studies indicate that the stromal cells in endometrial polyps contain acquired chromosomal rearrangements that are similar to those found in other benign mesenchymal tumors, suggesting that these lesions are best considered stromal neoplasms. Polyps are responsive to estrogen but show little or no response to progesterone. Endometrial polyps may occur in association with the administration of tamoxifen, which is often used in the therapy of

breast cancer due to its antiestrogenic activity on the breast. However, tamoxifen has weak proestrogenic effects in the endometrium. Atrophic polyps, which mainly occur in postmenopausal females, likely represent the atrophic remnants of polyps that developed prior to menopause. Rarely, adenocarcinoma arises within endometrial polyps.

ENDOMETRIAL HYPERPLASIA

Endometrial hyperplasia is an important cause of abnormal bleeding and a frequent precursor to the most common type of endometrial carcinoma. It is defined as an abnormal proliferation of the endometrial glands relative to the stroma, resulting in an increased gland-to-stroma ratio when compared with normal proliferative endometrium. Clinico-pathologic and epidemiologic studies have pointed to the malignant potential of endometrial hyperplasia, and molecular studies have confirmed this relationship, as endometrial hyperplasia and carcinoma share specific acquired driver mutations in cancer genes (described later).

Endometrial hyperplasia is associated with prolonged estrogenic stimulation of the endometrium, which can be due to anovulation, increased estrogen production from endogenous sources, or exogenous estrogen. Associated conditions include:

- Obesity (peripheral conversion of androgens to estrogens)
- Menopause
- Polycystic ovarian syndrome
- Functioning granulosa cell tumors of the ovary
- Excessive ovarian cortical function (cortical stromal hyperplasia)
- Prolonged administration of estrogenic substances (estrogen replacement therapy)

These are the same influences postulated to be of pathogenic significance in some endometrial carcinomas, discussed later.

Inactivation of the *PTEN* tumor suppressor gene is a common genetic alteration in both endometrial hyperplasias and endometrioid endometrial carcinoma. As discussed in Chapter 7, *PTEN* encodes a lipid phosphatase that is an important negative regulator of the phosphatidylinositol 3-kinase (PI3K)/AKT growth-regulatory pathway. When *PTEN* function is lost, the PI3K/AKT pathway becomes overactive. Mutations in *PTEN* are found in more than 20% of hyperplasias, both with and without atypia, and in 30% to 80% of endometrial carcinomas, suggesting that alterations in *PTEN* occur at an early stage of endometrial tumorigenesis (although they are not predictive of progression of hyperplasia to carcinoma). Of note, patients with *Cowden syndrome*, which is caused by germline mutations in *PTEN*, have a high incidence of endometrial carcinoma and certain other tumors, particularly breast cancer. As with many other tumor suppressors, it is not entirely clear why the loss of *PTEN* (which is expressed in many tissues) is so highly associated with particular tumors. It is interesting to note, however, that PI3K/AKT signaling enhances the ability of the estrogen receptor to turn on the expression of its target genes. Thus, loss of *PTEN* function may stimulate estrogen-dependent gene expression, leading to overgrowth of cell types that depend on estrogen for trophic signals, such as endometrial and mammary epithelial cells.

MORPHOLOGY

The classification of endometrial hyperplasia according to the World Health Organization (WHO) includes two major categories, hyperplasia without atypia and atypical hyperplasia (also referred to as endometrioid intraepithelial neoplasia), which differ in appearance and propensity to progress to carcinoma. **Hyperplasia without atypia** has a wide range of appearances, but the

cardinal feature is an increased gland-to-stroma ratio. The glands show variation in size and shape and may be dilated (Fig. 22.23A). Although there may be back-to-back glands focally, some intervening stroma is usually retained (Fig. 22.23B). These lesions are caused by persistent estrogen stimulation and rarely progress to adenocarcinoma (approximately 1% to 3%). Hyperplasia may evolve into cystic atrophy when estrogen is withdrawn.

Atypical hyperplasia (endometrioid intraepithelial neoplasia) is composed of complex patterns of proliferating glands displaying nuclear atypia. The glands are commonly back-to-back and often have complex outlines due to branching structures and represent clonal outgrowths of cells with numerous genetic alterations. Individual cells are rounded and lose the normal perpendicular orientation to the basement membrane. In addition, the nuclei have open (vesicular) chromatin and conspicuous nucleoli. The morphologic and genetic features of atypical hyperplasia have considerable overlap with those of well-differentiated endometrioid adenocarcinoma (discussed later), and accurate distinction from cancer may not be possible without hysterectomy (Fig. 22.23C–D). Indeed, up to 40% of females with a diagnosis of atypical hyperplasia are found to have carcinoma when a hysterectomy is performed.

Currently, atypical hyperplasia is managed by hysterectomy or, in young females who desire fertility, a trial of progestin therapy and close follow-up. Most often, hopefully after a successful pregnancy, lack of regression prompts removal of the uterus.

KEY CONCEPTS

Endometrial Hyperplasia

- Endometrial hyperplasia is defined as an increase in the number of glands relative to the stroma, appreciated as crowded glands, often with abnormal shapes.
- It is most caused by unopposed estrogen stimulation and is an important cause of abnormal vaginal bleeding.
- It is divided into hyperplasia without atypia and atypical hyperplasia based on nuclear morphology. Atypical hyperplasia is associated with an increased risk of endometrial carcinoma.
- The *PTEN* tumor suppressor gene is mutated in approximately 20% of endometrial hyperplasias.

MALIGNANT TUMORS OF THE ENDOMETRIUM

Carcinoma of the Endometrium

Endometrial carcinoma is the most common invasive cancer of the female genital tract in the United States. It accounts for 7% of all invasive cancer in US females, excluding skin cancer. At one time endometrial carcinoma was far less common than cancer of the cervix, but prevention and earlier detection and eradication of the precursor lesions of cervical carcinoma, coupled with an increase in endometrial carcinomas in younger females, have reversed this ratio.

Pathogenesis

Endometrial carcinoma is currently classified based on several defining molecular genetic abnormalities. Genomic sequencing of endometrial carcinoma has revealed four major molecular subtypes among the most common histologic types of endometrial carcinoma, endometrioid and serous carcinoma. The salient molecular features of these four subtypes are:

- **Ultramutated/*POLE* tumors**, which are defined by the presence of mutations in DNA polymerase ϵ (*POLE*) that produce an exceptionally

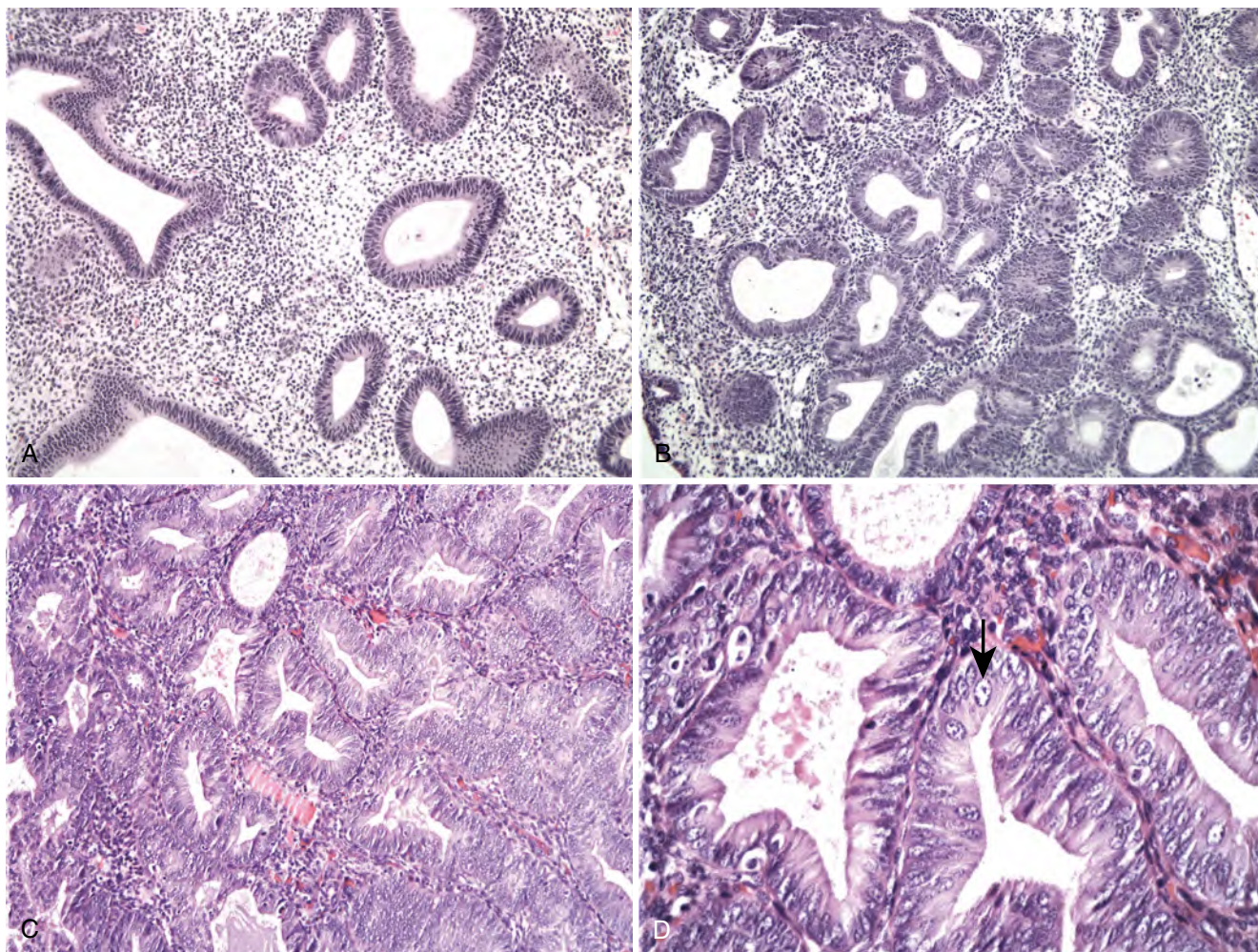


Fig. 22.23 Endometrial hyperplasia. (A) Hyperplasia without atypia. Note the architectural abnormalities including mild glandular crowding and cystic glandular dilation. (B) Hyperplasia without atypia demonstrating increased glandular crowding with areas of back-to-back glands and cytologic features similar to proliferative endometrium. (C) Atypical hyperplasia with further increase in glandular crowding and abnormal cytologic features. (D) High magnification of atypical hyperplasia showing rounded, vesicular nuclei with prominent nucleoli (arrow).

high burden of somatic mutations, the large majority of which are passenger mutations.

- *Hypermutated/MSI (microsatellite instability) tumors*, defined by mutations in or epigenetic silencing of DNA mismatch repair genes, also leading to genomic instability and a high burden of somatic mutations.
- *Copy number low/nonspecific molecular phenotype tumors*, a common subtype also associated with endometrioid morphology that is frequently associated with mutations that upregulate signaling through the PI3K/AKT pathway.
- *Copy number high/TP53 mutated*, aggressive tumors with serous or high-grade endometrioid morphology that are often associated with TP53 mutations and numerous genomic copy number variants.

Endometrioid Endometrial Carcinoma

This is the most common type of endometrial carcinoma, accounting for approximately 80% to 85% of cases. Most are well differentiated and mimic proliferative endometrial glands, features that are the basis for their name. As discussed earlier, it typically arises in the setting of endometrial hyperplasia, and, like endometrial hyperplasia, is associated with conditions in which unopposed estrogenic stimulation of the

endometrium occurs. Most notable of these is obesity, a major problem in high-resource countries and increasingly in lower resource countries that have adopted a Western diet. The association with obesity underlies other conditions that commonly co-occur with endometrial carcinoma, particularly hypertension and type 2 diabetes.

As with other cancers, development of endometrioid carcinoma stems from the stepwise acquisition of genetic alterations in tumor suppressor genes and oncogenes. In hysterectomy specimens containing both atypical hyperplasia and carcinoma, identical mutations in several tumor suppressor genes and oncogenes (e.g., *PTEN*, *ARID1A*, *PIK3CA*, *KRAS*, etc.) have been identified in each component, supporting the view that atypical hyperplasia is a precursor to carcinoma and that mutations occur before the development of overt carcinoma.

Sequencing of the genomes of endometrioid carcinomas has shown that the most common mutations act to increase signaling through the PI3K/AKT pathway, which is a hallmark of this tumor. As mentioned earlier, PI3K/AKT signaling augments expression of estrogen receptor–dependent target genes in endometrial cells. Endometrioid carcinomas are somewhat unique in that individual tumors may harbor multiple mutations that increase PI3K/AKT signaling,

suggesting that tumor development and progression is fostered by successive increases in signal strength. Among the mutations that impact the PI3K/AKT pathway in endometrial carcinomas are:

- Mutations in the *PTEN* tumor suppressor gene are found in 30% to 80% of endometrioid carcinomas.
- *PIK3CA*, an oncogene that encodes the catalytic subunit of PI3K, harbors activating mutations in approximately 40% of endometrioid carcinomas.
- Mutations that activate *KRAS*, which also stimulates PI3K/AKT signaling, are found in approximately 25% of cases.
- Loss-of-function mutations in *ARID1A*, a regulator of chromatin structure, occur in approximately one-third of tumors. Of interest, *ARID1A* is also frequently mutated in ovarian endometrioid and clear cell carcinomas, tumors that arise within endometriosis (endometriosis-associated ovarian cancer). Through unclear mechanisms, it appears that loss of *ARID1A* function also enhances prooncogenic changes in gene expression that are mediated by PI3K/AKT signaling.

Other commonly observed mutations in endometrioid carcinoma disrupt genes that are required for the maintenance of genomic stability, presumably acting in part to create a mutator phenotype that increases the rate of acquisition of mutations in oncogenes and tumor suppressor genes.

- *Defects involving DNA mismatch repair genes* are found in about 20% of sporadic tumors and are particularly prevalent in endometrial carcinomas arising in females with Lynch syndrome (discussed in Chapter 17). In sporadic endometrioid carcinoma, loss of expression of the *MLH1* DNA mismatch repair gene is commonly caused by epigenetic silencing (via promoter hypermethylation).
- *A small subset (less than 10%) of endometrial carcinomas has mutations that disrupt the proofreading function of DNA polymerase ϵ .* These tumors have a remarkably high burden of somatic point mutations, possibly the highest of any human cancer.
- *In approximately 50% of poorly differentiated carcinomas, loss-of-function mutations in *TP53* are present.* Because *TP53* mutations are lacking in the vast majority of well-differentiated endometrioid carcinomas, these mutations are thought to be late events involved in tumor progression.

MORPHOLOGY

Endometrioid carcinoma can take the form of a localized polypoid mass or diffusely involve the endometrial lining (Fig. 22.24A). Spread generally occurs by myometrial invasion followed by direct extension to adjacent structures/organs. Invasion of the broad ligaments may create a palpable mass. Eventually, dissemination to regional lymph nodes occurs, and in the late stages, distant metastatic spread to the lungs, liver, bones, and other organs may be seen.

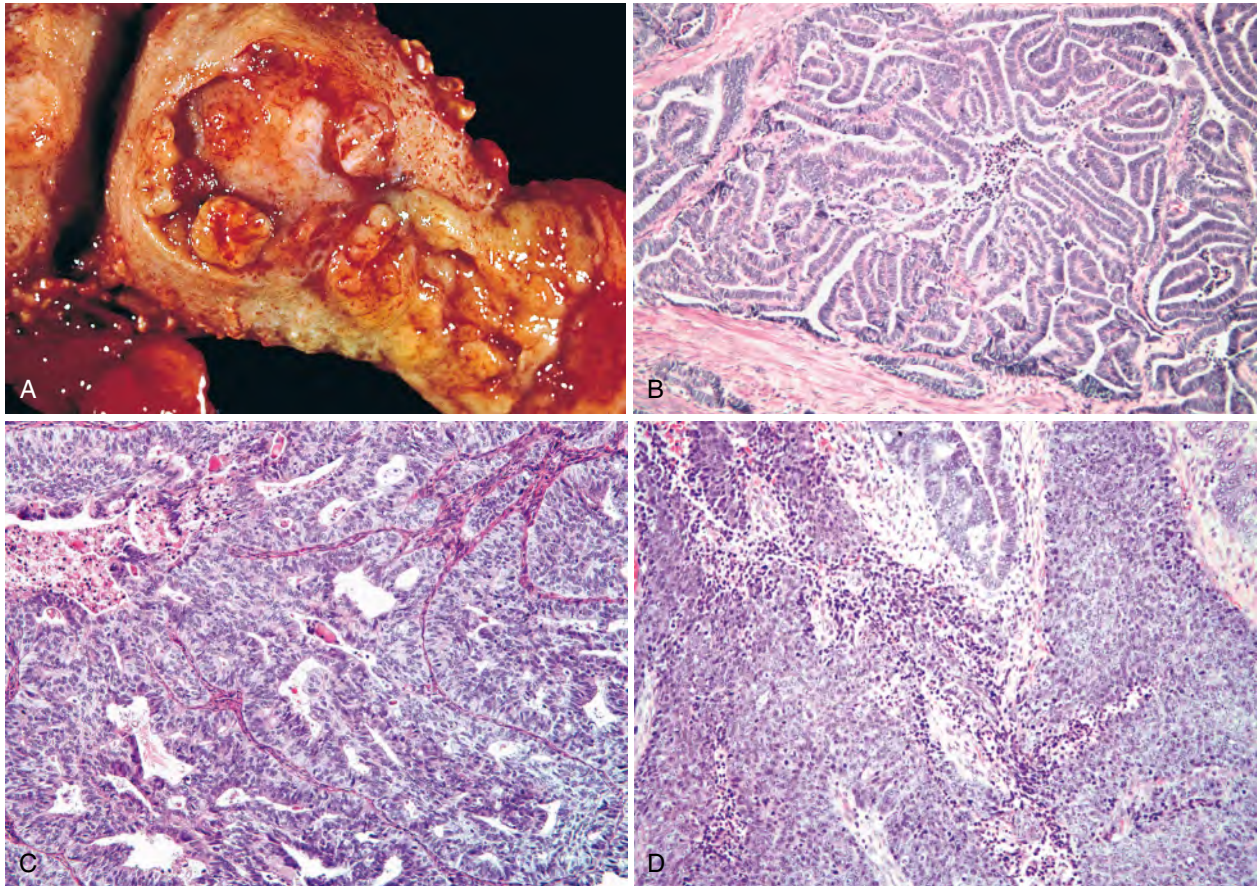


Fig. 22.24 Endometrioid carcinoma. (A) Endometrial adenocarcinoma presenting as a fungating mass in the fundus of the uterus. (B) Well-differentiated (grade 1) endometrioid adenocarcinoma with preserved glandular architecture but lack of intervening stroma. (C) Moderately differentiated (grade 2) endometrioid adenocarcinoma with glandular architecture admixed with solid areas. (D) Poorly differentiated (grade 3) endometrioid adenocarcinoma with a predominantly solid growth pattern.

Endometrioid adenocarcinomas demonstrate glandular growth patterns that fall into three histologic grades: **well differentiated** (grade 1) (Fig. 22.24B), composed almost entirely of well-formed glands; **moderately differentiated** (grade 2) (Fig. 22.24C), showing well-formed glands mixed with areas composed of solid sheets of cells, which by definition make up 50% or less of the tumor; and **poorly differentiated** (grade 3) (Fig. 22.24D), characterized by greater than 50% solid growth pattern. Well-differentiated tumors may be distinguished from hyperplasias by the presence of desmoplastic stroma or complex epithelial growth patterns (e.g., confluent glandular or papillary growth patterns). Many pathologists advocate a binary grading system in which grade 1 and 2 tumors can be combined into a low-grade category whereas grade 3 tumors are considered high-grade tumors.

Tumors with polymerase ϵ mutations or defects in DNA mismatch repair are frequently associated with large numbers of infiltrating T cells. It is believed that these cells represent an ineffective host response to neoantigens created by mutations that lead to tumor-specific amino acid substitutions in proteins.

Up to 20% of endometrioid carcinomas contain foci of squamous differentiation. Squamous elements may be histologically benign appearing when they are associated with low-grade tumors. Less commonly, poorly differentiated endometrioid carcinomas contain squamous elements that appear frankly malignant. Current classification systems grade the carcinomas based on glandular differentiation alone and ignore areas of squamous differentiation.

Serous Endometrial Carcinoma

Serous endometrial carcinoma generally occurs in females who are about 10 years older than those with endometrioid carcinomas, and in contrast with endometrioid carcinoma, they usually arise in the setting of endometrial atrophy. Serous tumors are poorly differentiated/high-grade tumors. They account for approximately 15% of cases of endometrial carcinoma. This subtype shows significant morphologic and biologic overlap with ovarian serous carcinoma.

Pathogenesis

Serous endometrial carcinoma is highly associated with disruptive mutations in the *TP53* tumor suppressor gene. *TP53* mutations are present in greater than 90% of these tumors. Most consist of missense mutations that result in the accumulation of the altered p53 protein (Fig. 22.25B and D). Genomic sequencing studies have detected driver mutations in several other cancer genes. In addition to point mutations, serous carcinomas typically are associated with significant chromosomal instability and numerous copy number alterations.

The tumor presumably arises as a surface epithelial neoplasm that then extends into adjacent gland structures and later invades endometrial stroma; mutations in *TP53* occur relatively early during tumor development as they are found in noninvasive precursor lesions. The poor prognosis of serous carcinomas is thought to be a consequence of their propensity to exfoliate, travel through the fallopian tubes, and implant on peritoneal surfaces like their ovarian counterparts. As a result, they have often spread outside of the uterus at the time of diagnosis.

MORPHOLOGY

Generally, serous carcinomas arise in small atrophic uteri and are often large bulky tumors or deeply invasive into the myometrium. The precursor lesion, **endometrial intraepithelial carcinoma**, consists of malignant cells identical to those of serous carcinoma that are confined to the epithelial surface (Fig. 22.25A–B). The **invasive lesions** may have a papillary growth pattern

composed of cells with marked cytologic atypia including high nuclear-to-cytoplasmic ratio, atypical mitotic figures, hyperchromasia, and prominent nucleoli (Fig. 22.25C–D). However, they can also have a predominantly glandular growth pattern; in such cases, they are distinguished from endometrioid carcinoma by the marked cytologic atypia. All the tumors in this category are classified as high-grade tumors irrespective of architectural pattern. Serous carcinoma, despite relatively superficial endometrial involvement, may be associated with extensive peritoneal disease, suggesting spread by routes other than direct invasion (i.e., tubal regurgitation).

There are other more uncommon morphologic variants of endometrial carcinoma (e.g., clear cell carcinoma, undifferentiated/dedifferentiated carcinoma, mesonephric-like carcinoma, etc.) which DNA sequencing suggests often fall into one of the four molecular subtypes of endometrial carcinoma described earlier.

Clinical Features

Historically, carcinoma of the endometrium has been uncommon in females younger than 40 years of age and has a peak incidence in postmenopausal females 55 to 65 years of age, but the incidence in younger females is increasing. There is currently no available screening test. Although it may be asymptomatic for a period, it usually produces irregular or postmenopausal vaginal bleeding. Fortunately, postmenopausal bleeding often leads to early detection, and cures are possible in most patients. The diagnosis must be established by histologic examination of tissue obtained by biopsy or curettage. At the time of diagnosis, endometrioid carcinomas are analyzed for evidence of DNA mismatch repair defects because approximately 3% to 5% of females with endometrial cancer have Lynch syndrome and are at high risk for colon carcinoma.

As would be anticipated, the prognosis depends heavily on the stage at diagnosis, as well as histologic grade and subtype. The staging system for endometrial adenocarcinoma is:

Stage I—Carcinoma is confined to the corpus uteri itself.

Stage II—Carcinoma involves the corpus and the cervical stroma.

Stage III—Carcinoma extends outside the uterus but not outside the true pelvis.

Stage IV—Carcinoma extends outside the true pelvis or involves the mucosa of the bladder or the rectum.

In the United States, most tumors (about 80%) are stage I well-differentiated or moderately differentiated endometrioid carcinomas. Surgery, alone or in combination with irradiation, gives about 90% 5-year survival in stage I (grade 1 or 2) disease. This rate drops to approximately 75% for stage I/grade 3 tumors and to 50% or less for stage II and III endometrial carcinomas.

As mentioned, serous carcinoma has a propensity for extrauterine (lymphatic or transtubal) spread. For unknown reasons, it occurs more frequently in females of African descent, a difference that accounts for a two-fold higher mortality rate from endometrial carcinoma compared with females of European descent. Overall, the 5-year survival for females with serous carcinoma is 18% to 27%, and even when it is confined to the uterus the recurrence rate is as high as 80%.

Treatment varies according to tumor type. Females with endometrioid carcinoma often receive adjuvant radiation to reduce local recurrence and are treated with chemotherapy when the tumor has spread beyond the uterus. By contrast, because of the aggressive nature of serous carcinoma, patients may be treated with chemotherapy even in the absence of detectable extrauterine spread. Immune checkpoint inhibitors are currently used in females with tumors in the hypermutated MSI and “ultramutated” polymerase ϵ molecular categories. In addition, HER2 is amplified in up to 25% to 30% of endometrial carcinoma,

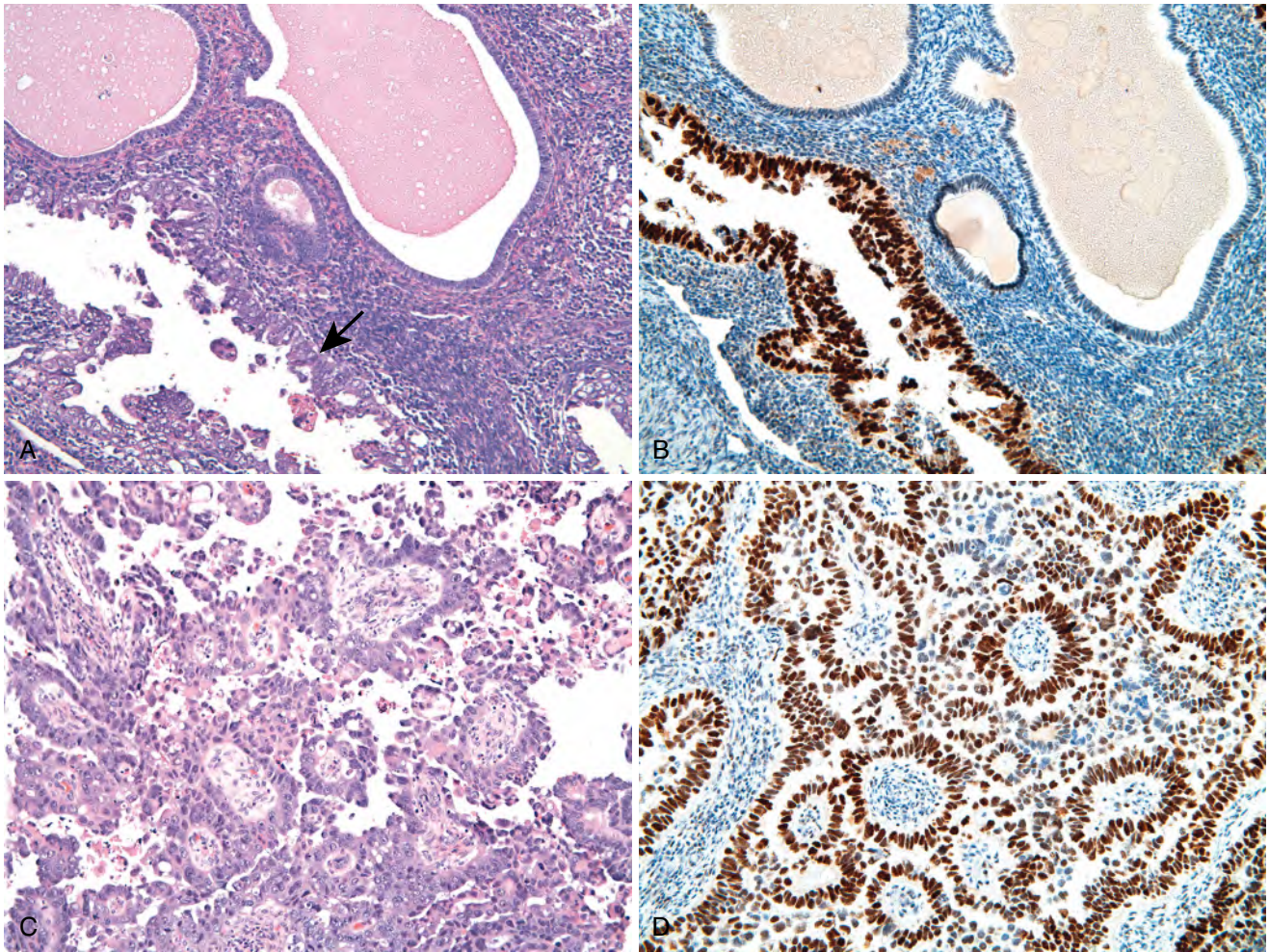


Fig. 22.25 Serous carcinoma. (A) Endometrial intraepithelial carcinoma, a noninvasive form of serous carcinoma showing malignant cells (*arrow*) with morphologic features of serous carcinoma lining the surfaces of the endometrial glands without obvious stromal invasion. (B) Strong, diffuse expression of p53 as detected by immunohistochemistry in endometrial intraepithelial carcinoma. (C) Serous carcinoma of the endometrium with papillary growth pattern consisting of malignant cells with marked cytologic atypia including high nuclear-to-cytoplasmic ratio, atypical mitotic figures, and hyperchromasia. (D) As with the previous lesion, there is an accumulation of p53 protein in the nucleus.

predominantly in *TP53* mutated tumors; thus, HER2 targeting agents are being explored as a possible treatment for a subset of high-stage or recurrent endometrial carcinoma.

Carcinosarcoma

Carcinosarcomas are biphasic tumors with malignant epithelial and mesenchymal components. The epithelial component most often resembles poorly differentiated endometrioid or serous carcinoma, while the mesenchymal component can take several forms. Some contain uterine mesenchymal elements (stromal sarcoma, leiomyosarcoma), while others contain heterologous malignant cell types (rhabdomyosarcoma, chondrosarcoma). Based on molecular studies showing the presence of shared driver mutations, the epithelial and mesenchymal elements appear to be derived from a single founding cancer cell. The mutations found in carcinosarcomas involve the same genes that are mutated in endometrial carcinoma, such as *PTEN*, *TP53*, and *PIK3CA*, whereas alterations typical of sarcomas are absent, suggesting that these tumors are carcinomas that have acquired the capacity for mesenchymal differentiation. At present, the mechanisms underlying the sarcomatous

transformation are unknown, but an abnormality of epigenetic regulation seems likely. The vast majority of carcinosarcomas belong to the copy number high/*TP53* mutated category of endometrial carcinoma.

MORPHOLOGY

Carcinosarcomas are often bulky and polypoid, and they may protrude through the cervical os. These tumors usually contain areas of adenocarcinoma (endometrioid, serous, or clear cell) mixed with the malignant mesenchymal (sarcomatous) elements (Fig. 22.26A); alternatively, the tumor may contain two distinct and separate epithelial and mesenchymal components. Sarcomatous components may also mimic extrauterine tissues (e.g., striated muscle, cartilage, adipose tissue, and bone). Metastases usually contain only epithelial components (Fig. 22.26B).

Carcinosarcomas occur in postmenopausal females and present with bleeding. Outcome is determined primarily by depth of invasion and stage. The only other known prognostic factor is the differentiation of the mesenchymal component; patients with tumors that have heterologous mesenchymal components have a poorer prognosis than

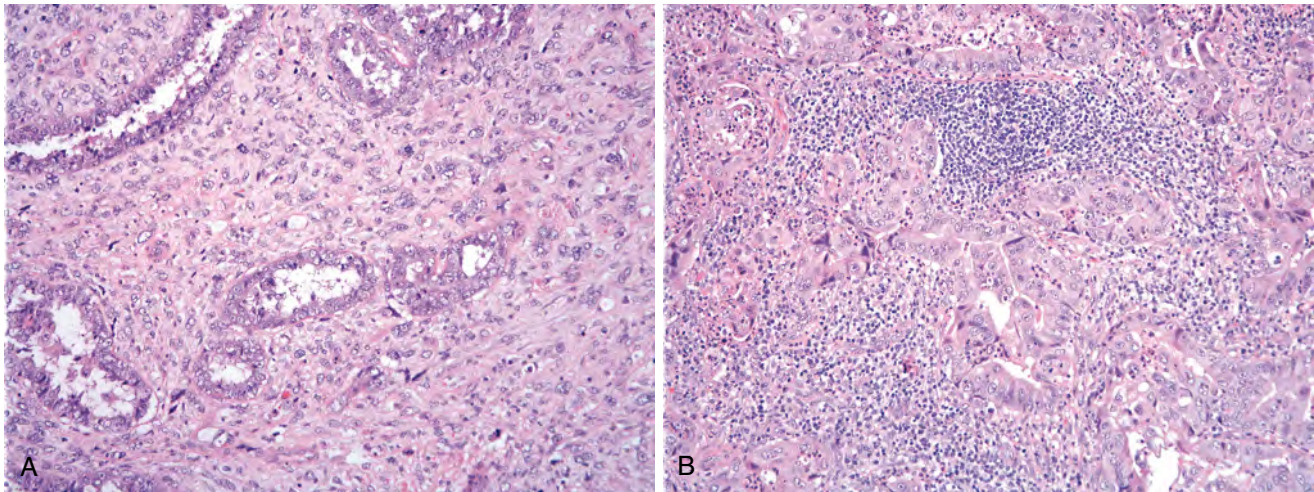


Fig. 22.26 Carcinosarcoma. (A) Micrograph showing both malignant epithelial and stromal components. (B) Lymph node metastasis from a carcinosarcoma showing only the epithelial component, as is typically the case.

those whose tumors do not. Overall 5-year survival rates are 25% to 30% for patients with high-stage disease.

KEY CONCEPTS

Endometrial Carcinoma

- Endometrial carcinoma is the most common malignancy of the female genital tract.
- Four molecular subtypes of endometrioid and serous carcinoma are currently recognized.
- Endometrioid carcinoma is often preceded by atypical hyperplasia and commonly has mutations that upregulate PI3K/AKT signaling.
- Serous carcinoma has a high frequency of mutations in *TP53*. *TP53* mutations are also found in the preinvasive lesion.
- Stage remains the most important factor in outcome; serous tumors are much more likely to present at advanced stage and have a decidedly worse prognosis.
- Carcinosarcomas are aggressive tumors that resemble endometrial carcinoma genetically and have poor outcomes with current therapies.

TUMORS OF ENDOMETRIAL STROMA

These relatively uncommon tumors comprise less than 5% of endometrial cancers and include stromal neoplasms admixed with benign glands (adenosarcomas) and pure stromal neoplasms.

Adenosarcoma

Adenosarcoma presents most commonly as large broad-based endometrial polypoid growths that may prolapse through the cervical os. The diagnosis is based on the presence of malignant-appearing stroma, which coexists with benign but abnormally shaped endometrial glands. These tumors predominate in females between the fourth and fifth decades and are generally considered to be a low-grade malignancy; recurrences develop in one-quarter of cases and are nearly always confined to the pelvis. The principal diagnostic dilemma is distinguishing these tumors from large benign polyps.

Stromal Tumors

The endometrium occasionally gives rise to neoplasms that resemble normal stromal cells. Endometrial stromal neoplasms are divided into

two categories: (1) benign stromal nodules and (2) endometrial stromal sarcoma. Stromal sarcoma may be further divided into low-grade and high-grade types depending on their morphologic appearance.

Clues to the pathogenesis of stromal sarcoma have come from the identification of several recurrent chromosomal aberrations that are quite specific for these malignancies. As with many sarcomas, stromal sarcoma is associated with chromosomal translocations that create fusion genes. Low-grade endometrial stromal sarcoma usually has a translocation in which a portion of the *JAZF1* gene, which encodes a transcriptional repressor, is fused to a second gene belonging to the polycomb gene family, such as *SUZ12*. Polycomb proteins participate in complexes that introduce repressive histone marks into chromatin, thereby silencing genes, and it is hypothesized that the *JAZF1* fusion proteins act by disrupting the function of the polycomb complex, leading to abnormal expression of oncogenic genes. Recently, high-grade endometrial stromal sarcomas have been observed to contain different chromosomal translocations including *YWHAE-NUTM2A/B* and *ZC3H7B-BCOR* fusion genes or internal tandem duplication of *BCOR*, which are presumed to be pathogenically significant but are currently of unknown function.

Unfortunately, relapse is not reliably predicted by either mitotic index or the degree of cytologic atypia. Distant metastases may announce their presence decades after the initial diagnosis, and death from metastatic tumor occurs in about 15% of cases. The 5-year survival rates average 90% for stage I/II, 50% for stage III/IV low-grade tumors, and lower for high-grade tumors.

TUMORS OF THE MYOMETRIUM

Leiomyoma

Uterine leiomyoma (commonly called fibroid) is perhaps the most common tumor in females. They are benign smooth muscle neoplasms that may occur singly, but more often are multiple. Most leiomyomas have normal karyotypes, but approximately 40% have a simple chromosomal abnormality. Several cytogenetic subgroups are recognized, including tumors with rearrangements of chromosomes 12q14 and 6p involving the *HMGIC* and *HMGII* genes, respectively, which are also implicated in a variety of other benign neoplasms. Both genes encode closely related DNA-binding factors that regulate chromatin structure. Additionally, mutations in the gene *MED12* occur in roughly 70% of uterine leiomyomas. *MED12* encodes a component of Mediator,

a multiprotein complex that stimulates gene expression by serving as a bridge between transcription factors and RNA polymerase. Mice expressing mutated forms of *MED12* develop uterine leiomyomas, proving that such mutations are causative, but precisely how they contribute to tumor development remains to be defined. Leiomyomas can also be associated with hereditary leiomyomas and renal cell carcinoma (HLRCC) syndrome caused by germline mutations in the fumarate hydratase (*FH*) gene. Acquired mutations in *FH* can also be seen in sporadic leiomyomas of the uterus.

MORPHOLOGY

Leiomyomas are sharply circumscribed, discrete, round, firm, gray-white tumors varying in size from small, barely visible nodules to massive tumors that fill the pelvis. Except in rare instances, they are found within the myometrium of the corpus, where they may be located intramurally or just beneath the endometrium (submucosal) or the serosa (subserosal) (Fig. 22.27A). Only infrequently do they involve the uterine ligaments, lower uterine segment, or cervix. The characteristic whorled pattern of smooth muscle bundles on cut section usually makes these lesions readily identifiable. Large tumors may develop areas of yellow-brown to red softening.

Leiomyomas are composed of bundles of smooth muscle cells that resemble the uninvolved myometrium (Fig. 22.27B). Usually, the individual muscle cells are uniform in size and shape and have a characteristic oval nucleus and long, slender bipolar cytoplasmic processes. Mitotic figures are scarce. Morphologic variants include leiomyoma with bizarre nuclei, which has nuclear atypia and giant cells (these may be associated with somatic or germline mutations in *FH*) and cellular leiomyomas. Both have a low mitotic index, helping to distinguish these benign tumors from leiomyosarcoma. An extremely rare variant, **intra-venous leiomyomatosis**, is a uterine leiomyoma that extends into vessels and spreads hematogenously to other sites, most commonly the vena cava and the right atrium. Another variant, **disseminated peritoneal leiomyomatosis**, presents as multiple small peritoneal nodules. Both are considered benign despite their unusual behavior.

Uterine leiomyomas, even when large or numerous, may be asymptomatic. Common signs and symptoms include abnormal bleeding, urinary frequency due to compression of the bladder, sudden pain from infarction of a large or pedunculated tumor, and impaired fertility. In pregnant females, leiomyomas may increase the frequency of

spontaneous abortion, fetal malpresentation, uterine inertia (failure to contract with sufficient force), and postpartum hemorrhage. Malignant transformation to leiomyosarcoma is extremely rare.

Leiomyosarcoma

These uncommon malignant neoplasms are thought to arise from the myometrium or endometrial stromal precursor cells, rather than malignant transformation of leiomyomas. In contrast to leiomyomas, leiomyosarcomas have complex, highly variable karyotypes that frequently include deletions and mutations in tumor suppressor genes and oncogenes (e.g., *TP53*, *RB*, *PTEN*). Like leiomyomas, a subset contains *MED12* mutations, a genetic aberration that appears to be virtually unique to uterine smooth muscle tumors.

MORPHOLOGY

Leiomyosarcomas grow within the uterus in two somewhat distinctive patterns: (1) bulky, fleshy masses that invade the uterine wall or (2) polypoid masses that project into the uterine lumen (Fig. 22.28A). They exhibit a wide range of cytologic atypia, from extremely well differentiated to highly anaplastic (Fig. 22.28B). The distinction from leiomyoma is based on nuclear atypia, mitotic index, and tumor necrosis. With few exceptions, the presence of 10 or more mitoses per 10 high-power (400 $\times$) fields indicates malignancy, particularly if accompanied by cytologic atypia and/or coagulative necrosis. If the tumor contains nuclear atypia or large (epithelioid) cells, four or more mitoses per 10 high-power (400 $\times$) fields are sufficient to justify a diagnosis of malignancy. Rare exceptions include mitotically active leiomyomas in young or pregnant females, and caution must be exercised in interpreting such neoplasms as malignant. A proportion of smooth muscle neoplasms may be impossible to classify and are called smooth muscle tumors of uncertain malignant potential (STUMP).

Clinical Features

Leiomyosarcoma occurs both before and after menopause, with a peak incidence at 40 to 60 years of age. These tumors often recur following surgery and more than one-half eventually metastasize hematogenously to distant organs, such as lungs, bone, and brain. Dissemination throughout the abdominal cavity is also encountered. The overall 5-year survival rate is about 40%, but the anaplastic lesions have a 5-year survival rate of only 10% to 15%.

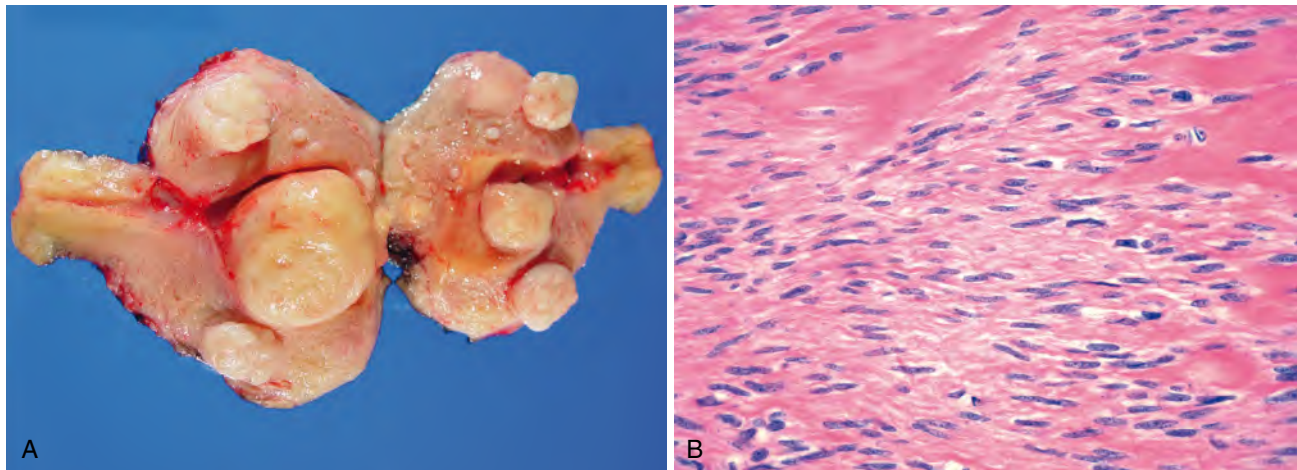


Fig. 22.27 Leiomyomas of the uterine myometrium (A) The uterus is opened to reveal multiple tumors in submucosal (bulging into the endometrial cavity), intramural, and subserosal locations that display a firm white appearance on sectioning. (B) Leiomyoma showing well-differentiated, regular, spindle-shaped smooth muscle cells associated with hyalinization.

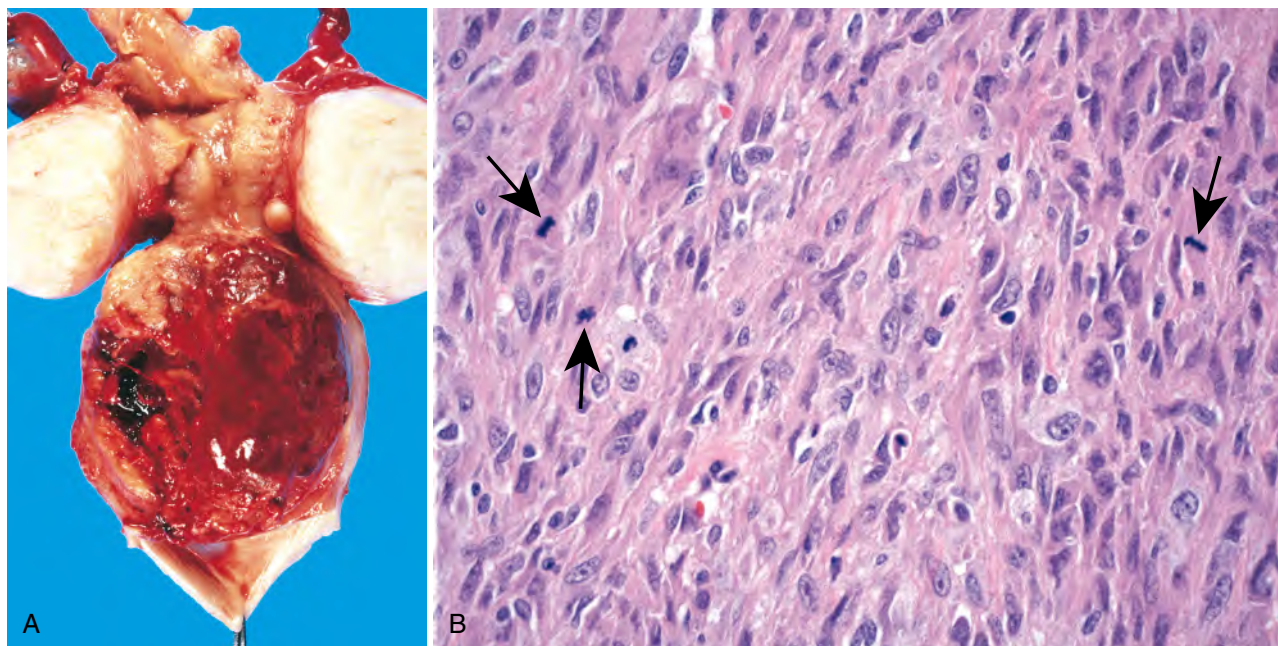


Fig. 22.28 Leiomyosarcoma. (A) A large hemorrhagic tumor mass distends the lower corpus and is flanked by two leiomyomas. (B) The tumor cells are irregular in size and have hyperchromatic nuclei. Numerous mitotic figures are present (arrows).

KEY CONCEPTS

Tumors of the Endometrial Stroma and Myometrium

- Endometrial stromal tumors include stromal nodules, low-grade stromal sarcoma, and high-grade stromal sarcoma.
 - Stromal nodules are benign, well-circumscribed tumors.
 - Low-grade stromal sarcoma resembles stromal nodules but infiltrates the surrounding myometrium. It is associated with fusion of the *JAZF1* gene and various polycomb factor genes, usually *SUZ12*.
 - High-grade stromal sarcoma shows marked atypia and is associated with other gene fusions.
 - Both low- and high-grade stromal sarcoma are prone to late recurrences.
- Leiomyomas are very common benign smooth muscle tumors that cause significant morbidity but are not prone to malignant transformation.
- Leiomyosarcoma (a malignant smooth muscle tumor) is an uncommon, highly malignant myometrial tumor that usually arises de novo.

FALLOPIAN TUBES

The most common disorders affecting the fallopian tube are infections and associated inflammatory conditions, followed in frequency by ectopic (tubal) pregnancy and endometriosis.

Inflammations

Suppurative salpingitis may be caused by any pyogenic organism; in some cases, more than one organism is involved. *Gonococcus* is the causative organism in more than 60% of cases, with *chlamydiae* being responsible for many of the remaining cases. These tubal infections are a part of pelvic inflammatory disease, described earlier in this chapter.

Tuberculous salpingitis is rare in the United States, accounting for no more than 1% to 2% of all forms of salpingitis. It is more common, however, in parts of the world where tuberculosis is prevalent and is an important cause of infertility in these areas.

Tumors and Cysts

The most common primary lesions of the fallopian tube (excluding endometriosis) are minute, 0.1- to 2-cm translucent cysts filled with clear serous fluid, called *paratubal cysts*. Larger varieties are found near the fimbriated end of the tube or in the broad ligaments and are referred to as *hydatis of Morgagni*. These cysts, lined by benign serous (tubal type) epithelium, are presumed to arise in remnants of the müllerian duct and are of little significance.

Benign tumors of the fallopian tube are rare and include *adenomatoid tumor*, which occurs subserosally on the tube or sometimes in the mesosalpinx. These small nodules are counterparts of the adenomatoid tumors that occur in the testes or epididymis (Chapter 21).

Historically, fallopian tube carcinoma was considered rare. However, it is now known that a majority of “serous ovarian cancers” actually arise from the epithelium of the fallopian tube (discussed later). This idea is supported by the frequent identification of serous tubal in situ carcinoma in females at risk for serous carcinoma (e.g., females with germline *BRCA1* mutations) and observations showing that failure to remove the fallopian tubes at the time of oophorectomy is associated with a significant residual risk of “ovarian” cancer. We will return to this issue when discussing ovarian cancer in the next section.

OVARIES

The most common lesions encountered in the ovary are functional or benign cysts and tumors. Neoplastic disorders can be grouped according to their origin from each of the three main ovarian cell types: (1) müllerian epithelium, (2) germ cells, and (3) sex cord–stromal cells. Primary inflammatory conditions of the ovary (oophoritis) are uncommon, and on rare occasions may have an autoimmune basis (autoimmune oophoritis); such autoimmune reactions affect the ovarian follicles and may lead to infertility.

NONNEOPLASTIC AND FUNCTIONAL CYSTS

Follicle and Luteal Cysts

Cystic follicles are very common in the ovary. They originate from unruptured graafian follicles or in follicles that have ruptured and immediately sealed.

MORPHOLOGY

These cysts are usually multiple. They range in size up to 2 cm in diameter, are filled with a clear serous fluid, and are lined by a gray, glistening membrane. On occasion, larger cysts exceeding 2 cm (follicle cysts) may be diagnosed by palpation or ultrasonography; these may cause pelvic pain. Granulosa lining cells are present if the intraluminal pressure has not been so great as to cause their atrophy. The outer theca cells may be conspicuous due to increased amounts of pale cytoplasm (a change referred to as luteinization). As discussed later, when luteinization is pronounced (hyperthecosis), it may be associated with increased estrogen production and endometrial abnormalities.

Luteal cysts (corpora lutea) are present in the normal ovaries of females of reproductive age. They are lined by a rim of bright yellow tissue containing luteinized granulosa cells and are prone to rupture, which may produce a peritoneal reaction. Sometimes the combination of old hemorrhage and fibrosis may make their distinction from endometriotic cysts difficult.

Polycystic Ovaries and Stromal Hyperthecosis

Polycystic ovarian syndrome (PCOS) is a complex endocrine disorder characterized by hyperandrogenism, menstrual abnormalities, polycystic ovaries, chronic anovulation, and decreased fertility. It affects 6% to 10% of reproductive age females worldwide. It is also associated with obesity, type 2 diabetes, and premature atherosclerosis, all of which are indicative of an underlying metabolic disorder. The etiology of PCOS remains incompletely understood. It is marked by a dysregulation of enzymes involved in androgen biosynthesis and excessive androgen production, which is a central feature of this disorder. In addition, females with PCOS show insulin resistance and altered adipose tissue metabolism, which contribute to the development of both diabetes and obesity.

The central morphologic abnormality of PCOS is numerous cystic follicles or follicle cysts that enlarge the ovaries. However, polycystic ovaries are detected in 20% to 30% of all females so this finding is not specific. In addition, due to an increase in free serum estrone levels, females with PCOS are at risk for endometrial hyperplasia and carcinoma.

Stromal hyperthecosis, also called cortical stromal hyperplasia, is a disorder of ovarian stroma most often seen in postmenopausal females, but it may overlap with PCOS in younger females. The disorder is characterized by uniform, usually bilateral, enlargement of the ovary (up to 7 cm), which has a white to tan appearance on sectioning. Microscopic examination shows hypercellular stroma and luteinization of the stromal cells, which are visible as discrete nests of cells with vacuolated cytoplasm. The clinical presentation and effects on the endometrium are similar to those of PCOS, although virilization may be even more striking.

OVARIAN TUMORS

There are numerous types of ovarian tumors. About 80% are benign, and these occur mostly in young females between 20 and 45 years of age. So-called borderline tumors (tumors of indeterminate malignancy)

occur at slightly older ages. Malignant tumors are more common in females between 45 and 65 years of age. Ovarian cancer accounts for 3% of cancers in females and is the fifth most common cause of death due to cancer in females in the United States. **Because most ovarian cancers have spread beyond the ovary and fallopian tube by the time of diagnosis, they account for a disproportionate number of deaths from cancer of the female genital tract.**

Classification

The classification of ovarian tumors given in Table 22.3 is a simplified version of the World Health Organization Histological Classification, which separates ovarian neoplasms according to the most probable tissue of origin. It is now believed that most tumors of the ovary arise ultimately from one of three ovarian components:

- Surface/fallopian tube epithelium and endometriosis

TABLE 22.3 WHO Classification of Ovarian Neoplasms

Surface Epithelial-Stromal Tumors

Serous tumors

- Benign (cystadenoma, cystadenofibroma)
- Borderline (serous borderline tumor)
- Malignant (low- and high-grade serous adenocarcinoma)

Mucinous tumors, endocervical-like and intestinal type

- Benign (cystadenoma, cystadenofibroma)
- Borderline (mucinous borderline tumor)
- Malignant (mucinous adenocarcinoma)

Endometrioid tumors

- Benign (cystadenoma, cystadenofibroma)
- Borderline (endometrioid borderline tumor)
- Malignant (endometrioid adenocarcinoma)

Clear cell tumors

- Benign
- Borderline
- Malignant (clear cell adenocarcinoma)

Brenner tumors

- Benign Brenner tumor
- Borderline Brenner tumor
- Malignant Brenner tumor

Epithelial-stromal

- Adenosarcoma
- Carcinosarcoma

Sex Cord–Stromal Tumors

Granulosa cell tumors

- Fibromas
- Fibrothecomas
- Thecomas
- Sertoli-Leydig cell tumors
- Steroid (lipid) cell tumors

Germ Cell Tumors

Teratoma

- Immature
- Mature
- Solid
- Cystic (dermoid cyst)
- Monodermal (e.g., struma ovarii, carcinosarcoma)

Dysgerminoma

Yolk sac tumor

Mixed germ cell tumors

Metastatic Cancer From Nonovarian Primary

- Colonic, appendiceal
- Gastric
- Pancreaticobiliary
- Breast

- Germ cells, which migrate to the ovary from the yolk sac and are pluripotent
- Stromal cells, including the sex cords, which are forerunners of the endocrine apparatus of the postnatal ovary

There is also a group of miscellaneous tumors, and finally there are secondary or metastatic tumors to the ovary.

Although some of the specific tumors have distinctive features and are hormonally active, most are nonfunctional and produce few symptoms until they reach a large size. Some, principally epithelial tumors, are often bilateral. Table 22.4 lists the tumors and their subtypes. Abdominal pain and distention, urinary and gastrointestinal tract symptoms due to compression by the tumor or cancer invasion, and vaginal bleeding are the most common symptoms. The benign forms may be entirely asymptomatic and occasionally are found incidentally on abdominal or pelvic examination or during surgery.

Epithelial Tumors

Most primary ovarian neoplasms arise from müllerian epithelium. The classification of these tumors is based on both differentiation and extent of proliferation of the epithelium. There are three major histologic types based on the differentiation of the neoplastic epithelium: serous, mucinous, and endometrioid tumors. These epithelial proliferations are classified as benign, borderline, and malignant. The benign tumors are often further subclassified based on the components of the tumors, which may include cystic areas (cystadenoma), cystic and fibrous areas (cystadenofibroma), and predominantly fibrous areas (adenofibroma). The borderline tumors and malignant tumors often have a cystic component. The tumors can be relatively small, or they can grow to fill the entire pelvis before they are detected.

Molecular studies have shown that the different morphologic types of ovarian carcinoma harbor distinct, but overlapping, genetic alterations

that shed light on their site of origin and pathogenesis (Fig. 22.29). Low-grade carcinomas (type I) often arise in association with borderline tumors or endometriosis. These cancers encompass several histologic subtypes, including serous, endometrioid, and mucinous, as discussed later. On the other hand, high-grade serous carcinomas (type II) most often arise in the fallopian tube fimbriae and are preceded by a serous tubal intraepithelial carcinoma (see later).

Serous Tumors

These neoplasms include the most common malignant ovarian tumors and account for approximately 50% of all cancers of the ovary. Although the term *serous* appropriately describes the cyst fluid, it has become synonymous with the tubal-like epithelium in these tumors. About 75% are benign or borderline, and 25% are malignant. Benign and borderline tumors are most common between 20 and 45 years of age. Serous carcinomas occur later in life on average, but often occur at earlier ages in familial cases.

Pathogenesis

Little is known about the risk factors for benign and borderline tumors. Risk factors for serous carcinomas are also not completely understood, but nulliparity, family history, and heritable mutations play a role. There is a higher frequency of carcinoma in females with low parity. Females 40 to 59 years of age who have taken oral contraceptives or undergone tubal ligation have a reduced risk of developing ovarian cancer. The most intriguing risk factors are genetic. As discussed in Chapters 7 and 23, inherited germline mutations in *BRCA1* or *BRCA2* increase susceptibility to both ovarian cancer and breast cancer. *BRCA1* mutations are present in about 5% of patients younger than 70 years with ovarian cancer. The estimated risk of ovarian cancer in females bearing *BRCA1* or *BRCA2* mutations is 20% to 60% by 70 years of age.

TABLE 22.4 Frequency of Major Ovarian Tumors

Type	Percentage of Malignant Ovarian Tumors	Percentage That Are Bilateral
Serous	47	
Benign (60%)		25
Borderline (15%)		30
Malignant (25%)		65
Mucinous	3	
Benign (80%)		5
Borderline (10%)		10
Malignant (10%)		<5
Endometrioid carcinoma	20	40
Undifferentiated carcinoma	10	—
Clear cell carcinoma	6	40
Granulosa cell tumor	5	5
Teratoma	1	15
Benign (96%)		Rare
Malignant (4%)		
Metastatic	5	>50
Others	3	—

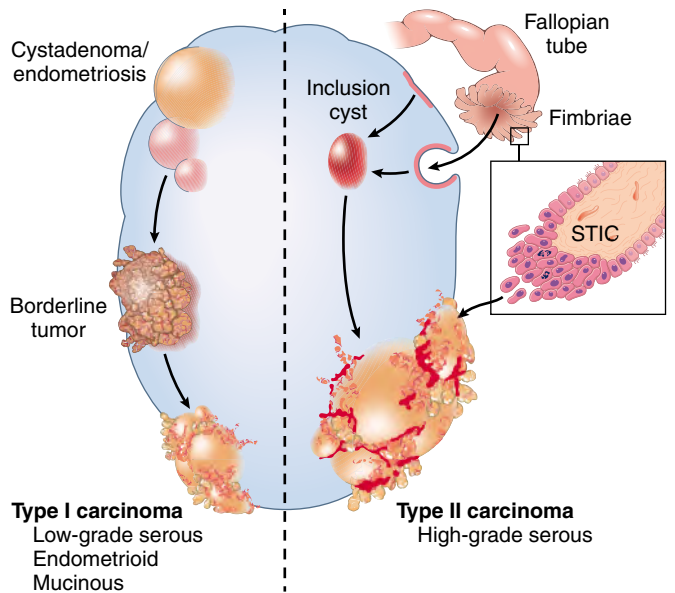


Fig. 22.29 Schematic diagram of the pathogenesis of ovarian epithelial tumors. Type I tumors progress from benign tumors through borderline tumors that may give rise to a low-grade carcinoma. These include low-grade serous, endometrioid, and mucinous carcinomas. Type II tumors arise from inclusions cysts/fallopian tube epithelium via intraepithelial precursors that are often not identified. They demonstrate high-grade features and are most commonly of serous histology. *STIC*, Serous tubal intraepithelial carcinoma.

Serous ovarian carcinoma is divided into two major groups: (1) low-grade carcinoma and (2) high-grade carcinoma. This distinction is made based on the degree of nuclear atypia and correlates with patient survival. Low-grade carcinomas may arise in association with serous borderline tumors, while high-grade carcinomas arise from in situ lesions in the fallopian tube fimbriae or from serous inclusion cysts within the ovary.

The concept of a fallopian tube origin for high-grade serous carcinomas was initially sparked by the observation that females with *BRCA1/2* germline mutations were often discovered at the time of prophylactic salpingo-oophorectomy to have areas of marked epithelial atypia in their fallopian tubes. The lesions, called *serous tubal intraepithelial carcinoma* (STIC), have since been described in association with sporadic high-grade serous ovarian cancers, suggesting that at least some high-grade serous carcinomas arise from the fallopian tube. What then is the origin of high-grade serous carcinomas that involve the ovary, without concomitant involvement of the fallopian tube? Historically, the source of these tumors was hypothesized to be cortical inclusion cysts (Fig. 22.30), which were thought to arise through invagination of the surface epithelium, followed by serous metaplasia. Recently, an alternative idea has been proposed that the cysts arise from implantation of detached fallopian tube epithelium at sites where ovulation has disrupted the surface of the ovary (see Fig. 22.29).

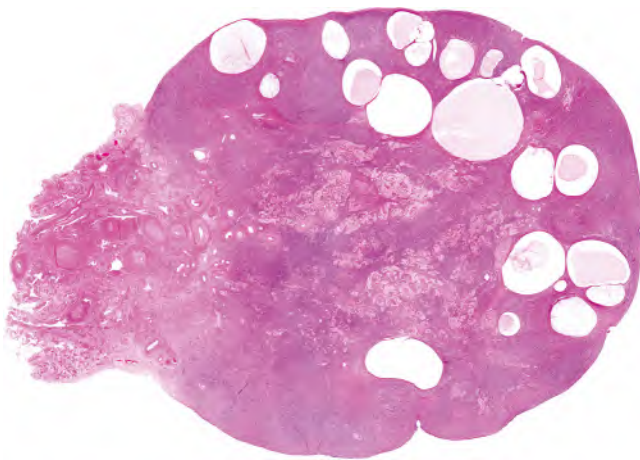


Fig. 22.30 Cortical inclusion cysts of the ovary.

Although the exact percentage of sporadic high-grade serous carcinomas that arise in the fallopian tube is not known, it is thought most have their origin in the fallopian tube with fewer coming from ovarian inclusion cysts. This paradigm shift has already altered the management of females at high-risk for ovarian carcinoma (*BRCA* mutation carriers and females with a strong family history of breast/ovarian cancer), as these individuals now undergo salpingo-oophorectomy, instead of simple oophorectomy. There is also increasing use of simple salpingectomy, particularly in young females, to reduce serous ovarian carcinoma risk. Because ovarian cancer arises in older females, it will be some time before the full impact of this intervention is understood.

Regardless of their origin, studies have shown that low- and high-grade serous carcinoma have distinct mutational profiles:

- Low-grade tumors arising in serous borderline tumors have mutations in the *KRAS*, *BRAF*, or *ERBB2* oncogenes, and usually lack mutations in *TP53*.
- High-grade tumors have a high frequency of *TP53* mutations and lack mutations in either *KRAS* or *BRAF*. Genomic imbalances are very common and include amplifications of several oncogenes (e.g., *PIK3CA*) and deletions of tumor suppressor genes (e.g., *RB*). Almost all ovarian carcinomas arising in females with *BRCA1* or *BRCA2* mutations are high-grade serous carcinomas with *TP53* mutations. Curiously, *BRCA1* and *BRCA2* mutations are rare in sporadic high-grade serous carcinoma.

The biologic behavior of serous tumors depends on the degree of differentiation and the extent and distribution of peritoneal disease. Serous tumors may occur on the surface of the ovaries; if unencapsulated, such tumors are likely to spread to the peritoneum. As discussed, most high-grade serous carcinomas originate from the fallopian tube, from which they may exfoliate and spread to the peritoneum. In rare instances, they appear to originate on the peritoneal surface (referred to as primary peritoneal serous carcinoma).

MORPHOLOGY

Serous tumors may present as multicystic lesions in which papillary epithelium is contained within fibrous walled cysts (intracystic) (Fig. 22.31A) or as a mass projecting from the ovarian surface. Benign tumors typically have a smooth glistening cyst wall with no epithelial thickening or with small papillary projections. Borderline tumors contain an increased number of papillary projections (Fig. 22.31A–B) and may be found within cysts or growing on the surface of the ovary. Larger areas of solid or papillary tumor growth, tumor irregularity, and

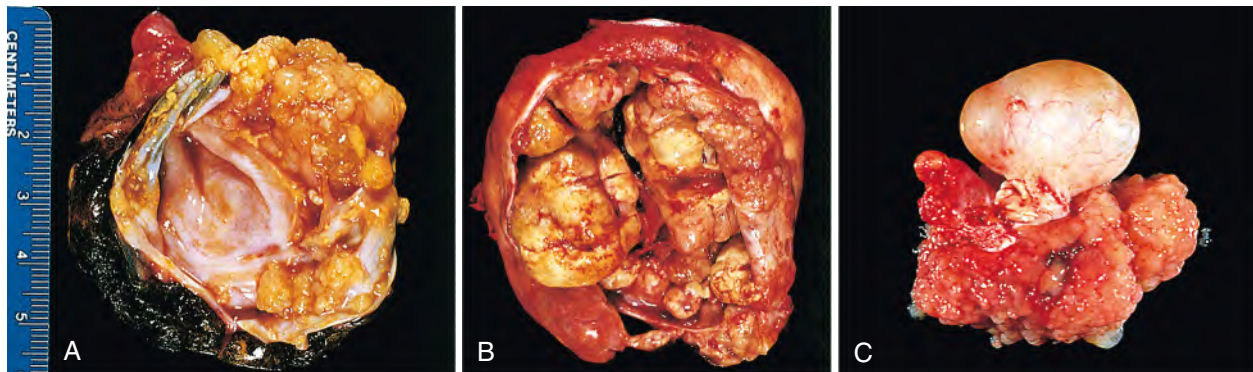


Fig. 22.31 Gross appearances of serous tumors of the ovary. (A) Serous borderline tumor opened to display a cyst cavity lined by delicate papillary tumor growths. (B) Another borderline tumor growing on the ovarian surface (lower). (C) Carcinoma. The cyst is opened to reveal a large, bulky tumor mass.

fixation or nodularity of the capsule are features associated with malignancy (Fig. 22.31C). Bilaterality is common, occurring in 20% of benign serous cystadenomas, 30% of serous borderline tumors, and approximately 66% of serous carcinomas.

Microscopically, the cysts are lined by columnar epithelium. In benign tumors (Fig. 22.32A), the epithelial cells retain abundant cilia, and microscopic papillae may be found. **Serous borderline tumors** exhibit increased complexity of the stromal papillae, stratification of the epithelium, and mild nuclear atypia, but invasion of the stroma is not seen (Fig. 22.32B). The epithelial cells often grow in a delicate, papillary pattern referred to as “micropapillary carcinoma,” which is thought to be the precursor to **low-grade serous carcinoma** (Fig. 22.32C). **High-grade serous carcinoma** is distinguished by having more complex growth patterns and widespread infiltration or effacement of the underlying stroma (Fig. 22.32D). The individual tumor cells display marked nuclear atypia, including pleomorphism (three-fold variation of nuclear size), numerous mitotic figures (greater than 12 mitoses per 10 high-power fields), atypical mitotic figures, and multinucleation; occasionally, tumors may be so undifferentiated that serous features are no longer recognizable. **Serous tubal intraepithelial carcinoma** consists of cells morphologically identical to high-grade serous carcinoma that do not invade the underlying stroma of the fallopian tube. Concentric calcifications (psammoma bodies) are common in all types of serous tumors but are not specific for neoplasia.

Ovarian serous tumors, both low- and high-grade, have a propensity to spread to the peritoneal surfaces and omentum and are commonly associated with the presence of ascites. As with other tumors, the extent of the spread outside the ovary determines the stage of the disease.

Prognosis is closely related to the histologic appearance of the tumor and the presence and extent of peritoneal disease. Borderline serous tumors may arise from or secondarily involve the peritoneal surfaces as noninvasive implants, remaining localized and causing no symptoms, or slowly spread, producing intestinal obstruction or other complications after many years. Similarly, low-grade carcinomas, even after spread outside the ovary, often progress slowly, and patients may survive for relatively long periods before dying of disease. In contrast, high-grade tumors are often widely metastatic throughout the abdomen at the time of presentation, a picture associated with rapid clinical deterioration. Consequently, grading of the tumor, even after extension to the peritoneum, impacts both prognosis and selection of therapy. The 5-year survival rate for borderline and malignant tumors confined to the ovary is 100% and 70%, respectively, whereas the 5-year survival rate for the same tumors involving the peritoneum is about 90% and 25%, respectively. Because of their protracted course, borderline tumors may recur after many years, and 5-year survival is not synonymous with cure.

Mucinous Tumors

Mucinous tumors account for about 20% to 25% of all ovarian neoplasms. They occur principally in middle adult life and are rare before puberty and after menopause. The vast majority are benign or borderline tumors. Primary ovarian mucinous carcinomas are uncommon and account for approximately 3% of all ovarian cancers.

Pathogenesis

Mutation of the *KRAS* proto-oncogene is a consistent genetic alteration in mucinous tumors of the ovary, including most benign mucinous

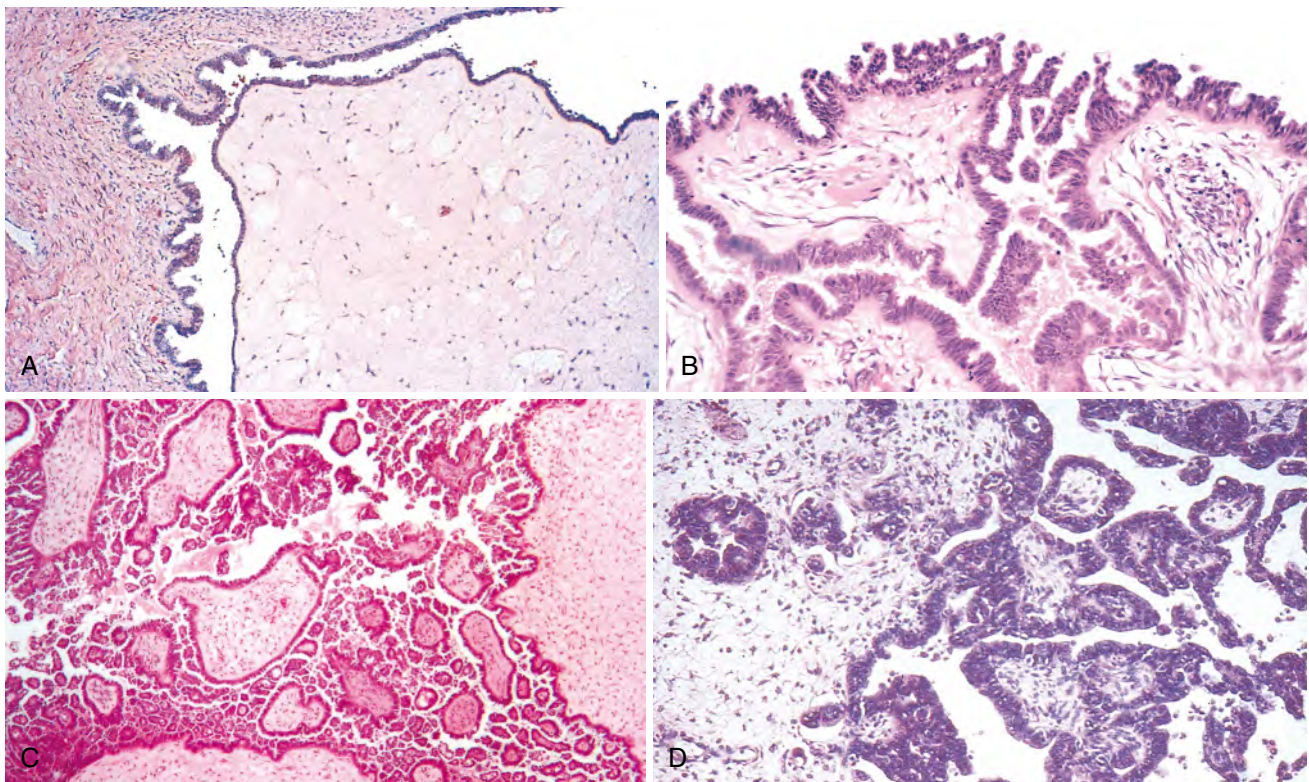


Fig. 22.32 Microscopic appearances of serous tumors of the ovary. (A) Serous cystadenoma revealing stromal papillae with a columnar epithelium. (B) Borderline serous tumor showing increased architectural complexity and epithelial cell stratification. (C) Complex micropapillary growth defines a low-grade “micropapillary” serous carcinoma. (D) High-grade serous carcinoma of the ovary with invasion of underlying stroma.

cystadenomas (58%), mucinous borderline tumors (75% to 86%), and ovarian mucinous carcinomas (85%). Interestingly, one study showed that several tumors containing distinct areas with benign, borderline, or malignant morphologies had identical *KRAS* mutations in each area. Thus, *KRAS* mutations may initiate the development of these neoplasms. The lack of benign mucinous epithelium in the ovary has long created curiosity about their origin. Current data suggest that some primary mucinous tumors may have their origin in mucinous glands of teratomas and Brenner tumors, while others may arise from müllerian epithelium that undergoes mucinous differentiation.

MORPHOLOGY

Mucinous tumors differ from the serous variety in several ways. The surface of the ovary is rarely involved, and only 5% of primary mucinous cystadenomas and mucinous carcinomas are bilateral. Mucinous tumors also tend to produce larger cystic masses; some have been recorded with weights of more than 25 kg. They are multiloculated tumors filled with sticky, gelatinous fluid rich in glycoproteins (Fig. 22.33A).

Microscopically, benign mucinous tumors are characterized by a lining of tall, columnar epithelial cells with apical mucin that lack cilia. The vast majority demonstrate gastric or intestinal type differentiation; uncommonly, tumors may show endocervical type mucinous differentiation instead (Fig. 22.33B). Mucinous borderline tumors are distinguished from cystadenomas by epithelial stratification, tufting, and/or papillary intraglandular growth, often producing an appearance strikingly similar to tubular adenomas or villous adenomas of the intestine. **Mucinous carcinoma** characteristically demonstrates confluent glandular growth that is now recognized as a form of “expansile” invasion; however, others show desmoplastic stromal invasion, and the two invasive patterns can coexist. Some authors use the term intraepithelial carcinomas for tumors with marked epithelial atypia that lack invasive features. Approximate 10-year survival rates for stage I noninvasive “intraepithelial carcinoma” and invasive malignant tumors are greater than 95% and 90%, respectively. Mucinous carcinomas that have spread beyond the ovary are usually fatal, but as stated earlier, these tumors are uncommon and must be distinguished from metastatic mucinous adenocarcinoma.

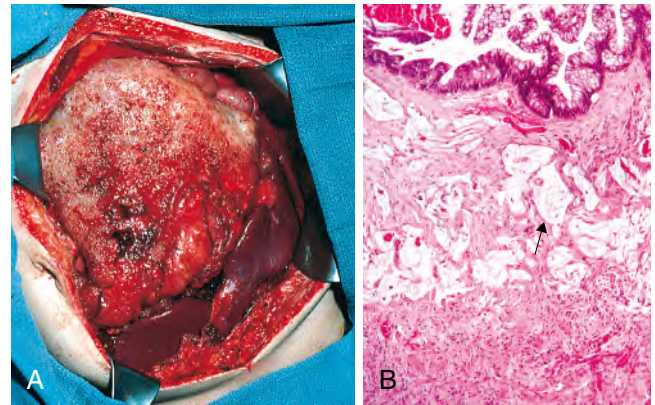


Fig. 22.34 Pseudomyxoma peritonei. (A) View at laparotomy revealing massive overgrowth of a gelatinous metastatic tumor. (B) Histology of peritoneal implants from an appendiceal tumor, showing mucin-producing epithelium and free mucin (arrow). (A, Courtesy Dr. Paul H. Sugarbaker, Washington Hospital Cancer Center, Washington, DC.)

A clinical condition referred to as *pseudomyxoma peritonei* is marked by mucinous ascites, cystic epithelial implants on the peritoneal surfaces, adhesions, and frequent involvement of the ovaries (Fig. 22.34). Pseudomyxoma peritonei, if extensive, may result in intestinal obstruction and death. Historically, it was thought that many cases in females were due to the spread of primary ovarian mucinous neoplasms. However, recent evidence points in almost all cases to an extraovarian source, usually the appendix (Chapter 17). Because most primary mucinous ovarian tumors are unilateral, bilateral presentation of mucinous tumors always requires exclusion of metastasis from a nonovarian primary tumor.

Endometrioid Ovarian Tumors

Endometrioid carcinoma accounts for approximately 20% of all ovarian cancers. Benign endometrioid tumors, called *endometrioid adenofibroma*, and borderline endometrioid tumors also occur, but are uncommon. Approximately 15% to 30% of ovarian endometrioid carcinomas are accompanied by carcinoma of the endometrium; although such cases have a relatively good prognosis, molecular data

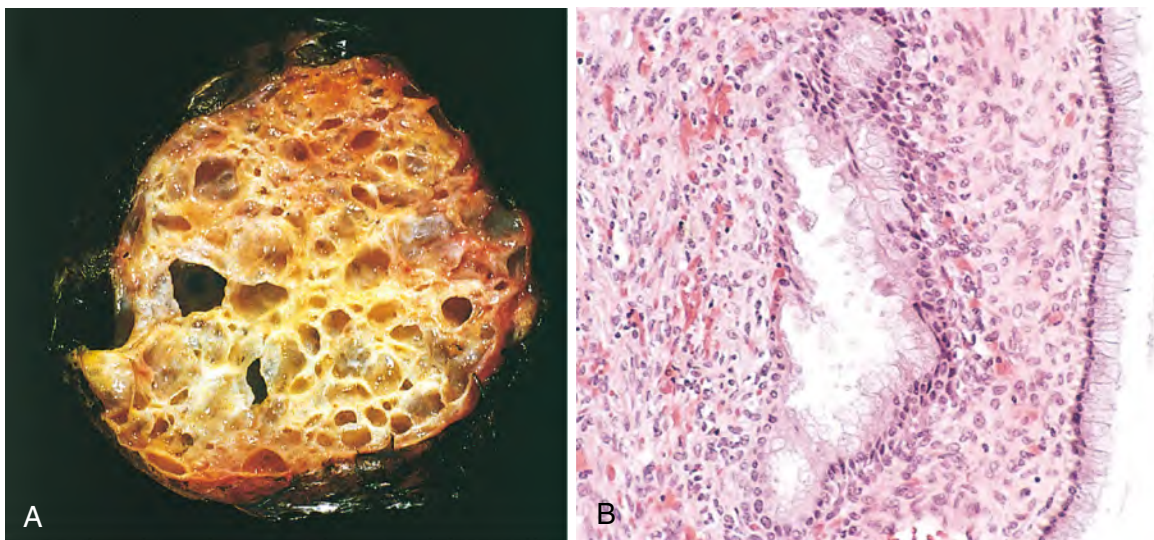


Fig. 22.33 Mucinous cystadenoma (A) Note the multicystic appearance, delicate septa, and the presence of glistening mucin within the cysts. (B) Columnar cells lining the cysts.

suggest that in most instances of concomitant endometrial and ovarian disease, the ovarian disease arises from metastatic spread of a primary endometrial tumor.

Pathogenesis

In about 15% to 20% of cases, endometrioid carcinoma coexists with endometriosis. The peak incidence of tumors associated with endometriosis is a decade earlier than that of endometrioid carcinomas that are not associated, suggesting that ovarian endometriosis serves as a precursor to ovarian endometrioid carcinoma in some instances. Molecular studies have found striking similarities to endometrial endometrioid carcinoma; shared features include relatively frequent alterations that increase PI3K/AKT pathway signaling (mutations in *PTEN*, *PIK3CA*, *ARID1A*, and *KRAS*) and mutations in mismatch DNA repair genes. Also similar to endometrioid carcinomas of the endometrium, *TP53* mutations are common in poorly differentiated tumors.

MORPHOLOGY

Endometrioid tumors are distinguished from serous and mucinous tumors by the presence of tubular glands resembling benign or malignant endometrium. Endometrioid carcinomas typically present with solid and cystic areas of growth and as in the endometrium can show areas of squamous and/or mucinous differentiation. Forty percent involve both ovaries, and such bilaterality usually implies extension of the neoplasm beyond the genital tract. These are low-grade tumors that reveal glandular patterns bearing a strong resemblance to those of endometrial origin. The 5-year survival rate for patients with stage I tumors is approximately 75%.

Clear Cell Carcinoma

Benign and borderline clear cell tumors are exceedingly rare, and clear cell carcinomas are uncommon. They are composed of large epithelial cells with abundant clear cytoplasm, an appearance that resembles hypersecretory gestational endometrium. Because these tumors sometimes occur in association with endometriosis or endometrioid carcinoma of the ovary and resemble clear cell carcinoma of the endometrium, they are now thought to be variants of endometrioid adenocarcinoma. In line with this idea, the most common genetic aberrations (*PIK3CA*, *ARID1A*, *KRAS*, *PTEN*, and *TP53*) are shared with endometrioid carcinoma, albeit at somewhat different frequencies. Clear cell tumors of the ovary can be predominantly solid or cystic. In the solid neoplasms, the clear

cells are arranged in sheets or tubules, while in the cystic variety, the neoplastic cells line the spaces. Clear cell carcinoma confined to the ovaries has a 90% 5-year survival, but in advanced stage disease it appears that clear cell morphology portends a poor outcome. Clear cell carcinoma is treated like other types of ovarian carcinoma.

Transitional Cell Tumors (Brenner Tumors)

Transitional cell tumors contain neoplastic epithelial cells resembling urothelium and are usually benign. They comprise roughly 10% of ovarian epithelial tumors and are also referred to as *Brenner tumors*. Uncommon transitional cell carcinomas also occur in the ovary.

MORPHOLOGY

Brenner tumors may be solid or cystic, are usually unilateral (approximately 90%), and vary in size from small lesions less than 1 cm in diameter to massive tumors up to 20 to 30 cm in diameter (Fig. 22.35A). The fibrous stroma, resembling that of the normal ovary, is marked by sharply demarcated nests of epithelial cells resembling the epithelium of the urinary tract, often with mucinous glands in their center (Fig. 22.35B). Infrequently, the stroma is composed of plump fibroblasts resembling theca cells; such neoplasms may have hormonal activity. Most Brenner tumors are benign, but borderline and malignant counterparts have been reported. Tumors with benign Brenner nests admixed with malignant tumor cells are referred to as malignant Brenner tumors, while tumors without a definitive benign Brenner component are considered transitional cell carcinomas of the ovary.

Brenner tumors are often detected incidentally, and even when large, behave in a benign fashion. Malignant Brenner tumors generally present in stage I and for prognostic purposes are equivalent to low-grade carcinomas. The uncommon transitional cell carcinomas are equivalent to high-grade ovarian carcinomas; these often present at advanced stage and are treated like high-grade serous carcinomas.

Clinical Course, Detection, and Prevention of Ovarian Epithelial Tumors

All ovarian carcinomas produce similar clinical manifestations, most commonly lower abdominal pain, and abdominal enlargement. Gastrointestinal complaints, urinary frequency, dysuria, pelvic pressure, and many other symptoms may appear. Benign lesions are easily resected and cured. The malignant tumors tend to cause progressive weakness,

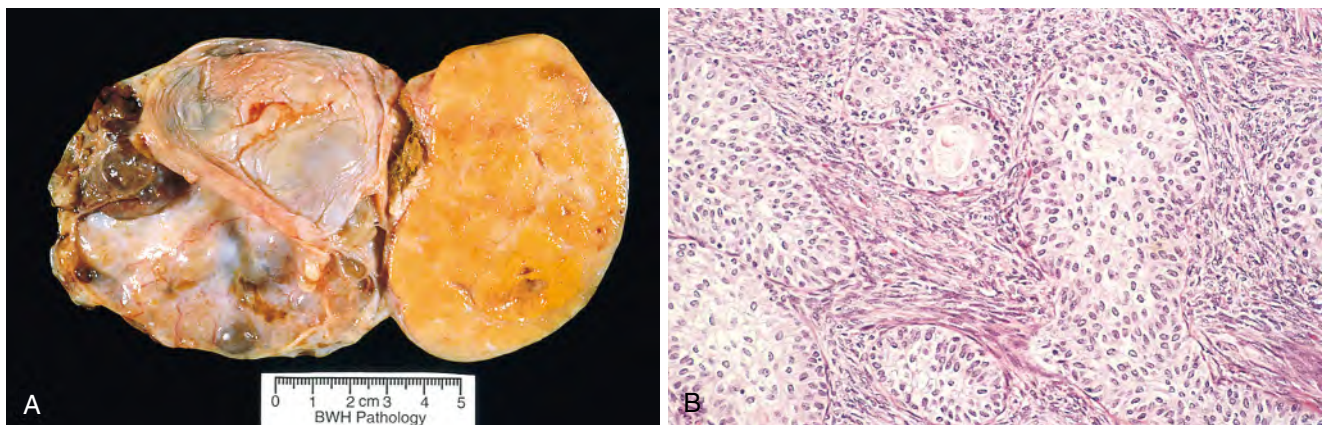


Fig. 22.35 Brenner tumor (A) Brenner tumor (right) associated with a benign cystic teratoma (left). (B) Histologic detail of characteristic epithelial nests within the ovarian stroma. (Courtesy Dr. M. Nucci, Brigham and Women's Hospital, Boston.)

weight loss, and cachexia. Once carcinomas extend through the capsule of the tumor and seed the peritoneal cavity, they often cause ascites, which may be massive. Characteristically, the ascitic fluid is filled with exfoliated tumor cells. The peritoneal pattern of spread is distinctive. Serosal surfaces are diffusely seeded with 0.1- to 0.5-cm nodules of tumor that only rarely invade deeply into the underlying parenchyma. The regional nodes are often involved, and metastases may be found in the liver, lungs, gastrointestinal tract, and elsewhere.

Most patients with ovarian carcinoma present with advanced disease. This is the primary reason for the relatively poor 5- and 10-year survival rates of patients with these tumors, as compared with rates for patients with cervical or endometrial carcinoma. For these reasons, development of new assays that permit early diagnosis is a top priority. Biochemical tests for tumor antigens or tumor products in the plasma of these patients are being sought vigorously, but none proposed to date has sufficient sensitivity and specificity to be useful. The serum marker CA-125 is used in patients with known disease to monitor disease recurrence/progression.

Prevention of ovarian cancer also remains an elusive goal. Screening to identify females at risk (positive for *BRCA* mutations or with strong family histories) and treatment with risk-reducing salpingectomy or salpingo-oophorectomy are currently available. The long-term impact of these approaches in females at high risk remains to be determined.

KEY CONCEPTS

Epithelial Ovarian Tumors

- Epithelial ovarian tumors are classified as benign, borderline, or malignant.
- About 80% of all ovarian epithelial tumors are benign and occur in young females. The malignant tumors occur most commonly in older females and account for approximately 3% of all cancers in the United States.
- Most of the malignant epithelial tumors are high-grade serous carcinomas, which have a poor prognosis in large part because they are usually detected after they have spread beyond the ovary and/or the fallopian tube.
- There are three major histologic types of epithelial ovarian tumors: serous, mucinous, and endometrioid, all of which have a benign, borderline, and malignant category.
- Benign tumors are composed of well-differentiated epithelial cells with minimal proliferation. Borderline tumors show increased cell proliferation but lack stromal invasion. Malignant tumors show increased epithelial atypia and are defined by the presence of stromal invasion.
- The origin of ovarian tumors is still under investigation, but it appears that *BRCA1*- and *BRCA2*-related tumors as well as most sporadic ovarian serous tumors likely arise from fallopian tube epithelium rather than ovarian epithelium.

Germ Cell Tumors

Germ cell tumors constitute 15% to 20% of all ovarian tumors and include multiple subtypes. Most are *benign cystic teratomas*, but others, found principally in children and young adults, may show malignant behavior, and pose problems in histologic diagnosis and in therapy. They bear a high degree of morphologic and molecular similarity to germ cell tumors in the male testis (Chapter 21).

Teratoma

Teratomas are divided into three categories: (1) mature (benign), (2) immature (malignant), and (3) monodermal or highly specialized.

Mature (Benign) Teratomas. Most benign teratomas are cystic and are often referred to as *dermoid cysts* because they are almost always lined by skinlike structures. Cystic teratomas are usually found in young adult females. They may be discovered incidentally, but are

occasionally associated with clinically important paraneoplastic syndromes, such as *inflammatory limbic encephalitis*, which may remit following removal of the tumor.

MORPHOLOGY

Benign teratoma is bilateral in 10% to 15% of cases. Characteristically it consists of a unilocular cyst containing hair and sebaceous material (Fig. 22.36). Sectioning reveals a cyst wall lined by opaque, gray-white, wrinkled epidermis, frequently with protruding hair shafts. Within the wall, it is common to find grossly evident tooth structures and areas of calcification.

Microscopically, the cyst wall is composed of stratified squamous epithelium with underlying sebaceous glands, hair shafts, and other skin adnexal structures (Fig. 22.37). In most cases, tissues from other germ layers can be identified, such as cartilage, bone, thyroid, and neural tissue. Dermoid cysts are sometimes incorporated within the wall of a mucinous cystadenoma. **About 1% of the dermoid cysts undergo malignant transformation**, most commonly to squamous cell carcinoma, but also to other cancers (e.g., thyroid carcinoma, melanoma).

In rare instances, a benign teratoma is solid and composed entirely of benign-looking heterogeneous collections of tissues and organized structures derived from all three germ layers. These tumors presumably have the same histogenetic origin as dermoid cysts but lack preponderant differentiation into ectodermal derivatives. Such neoplasms may be difficult to distinguish from malignant immature teratomas on gross inspection.

The origin of teratomas has been a matter of fascination for centuries. Some ancient common beliefs blamed witches, nightmares, or adultery with the devil. The karyotype of almost all benign ovarian teratomas is 46,XX. Genetic analyses indicate that the majority of teratomas arise from an ovum after the first meiotic division, while the remainder arise before the first division.

Immature (Malignant) Teratomas. These are rare tumors that differ from benign teratomas in that the component tissues resemble embryonal and immature fetal tissue. The tumor is found chiefly in prepubertal adolescents and young females, the mean age being 18 years.

MORPHOLOGY

Immature (malignant) teratomas are bulky, have a smooth external surface, and tend to be solid on sectioning. Hair, sebaceous material, cartilage, bone, and calcification may be present, along with areas of necrosis and hemorrhage. On microscopic examination, there are varying amounts of immature neuroepithelium, cartilage, bone, muscle, and other elements. An important risk for subsequent extraovarian spread is the histologic grade (I to III) of the tumor. Grading is based on the proportion of the tumor that is comprised of immature neuroepithelium (Fig. 22.38).

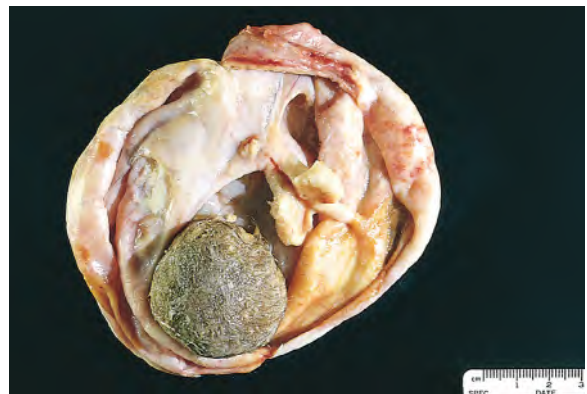


Fig. 22.36 Opened mature cystic teratoma (dermoid cyst) of the ovary. Hair (bottom) and a mixture of tissues are evident.

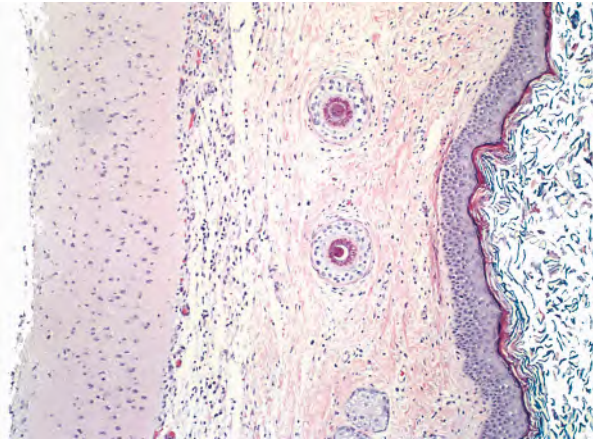


Fig. 22.37 Benign cystic teratoma. Low-power view of skin (right edge), beneath which there is brain tissue (left edge).

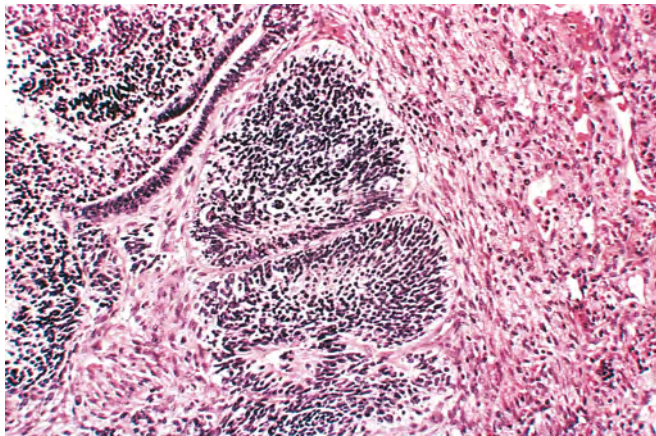


Fig. 22.38 Immature teratoma of the ovary illustrating primitive neuroepithelium.

Immature teratomas grow rapidly, frequently penetrate the capsule, and spread either locally or distantly. Stage I tumors, particularly those with low-grade (grade 1) histology, have an excellent prognosis. Higher-grade tumors (grade 2 and 3) confined to the ovary are generally treated with adjuvant chemotherapy. Most recurrences develop in the first 2 years, and absence of disease beyond this period carries an excellent chance of cure.

Monodermal or Specialized Teratomas. Specialized teratomas are a rare but remarkable group of tumors, the most common of which are struma ovarii and carcinoid. They are always unilateral, although a contralateral teratoma may be present. Struma ovarii is composed entirely of mature thyroid tissue, which may be functional and cause hyperthyroidism. Carcinoid tumors, which presumably arise from intestinal tissue found in teratomas, may also be functional; particularly if large (>7 cm), they can produce sufficient 5-hydroxytryptamine to cause the carcinoid syndrome even in the absence of hepatic metastases because ovarian veins connect directly to the systemic circulation. Primary ovarian carcinoid must be distinguished from metastatic intestinal carcinoid, which virtually always involves the ovaries bilaterally. Even rarer is strumal carcinoid, a combination of struma ovarii and carcinoid in the same ovary. Only about 2% of carcinoids in teratomas metastasize.

Dysgerminoma

Dysgerminoma is the ovarian counterpart of testicular seminoma. Dysgerminoma accounts for about 2% of ovarian cancers and roughly 50% of malignant ovarian germ cell tumors. They may occur in childhood, but 75% occur in the second and third decades of life. Some occur in patients with gonadal dysgenesis, including pseudohermaphroditism. Most of these tumors have no endocrine activity. A few produce elevated levels of chorionic gonadotropin, a finding that correlates with the presence of syncytiotrophoblastic giant cells. Like seminoma, dysgerminoma expresses stem cell markers such as OCT3, OCT4, and NANOG, transcription factors implicated in maintenance of pluripotency. They are also strongly associated with the presence of isochromosome 12p and expression of the receptor tyrosine kinase KIT, with approximately 30% to 50% have activating mutations in the *KIT* gene and 30% have amplification of *KIT*. These proteins are useful diagnostic markers, and mutated, constitutively active KIT also represents a potential therapeutic target.

MORPHOLOGY

Most dysgerminomas (80% to 90%) are unilateral tumors ranging in size from barely visible nodules to masses that virtually fill the abdomen. On cut surface they have a solid yellow-white to gray-pink appearance and are often soft and fleshy. Like seminoma, it is composed of large vesicular cells with clear cytoplasm, well-defined cell boundaries, and centrally placed regular nuclei. The tumor cells grow in sheets or cords separated by scant fibrous stroma (Fig. 22.39), which is infiltrated by lymphocytes and may contain noncaseating granulomas. On occasion, small nodules of dysgerminoma are encountered in the wall of an otherwise benign cystic teratoma; conversely, a predominantly dysgerminomatous tumor may contain a small cystic teratoma.

All dysgerminomas are malignant, but the degree of histologic atypia is variable, and only about one-third are aggressive. A unilateral tumor that has not broken through the capsule or spread outside the ovary has an excellent prognosis (>90% 10-year progression free survival) after simple salpingo-oophorectomy. These neoplasms are responsive to chemotherapy, and even those that have extended beyond the ovary can often be cured.

Yolk Sac Tumors

Though rare, yolk sac tumor (also known as *endodermal sinus tumor*) ranks as the second most common malignant ovarian tumor of germ cell

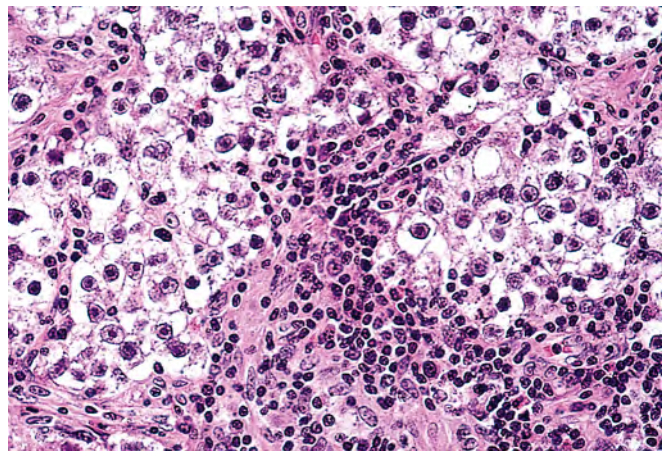


Fig. 22.39 Dysgerminoma showing polyhedral tumor cells with round nuclei and adjacent inflammation.

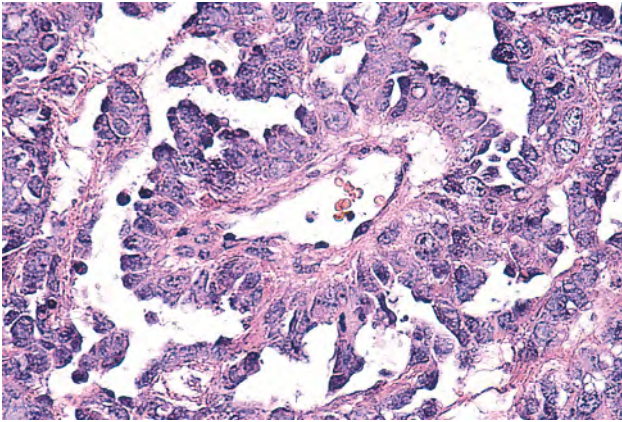


Fig. 22.40 A Schiller-Duval body in a yolk sac carcinoma.

origin. It is thought to be derived from malignant germ cells that are differentiating along the extraembryonic yolk sac lineage. Similar to the normal yolk sac, the tumor cells elaborate α -fetoprotein. RUNX3 gene hypermethylation and GATA-4 overexpression may be involved in the pathogenesis of yolk sac tumors. Its characteristic histologic feature is a glomerulus-like structure composed of a central blood vessel enveloped by tumor cells within a space that is also lined by tumor cells (*Schiller-Duval body*) (Fig. 22.40). Conspicuous intracellular and extracellular hyaline droplets are typically present, most of which stain for α -fetoprotein by immunoperoxidase techniques.

Most patients are children or young females presenting with abdominal pain and a rapidly growing pelvic mass that involves a single ovary. With combination chemotherapy, the 5-year survival rates are greater than 95% for stage I to II, 70% for stage III, and 50% for stage IV disease.

Choriocarcinoma

More commonly of placental origin, ovarian choriocarcinoma, like yolk sac tumor, is an example of a malignant germ cell that exhibits extraembryonic differentiation. Most ovarian choriocarcinomas exist in combination with other germ cell tumors, and pure ovarian choriocarcinoma is extremely rare. They are histologically identical to the more common placental lesions (described later). The ovarian choriocarcinomas are aggressive and have usually metastasized hematogenously to the lungs, liver, bone, and other sites by the time of diagnosis. Like all choriocarcinomas, they elaborate high levels of *chorionic gonadotropins*, which may be helpful in establishing the diagnosis or detecting recurrences. In contrast to choriocarcinoma arising in placental tissue, those arising in the ovary are generally unresponsive to chemotherapy and are often fatal.

Other Germ Cell Tumors

These include (1) *embryonal carcinoma*, a highly malignant tumor of primitive embryonal elements that is histologically similar to embryonal carcinoma arising in the testes (Chapter 21); (2) *polyembryoma*, a malignant tumor containing so-called embryoid bodies; and (3) *mixed germ cell tumors* containing various combinations of dysgerminoma, teratoma, yolk sac tumor, and choriocarcinoma.

- The remainder occur in young females and children; in these age groups, malignant tumors dominate.
- Immature teratomas are distinguished from mature teratomas by the presence of embryonal or fetal elements, most often consisting of primitive neuroepithelium.
- Germ cell tumors show various lines of differentiation toward oogonia (dysgerminoma), extraembryonic yolk sac (yolk sac tumors), placenta (choriocarcinoma), or multiple germ layers (teratoma).

Sex Cord–Stromal Tumors

These ovarian neoplasms arise from the ovarian stroma, which is derived from the sex cords of the embryonic gonad. The undifferentiated gonadal mesenchyme eventually produces specific types of cells in female (granulosa and theca) and male (Sertoli and Leydig) gonads, and tumors resembling each of these cell types can be identified in the ovary. Moreover, because some of these cells normally secrete estrogens (granulosa and theca cells) or androgens (Leydig cells), their corresponding tumors may be either feminizing (granulosa/theca cell tumors) or masculinizing (Leydig cell tumors).

Granulosa Cell Tumors

Granulosa cell tumors are composed of cells that resemble granulosa cells of a developing ovarian follicle. They are broadly divided into adult and juvenile granulosa cell tumors, based largely on the age of the patient. Collectively, these neoplasms account for about 5% of all ovarian tumors, and 95% of granulosa cell tumors are of the adult type. Although they may be discovered at any age, approximately two-thirds occur in postmenopausal females.

MORPHOLOGY

Granulosa cell tumors are usually unilateral and vary from microscopic foci to large, solid, and cystic encapsulated masses. Tumors that are hormonally active have a yellow coloration to their cut surfaces, due to intracellular lipids.

The granulosa cell component of these tumors has many histologic patterns. The small, cuboidal to polygonal cells may grow in anastomosing cords, sheets, or strands (Fig. 22.41A). In occasional cases, small, distinctive, glandlike structures filled with an acidophilic material recall immature follicles (**Call-Exner bodies**). When these structures are evident, the diagnosis is straightforward. Occasionally, there is a predominant thecoma component that consists of clusters or sheets of cuboidal to polygonal cells (see later). In some tumors, the granulosa or theca cells may appear plumper and have ample cytoplasm characteristic of luteinization (i.e., luteinized granulosa–theca cell tumors). Like normal granulosa cells, tumors express high levels of **inhibin** (Fig. 22.41B), a feedback inhibitor of follicle stimulating hormone release from the anterior pituitary gland.

Granulosa cell tumors are of clinical importance for two reasons: (1) they sometimes elaborate large amounts of estrogen, and (2) they may behave like low-grade malignancies. Functionally active tumors in prepubertal girls (juvenile granulosa cell tumors) may cause precocious sexual development. In adult females, they may be associated with proliferative breast disease, endometrial hyperplasia, and endometrial carcinoma, which eventually develops in about 10% to 15% of females with steroid-producing tumors. Occasionally, granulosa cell tumors produce androgens, masculinizing the patient.

The most common driver mutations in granulosa cell tumors involve the *FOXL2* gene, which is mutated in 97% of adult granulosa cell tumors. *FOXL2* encodes a transcription factor that is important in granulosa cell development, which presumably explains its strong

KEY CONCEPTS

Germ Cell Tumors

- Germ cell tumors constitute 15% to 20% of ovarian tumors.
- The majority are mature cystic teratomas (dermoid cysts) in females of reproductive age.

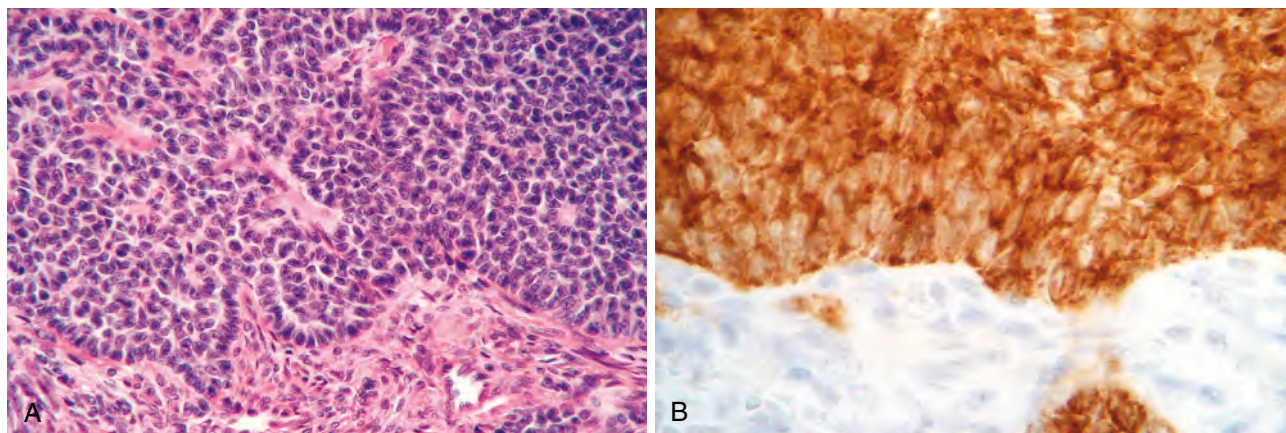


Fig. 22.41 Granulosa cell tumor. (A) Tumor cells are arranged in sheets punctuated by small follicle-like structures (Call-Exner bodies). (B) Strong immunohistochemical positivity with an antibody to inhibin characterizes these tumors.

association with this tumor type. Juvenile granulosa cell tumors show mutations in the oncogenes *AKT1* (60%) and *GNAS* (30%), and in some tumors, they occur in the setting of genetic syndromes (e.g., *DICER1* syndrome, Ollier disease, and Maffucci syndrome).

All granulosa cell tumors are potentially malignant. It is difficult to predict their biologic behavior from histology. The likelihood of malignant behavior (recurrence, extension) ranges from 5% to 25%. In general, malignant tumors pursue an indolent course in which local recurrences may be amenable to surgical therapy. Recurrences within the pelvis and abdomen may appear 10 to 20 years after removal of the original tumor. The 10-year survival rate is approximately 90% to 95% for stage 1 tumors. Tumors composed predominantly of theca cells are almost never malignant. Monitoring serum levels of inhibin is useful in gauging the response to treatment in affected patients.

Fibromas and Thecomas

Tumors arising in the ovarian stroma that are composed of either fibroblasts (fibromas) or plump spindle cells with lipid droplets (thecomas) are relatively common, accounting for about 4% of all ovarian tumors. Fibromas and thecomas may be hormonally active.

Fibromas of the ovary are unilateral in about 90% of cases and are usually solid, spherical or slightly lobulated, encapsulated, hard, gray-white masses covered by glistening, intact ovarian serosa (Fig. 22.42A). On histologic examination, they are composed of well-differentiated

fibroblasts and scant interspersed collagenous stroma (Fig. 22.42B). Focal areas of thecal differentiation may be identified.

Most of these tumors come to attention as a pelvic mass, sometimes accompanied by pain and two decidedly curious associations. The first is ascites, found in about 40% of cases in which the tumors measure more than 6 cm in diameter. Uncommonly there is also a pleural effusion, usually only on the right side. This combination of findings (ovarian tumor, pleural effusion, and ascites) is designated *Meigs syndrome*. Its genesis is unknown. The second association is with the basal cell nevus syndrome, described in Chapter 25. Most fibromas and thecomas are benign.

Sertoli-Leydig Cell Tumors

These tumors are often functional; they most commonly produce masculinization or defeminization, but a few have estrogenic effects.

The tumor cells recapitulate, to a certain extent, testicular Sertoli or Leydig cells at various stages of development. They occur in females of all ages, although the peak incidence is in the second and third decades. In over one-half of cases (intermediate and poorly differentiated tumors, see later), the tumor cells have mutations in *DICER1*, a gene that encodes an endonuclease that is essential for proper processing of micro-RNAs (Chapter 1). The presence of *DICER1* mutations suggests that genesis of these tumors involves abnormalities of gene expression related to dysregulation of micro-RNAs.

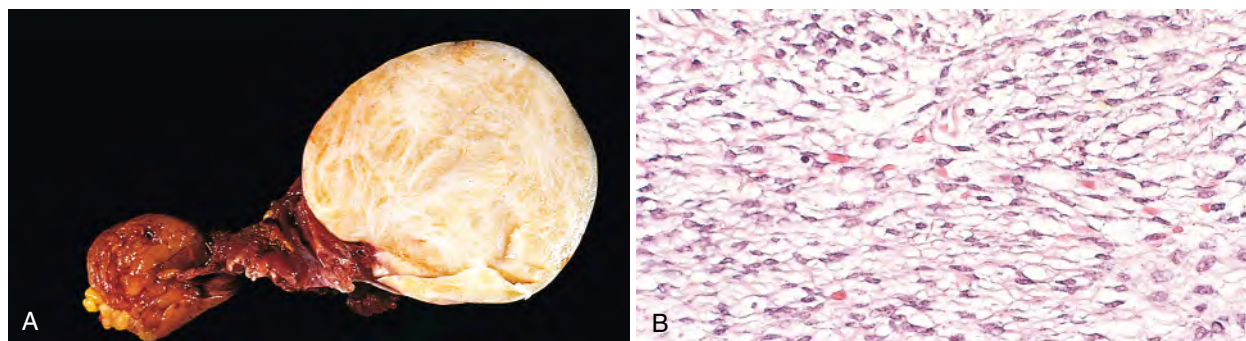


Fig. 22.42 Ovarian fibroma. (A) Large, bisected fibroma of the ovary apparent as a white, firm mass (right). The fallopian tube is attached. (B) Thecoma-fibroma composed of plump, differentiated stromal cells with thecal appearance.

MORPHOLOGY

Sertoli-Leydig tumors are unilateral and may resemble granulosa cell tumors grossly. The cut surface is usually solid and varies from gray to golden brown in appearance (Fig. 22.43A). Microscopically, a range of differentiation is seen. Well-differentiated tumors show tubules composed of Sertoli cells or Leydig cells interspersed with stroma (Fig. 22.43B). The intermediate forms show only immature tubules and large eosinophilic Leydig cells. The poorly differentiated tumors have a sarcomatous pattern with a disorderly disposition of epithelial cell cords. Leydig cells may be absent. Heterologous elements, such as mucinous glands, bone, and cartilage, may be present in some tumors.

The incidence of recurrence or metastasis by Sertoli-Leydig cell tumors is less than 5%. These neoplasms may block normal female sexual development in children and may cause defeminization of females, manifest by atrophy of the breasts, amenorrhea, sterility, and loss of hair. The syndrome may progress to striking virilization (hirsutism) associated with male hair distribution, hypertrophy of the clitoris, and voice changes.

Other Sex Cord–Stromal Tumors

There are several other uncommon but distinctive ovarian tumors of sex cord or stromal origin that often produce steroid hormones.

- *Leydig cell tumors* are rare, unilateral tumors located in the ovarian hilus comprised of large lipid-laden Leydig cells with distinct borders and characteristic cytoplasmic structures called *Reinke crystalloids*. Females with Leydig cell tumors usually present with evidence of masculinization (hirsutism, voice changes, and clitoral enlargement), but these changes are milder than those seen in association with Sertoli-Leydig cell tumors. The tumors produce predominantly testosterone. Treatment consists of surgical excision. True Leydig cell tumors are almost always benign.
- *Pregnancy luteoma* refers to a rare tumor that closely resembles the corpus luteum of pregnancy. These tumors may produce virilization in pregnant patients and their female infants.
- *Gonadoblastoma* is an uncommon tumor composed of germ cells and sex cord–stroma derivatives resembling immature Sertoli and granulosa cells. It occurs in individuals with abnormal sexual development and in gonads of indeterminate nature. Eighty percent of patients are phenotypic females, and 20% are phenotypic males with undescended testes and female internal secondary organs. A coexistent

dysgerminoma occurs in 50% of the cases. These tumors are classified as a noninvasive germ cell neoplasia, and prognosis is excellent if the tumor is completely excised.

KEY CONCEPTS

Sex Cord–Stromal Tumors

- Granulosa cells tumors are the most common malignant tumor in this category. They are indolent tumors but may recur 10 to 20 years after resection of the primary tumor. They are often hormonally active; such tumors are associated with hyperestrinism, leading to an elevated risk of endometrial carcinoma.
- Fibromas are relatively common benign tumors composed of fibroblasts. They are predominantly unilateral and are generally hormonally inactive.
- Pure thecomas are rare but may be hormonally active.
- Sertoli-Leydig cell tumors commonly present with masculinization; less than 5% recur or metastasize.

Metastatic Tumors

The most common metastatic tumors of the ovary are derived from tumors of müllerian origin: the uterus, fallopian tube, contralateral ovary, or pelvic peritoneum. The most common extra-müllerian tumors metastatic to the ovary are carcinomas of the breast and gastrointestinal tract, including colon, stomach, biliary tract, and pancreas. Also included in this group are rare cases of pseudomyxoma peritonei, derived from appendiceal tumors. A classic metastatic gastrointestinal carcinoma involving the ovaries is termed *Krukenberg tumor*, characterized by bilateral metastases composed of mucin-producing cancer cells with a “signet-ring” appearance, most often of gastric origin.

GESTATIONAL AND PLACENTAL DISORDERS

Diseases of pregnancy and pathologic conditions of the placenta are important causes of fetal intrauterine or perinatal death, congenital malformations, intrauterine growth restriction, mortality, and morbidity for both the mother and child. The value of placental pathology lies in its ability to explain an adverse outcome, predict recurrence in future pregnancies, impact immediate neonatal and maternal care, and provide insights into disorders of the perinatal period. Only disorders in which recognizable morphologic features contribute to an understanding of their pathogenesis are discussed here. These include

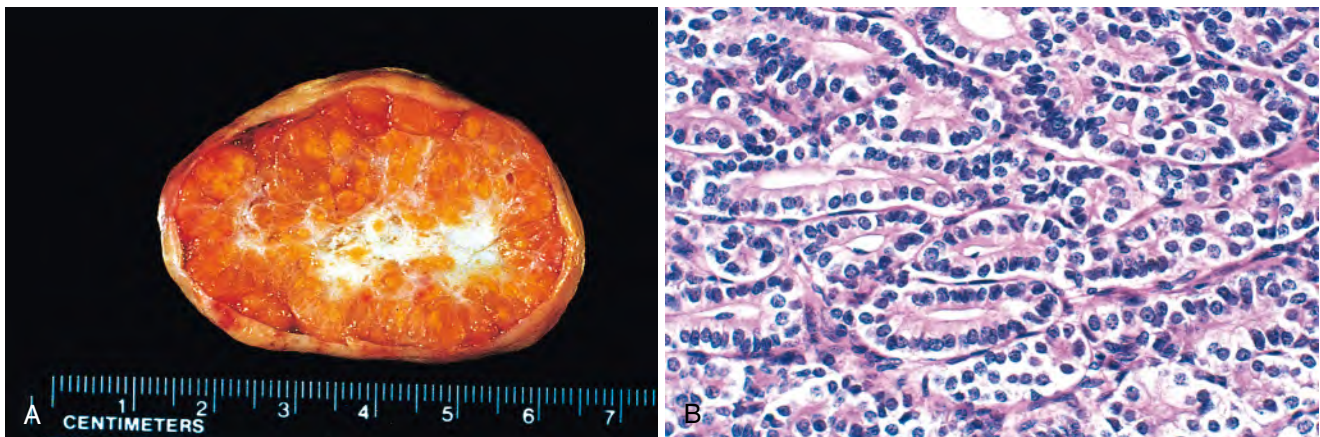


Fig. 22.43 Sertoli cell tumor. (A) Gross photograph illustrating characteristic golden-yellow appearance of the tumor. (B) Photomicrograph showing well-differentiated Sertoli cell tubules. (Courtesy Dr. William Welch, Brigham and Women's Hospital, Boston.)

pathologic changes associated with each trimester of pregnancy and trophoblastic neoplasia.

Understanding placental disorders requires a working knowledge of placental anatomy and physiology. The principal cells of the placenta are *trophoblast*, specialized epithelia that include cytotrophoblast and syncytiotrophoblast cells with roles in hormone secretion, gas and nutrient transport, and barrier function; and *extravillous trophoblasts*, invasive cells that anchor the placenta to the uterus and remodel uterine vessels to create the low pressure-high capacitance vessels required for placental perfusion. The parenchyma of the placenta is composed of a branching network of structures covered by cytotrophoblasts and syncytiotrophoblasts termed *chorionic villi* (Fig. 22.44) that provide a large surface area for gas and nutrient exchange between the fetal and maternal circulations (Fig. 22.45). Maternal blood first enters the intervillous space through the endometrial arteries (the *spiral arteries*) starting at 10 to 12 weeks gestational age; by later stages of pregnancy, approximately one-third of the maternal cardiac output and over half of the fetal cardiac output perfuses the placenta. Deoxygenated fetal blood enters the placenta through two umbilical arteries that branch radially over the chorionic plate to form chorionic arteries, which branch further as they dive down from the chorionic plate and enter the villi. In the mature placenta, the vessels form an extensive fetal capillary system separated from maternal blood by only capillary endothelial cells and a thin layer of syncytiotrophoblast and cytotrophoblast. Under normal circumstances, there is little to no mixing of fetal and maternal blood, though sufficient free fetal DNA (likely from syncytiotrophoblast shedding) reaches the maternal circulation to permit prenatal genetic testing (Chapter 5). Blood oxygenated in the placenta returns to the fetus through the single umbilical vein.

DISORDERS OF FIRST AND EARLY SECOND TRIMESTER GESTATIONS

Spontaneous Abortion

Spontaneous abortion, or “miscarriage,” is defined as pregnancy loss before 20 weeks of gestation. Most of these occur before 12 weeks. Ten to fifteen percent of clinically recognized pregnancies terminate in spontaneous abortion. However, using sensitive chorionic gonadotropin assays, it has been determined that an additional 20% of early pregnancies in otherwise healthy females terminate spontaneously. In most individual instances, the mechanisms leading to early loss of pregnancy are unknown, but multiple fetal and maternal causes of spontaneous abortion have been identified. Among the most important are:

- *Fetal chromosomal anomalies*, such as aneuploidy, polyploidy, and translocations, are present in approximately 50% of early abortuses. More subtle genetic defects, for which routine genetic testing may be unavailable, account for an additional fraction of abortions.
- *Maternal endocrine factors*, including luteal-phase defect, poorly controlled diabetes, and other endocrine disorders.
- *Physical anomalies of the uterus*, such as submucosal leiomyomas, uterine polyps, or uterine malformations, may prevent or disrupt implantation.
- *Systemic disorders affecting the maternal vasculature*, such as antiphospholipid antibody syndrome, coagulopathies, and hypertension.
- *Infections* with protozoa (*Toxoplasma*), bacteria (*Mycoplasma*, *Listeria*), or certain viruses. These may reach the placenta and fetus by either ascending or hematogenous routes. Ascending infections from the cervicovaginal vault are a particularly common cause of second trimester losses.



Fig. 22.44 Normal placenta. (A) Gross specimen of a normal term placenta seen from the fetal side. (B) First-trimester chorionic villi composed of delicate mesh of central stroma surrounded by two discrete layers of trophoblastic epithelium—the outer layer consisting of syncytiotrophoblast (*double arrows*) and the inner layer consisting of cytotrophoblast (*arrow*). (C) Third trimester chorionic villi composed of stroma with a dense network of dilated capillaries surrounded by markedly thinned syncytiotrophoblast and endothelium, the vasculosyncytial membrane (*arrows*). The cytotrophoblast layer is significantly diminished.

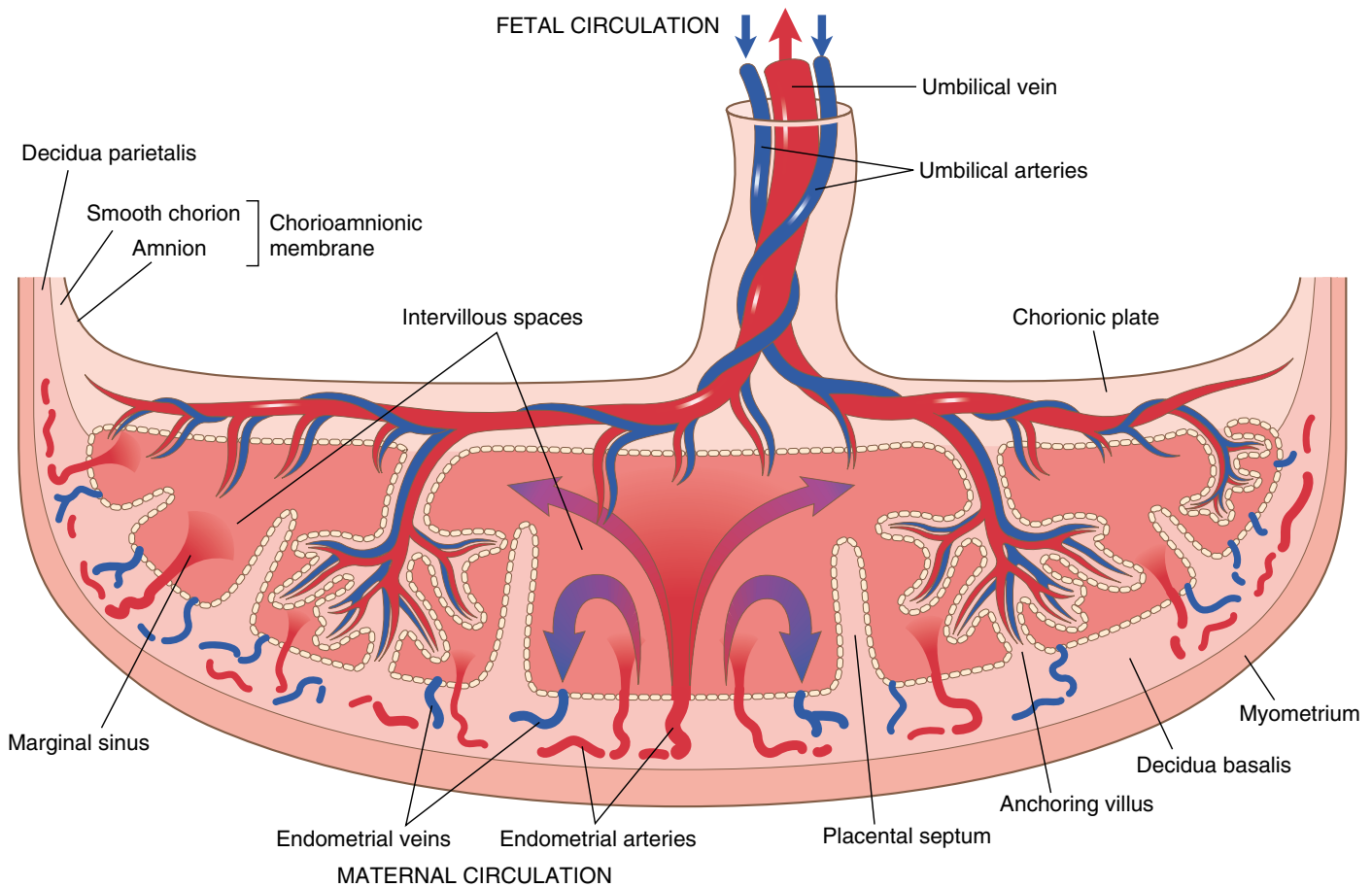


Fig. 22.45 Placental anatomy. Within the outer boundary of myometrium is a layer of decidua, from which the maternal vessels originate and deliver blood to and from the intervillous spaces. Umbilical vessels branch and terminate in placental villi, where nutrient exchange takes place.

- **Fetal-maternal immune imbalance.** As the fetal-placental unit is a hemi-allograft, immune mediated rejection must be kept in check for pregnancy to be successful. Invasion of the maternal uterus by extravascular trophoblasts requires the maternal immune system to direct and tolerate the “foreign” fetal cells (trophoblast). Although mechanisms are poorly understood, there is evidence that disturbances of innate and adaptive immunity, including defective functions of regulatory T cells, may underlie recurrent spontaneous abortion in some females.

Ectopic Pregnancy

Ectopic pregnancy refers to implantation of the gestation in a site other than the normal intrauterine location; the most common site is the fallopian tube (approximately 90% of cases). Other sites include the ovary, the abdominal cavity, previous cesarean section scars, and the intrauterine portion of the fallopian tube (cornual pregnancy). Ectopic pregnancies account for 2% of confirmed pregnancies. The most important predisposing condition, present in 35% to 50% of patients, is intraluminal fallopian tube scarring (chronic salpingitis) from pelvic inflammatory disease. The risk of ectopic pregnancy is also increased with peritubal scarring and adhesions, which may be caused by appendicitis, endometriosis, and previous pelvic or abdominal surgery. In some cases, however, the fallopian tubes are apparently normal. Another risk

factor is use of an intrauterine contraceptive device, which causes a two-fold increase in ectopic pregnancy.

Ovarian pregnancy results from the fertilization and trapping of the ovum within the follicle just at the time of its rupture. Abdominal pregnancies occur when the fertilized ovum fails to enter or drops out of the fimbriated end of the tube. In each abnormal location, the fertilized ovum develops as usual, forming placental tissue, amniotic sac, and fetus. The extrauterine implantation site may also develop decidual changes.

MORPHOLOGY

Tubal pregnancy is the most common cause of **hematosalpinx (blood-filled fallopian tube)** and should always be suspected when a tubal hematoma is present. Initially the embryonal sac, surrounded by immature chorionic villi, implants within the lumen of the fallopian tube. Trophoblastic cells and chorionic villi then invade the wall of the fallopian tube, as in the uterus during normal pregnancy. With time, the growth of the gestational sac distends the fallopian tube, causing thinning of the wall that can lead to rupture (**Fig. 22.46**). **Rupture frequently results in massive intraperitoneal hemorrhage**, which can be fatal. Less commonly, the tubal pregnancy may undergo spontaneous regression and resorption, or may be extruded through the fimbriated end of the tube into the abdominal cavity (tubal abortion).

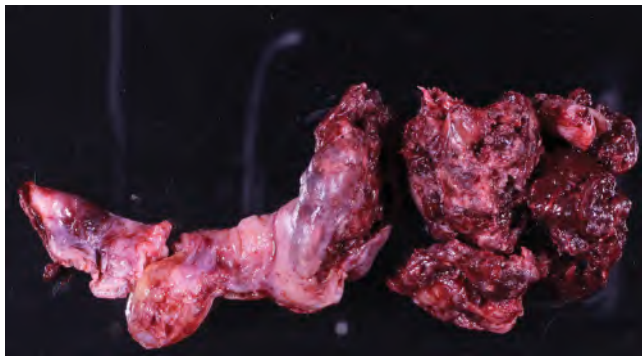


Fig. 22.46 Ectopic tubal pregnancy. The fallopian tube that has dilated and ruptured due to intratubal implantation.

Clinical Features

Rupture of a tubal pregnancy is a medical emergency. The clinical course of ectopic tubal pregnancy is characterized by the onset of moderate to severe abdominal pain and vaginal bleeding 6 to 8 weeks after the last menstrual period, correlating with distention and then rupture of the fallopian tube. In such cases, the patient may rapidly develop hemorrhagic shock and signs of an acute abdomen. Rapid diagnosis is critical and is based on determination of inappropriately rising chorionic gonadotropin titers, pelvic sonography, endometrial biopsy (which shows decidua without chorionic villi, trophoblast, or placental implantation site), and/or laparoscopy. Despite advances in early diagnosis and treatment, ectopic pregnancy still accounts for 4% to 10% of pregnancy-related deaths.

DISORDERS OF MID TO LATE SECOND TRIMESTER GESTATIONS

During the second trimester, fetal nutritional demands increase rapidly, and therefore disorders of the placenta take on increased importance. This is also the stage of fetal development during which congenital malformations take an increasing toll (Chapter 10). In severe cases, disorders that are more typical of third trimester gestations (discussed next), such as infections (e.g., amniotic fluid infection: acute chorioamnionitis, and viral infections as in CMV) and placental hypoplasia, may appear during the second trimester. When they do so, they are more likely to lead to fetal demise because the immature placenta has less reserve capacity for nutrient exchange. Another cause of second trimester loss are disorders causing diminished perfusion of the fetal side of the placental circulation. These are poorly understood but are marked morphologically by the presence of villous vascular thrombi and endothelial damage, or avascular villi.

DISORDERS OF THE THIRD TRIMESTER, PERIPARTUM, AND PUERPERIUM

In the third trimester, umbilical cord pathologies play a major role in poor pregnancy outcome. Interruption of blood flow through the umbilical cord from any cause (e.g., constricting knots (Fig. 22.47A), kinks, nuchal or body cords, prolapse or other causes of compression) causes significant morbidity and is often fatal. Acute umbilical cord obstruction associated with fetal demise leaves few morphologic clues, but chronic or intermittent interruption of blood flow leads to characteristic placental changes of fetal vascular malperfusion (such as avascular villi (Fig. 22.47B) or vascular thrombi). Other disorders of this period include complications related to twin pregnancies, maternal vascular malperfusion, abnormal placental implantation, and infections.

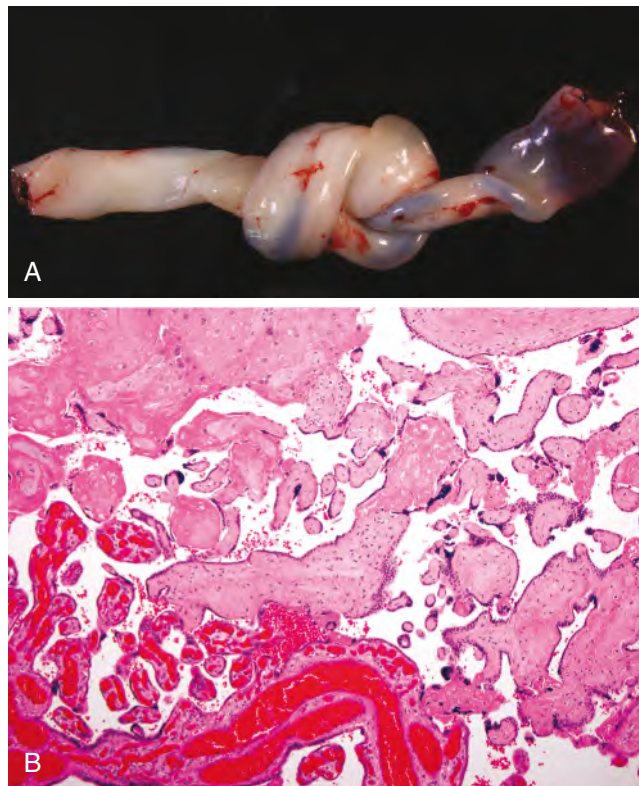


Fig. 22.47 Examples of pathologies associated with fetal vascular malperfusion. (A) Tight complex true knot of the umbilical cord. Occlusion of the umbilical vessels by the knot leads to downstream villous fetal vascular malperfusion. (B) Avascular villi caused by malperfusion. Compare with normally vascularized villi (bottom left) to the avascular villi (top right).

Twin Placentas

Twin pregnancies arise from fertilization of two ova (dizygotic) or from division of one fertilized ovum (monozygotic). There are three types of twin placentas (Fig. 22.48): diamniotic dichorionic (which may be fused or separate); diamniotic monochorionic; and monoamniotic monochorionic. Monochorionic placentas imply monozygotic (identical) twins, and the time at which splitting of the developing embryo occurs determines whether there is a dividing membrane and two amnions or no dividing membrane. Dichorionic placentation may occur with either monozygotic or dizygotic twins.

Multiple gestations carry many risks, and twins may be discordant for size, karyotype, malformations, and placental pathologies. One unique complication of monochorionic twin pregnancy is *twin-to-twin transfusion syndrome*. Monochorionic twin placentas have vascular anastomoses that connect the twin circulations. Twin-to-twin perfusion syndrome results when these shunts preferentially increase blood flow to one twin at the expense of the second, causing one twin (the donor) to be under perfused and the second twin (the recipient) to be fluid overloaded. If severe, it may result in the death of one or both fetuses.

Maternal Vascular Malperfusion

Maternal vascular malperfusion refers to inadequate blood supply to the intervillous space due to various abnormalities in the maternal circulation of the placenta. It may be precipitated by abnormal placental implantation or development, or maternal vascular disease; the effects range from mild intrauterine growth restriction to severe uteroplacental

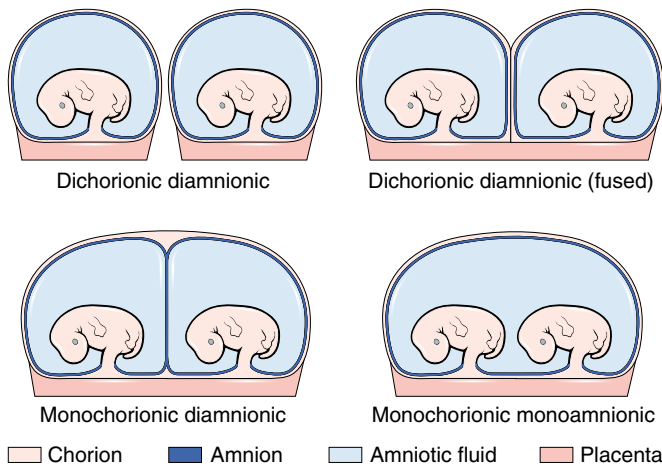


Fig. 22.48 Diagrammatic representation of the various types of twin placentation and their membrane relationships. (Modified from Gersell D, et al: Diseases of the placenta. In Kurman R, ed. *Blaustein's Pathology of the Female Genital Tract*. Springer-Verlag; 1994.)

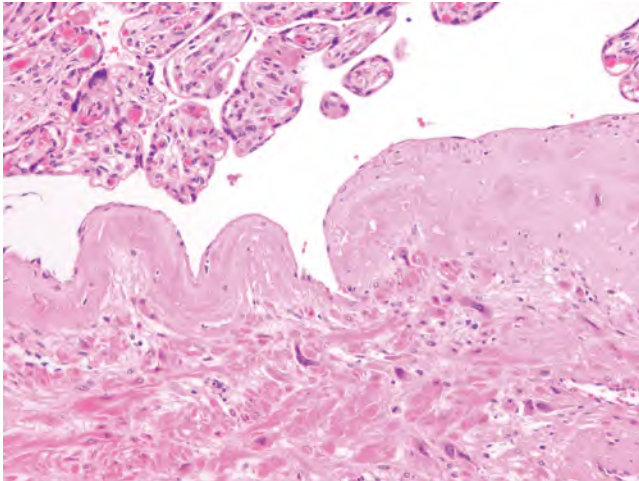


Fig. 22.49 Placental accreta. Note the absence of decidua and the presence of villi apposed to fibrin and myometrium.

ischemia and maternal preeclampsia (described later). Maternal vascular malperfusion is characterized by a constellation of findings, including any of the following: placental hypoplasia (placental weight less than the tenth percentile for gestational age); placental infarcts, maternal vascular pathology, typically in the form of abnormal remodeling of spiral arteries; ischemic changes of the chorionic villi; and placental abruption.

Abnormalities of Placental Implantation

Several types of abnormal placental implantations are associated with significant complications.

- **Placenta previa** is a condition in which the placenta implants in the lower uterine segment completely or partially covering the cervix, which can lead to serious third trimester bleeding. A complete placenta previa requires cesarean delivery to avert placental rupture and fatal maternal and fetal hemorrhage during vaginal delivery.
- **Placenta accreta spectrum disorder** is caused by complete absence of the decidua (at least focally), such that the placental villous tissue is in direct contact with the myometrium (Fig. 22.49) and hence invades, or penetrates completely through the myometrium, sometimes

even adhering to serosal surfaces or pelvic organs. It leads to a failure of placental separation at birth and is an important cause of severe, potentially life-threatening postpartum bleeding. Common predisposing factors are placenta previa (in up to 60% of cases) and previous cesarean delivery or other uterine surgery that breached the endometrial lining and caused scarring.

Placental Infections

Infections in the placenta develop by two pathways: (1) ascending infection through the birth canal and (2) hematogenous (transplacental) infection due to maternal sepsis, viremia, or parasitemia. Ascending infections are by far the most common and are virtually always bacterial; in many such instances, localized infection of the membranes produces premature rupture of membranes and preterm delivery. The amniotic fluid may be cloudy with purulent exudate, and histologically the chorion-amnion contains an infiltrate of maternal neutrophils (maternal inflammatory response) (Fig. 22.50A–B). The infection frequently elicits a fetal response consisting of a “vasculitis” of the umbilical vessels and/or fetal chorionic plate vessels (fetal inflammatory response). Of note, intense fetal inflammatory responses are associated with perinatal morbidity and an increased risk for the development of cerebral palsy. Uncommonly, bacterial infections can result from hematogenous spread to the placenta, leading to acute villitis (Fig. 22.50C).

Several microbes of the TORCH group (*toxoplasmosis* and others [syphilis, tuberculosis, listeriosis], rubella, cytomegalovirus (Fig. 22.50D), herpes simplex), can infect the placenta via the hematogenous route. They typically give rise to chronic inflammatory cell infiltrates in the chorionic villi (chronic villitis) and are associated with fetal growth restriction (Chapter 10).

Preeclampsia and Eclampsia

Preeclampsia is a systemic syndrome characterized by widespread maternal endothelial dysfunction that presents after 20 weeks of gestation with new onset hypertension and at least one of other findings, such as edema and proteinuria. Preeclampsia and eclampsia account for >50,000 maternal deaths yearly worldwide. Preeclampsia is most prevalent among Black and Hispanic patients, resulting in approximately 26% of maternal deaths among this population in the United States. It occurs in about 3% to 5% of pregnant females, usually in the last trimester and more commonly in females pregnant for the first time (*primiparas*). Some of these females become seriously ill, developing seizures; this particularly severe form of the disorder is termed **eclampsia**. Other complications stemming from systemic endothelial dysfunction include hypercoagulability, acute renal failure, and pulmonary edema. Approximately 10% of females with preeclampsia develop microangiopathic hemolytic anemia, elevated liver enzymes, and low platelets, referred to as the **HELLP syndrome** (Chapter 14). Preeclampsia should be distinguished from gestational hypertension that can develop in pregnancy without the other symptoms/signs of preeclampsia.

Pathogenesis

The placenta plays a central role in the pathogenesis of preeclampsia, as symptoms almost always disappear rapidly after its delivery. The critical abnormalities in preeclampsia are diffuse endothelial dysfunction, vasoconstriction (leading to hypertension), and increased vascular permeability (resulting in proteinuria and edema). These effects are most likely mediated by placenta-derived factor(s) released into the maternal circulation. Although the release of these factors and the clinical syndrome develop late in gestation, the pathogenesis of the disease appears to be closely tied to the earliest events of pregnancy and placentalation. The principal pathophysiologic aberrations are proposed to be:

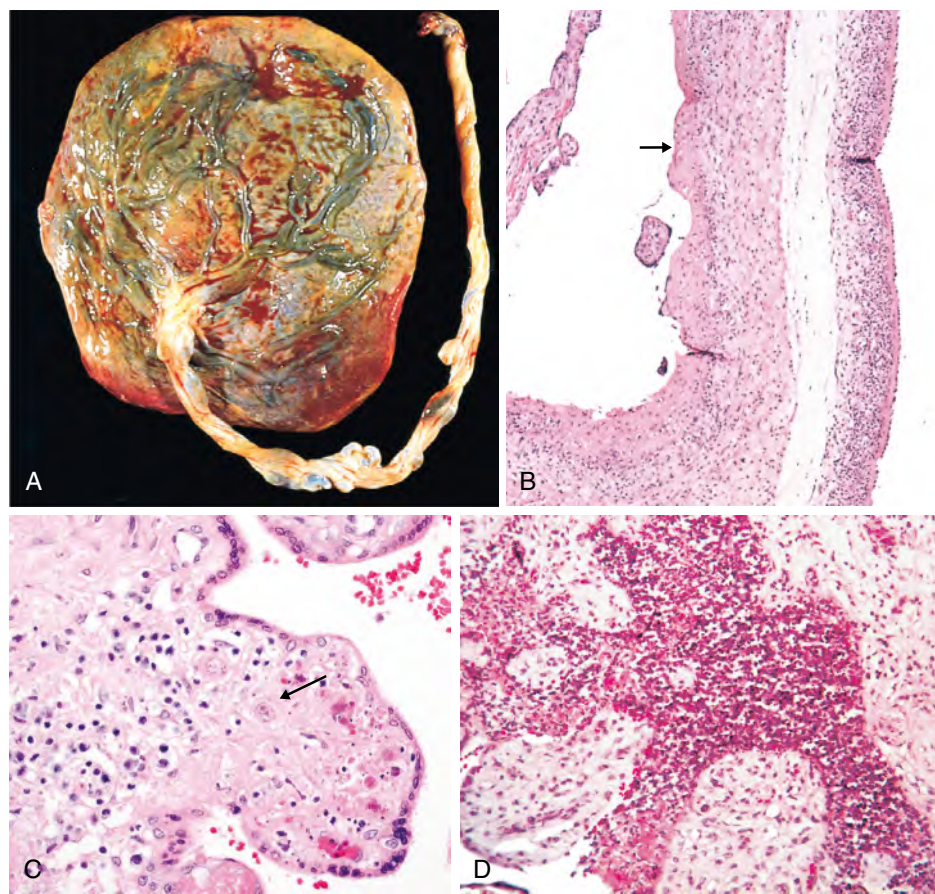


Fig. 22.50 Placental infection. (A) Gross photograph of a yellow/green cloudy discolored placenta from acute chorioamnionitis. (B) Histology of the placental membranes with maternal acute chorioamnionitis. A photomicrograph illustrates a dense bandlike inflammatory exudate on the amniotic surface (*arrow*). (C) A villus containing a cell with viral inclusions due to cytomegalovirus infection (*arrow*) in an area of chronic inflammation and necrosis. (D) Acute necrotizing intervillitis from *Listeria* infection. The space between the villi is filled with blood and acute inflammatory cells (neutrophils) and the villi appear necrotic.

- **Failure of the normal physiologic remodeling of the placental bed vessels.** The precipitating events in preeclampsia are abnormally shallow trophoblastic implantation and a failure of physiologic remodeling of the maternal vessels, which is required for adequate perfusion of the placenta. At the implantation site of a normal pregnancy, fetal extravillous trophoblast invade the maternal decidua and decidual vessels, destroy the vascular smooth muscle, and replace the maternal endothelial cells with fetal trophoblastic cells, forming hybrid fetomaternal blood vessels. This process converts the decidual spiral arteries from small-caliber resistance vessels to large-capacity low-resistance uteroplacental vessels lacking a smooth muscle coat (Fig. 22.51). In preeclampsia, this vascular remodeling fails to occur, leaving the placenta ill equipped to meet the increased circulatory demands of late gestation and setting the stage for the development of placental ischemia.
- **Endothelial dysfunction and imbalance of angiogenic and antiangiogenic factors.** It is postulated that in response to hypoxia, the ischemic placenta releases factors into the maternal circulation that cause an imbalance in circulating angiogenic and anti-angiogenic factors; this, in turn, leads to systemic maternal endothelial dysfunction and the clinical manifestations of the disease. In support of this, the blood levels of two placenta-derived antiangiogenic factors, soluble feline McDonough sarcoma (FMS)-like tyrosine

kinase (sFLT-1) and endoglin, which antagonize the effects of VEGF and TGF β , respectively, are several orders of magnitude higher in females with preeclampsia than in healthy controls. In preeclampsia, high levels of sFLT1 and soluble endoglin bring about a decrease in angiogenesis much earlier than in normal pregnancy, resulting in defective placental vascular development. Furthermore, TGF β induces endothelial production of nitric oxide, a potent vasodilator; thus, inhibition of TGF β by endoglin may directly contribute to systemic vasoconstriction, hypertension, and tissue hypoperfusion. Studies in animal models also implicate sFLT1 and soluble endoglin in the pathogenesis of endothelial dysfunction. When sFLT1 and endoglin are overexpressed together, pregnant rats develop nephrotic-range proteinuria, severe hypertension, as well as features of the HELLP syndrome, including elevated liver enzymes, decreased platelet count, and hemolysis, while their pups are growth restricted, all hallmarks of severe preeclampsia. Thus, it seems that sFLT1 and soluble endoglin are key mediators that link the placenta to the characteristic maternal endothelial dysfunction of preeclampsia.

- **Coagulation abnormalities.** Preeclampsia is associated with a hypercoagulable state that may lead to the formation of thrombi in arterioles and capillaries throughout the body, but particularly in the placenta, liver, kidneys, brain, and pituitary gland. Hypercoagulability is likely

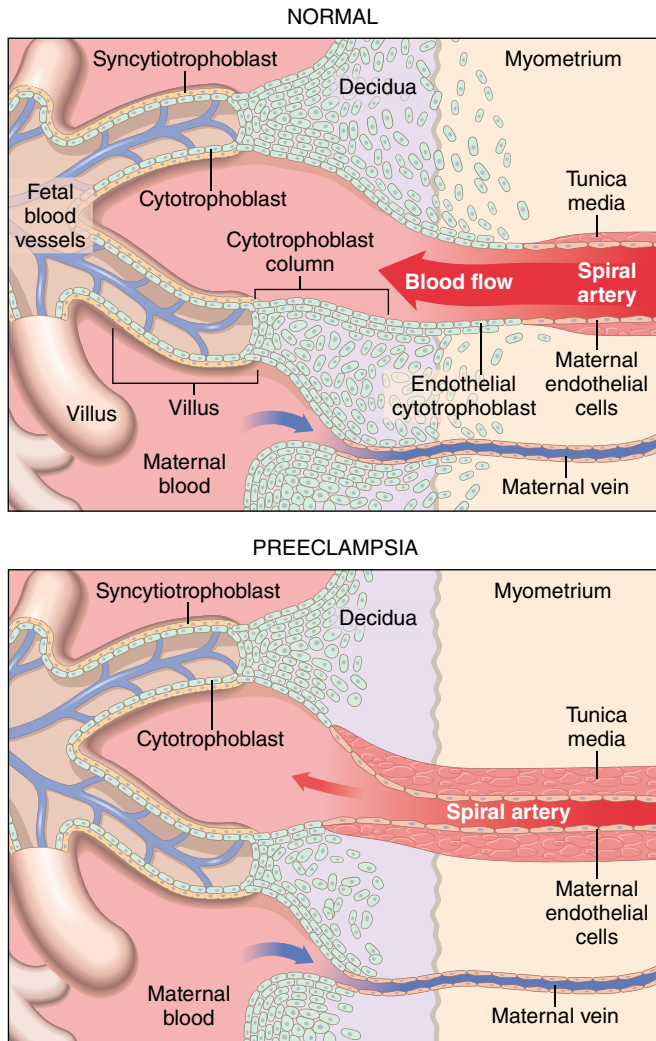


Fig. 22.51 The physiologic alterations in the uterine spiral arteries and the failure of their remodeling in preeclampsia. (Modified from Maynard S, Epstein FH, Karumanchi SA. Preeclampsia and angiogenic imbalance. *Annu Rev Med.* 2008;59:61-78.)

related to the reduced endothelial production of PGI_2 , a potent anti-thrombotic factor, and increased release of procoagulant factors. Production of PGI_2 is stimulated by VEGF, and females with preeclampsia have decreased endothelial production of PGI_2 , presumably due to antagonism of VEGF by sFlt1.

MORPHOLOGY

The placenta reveals several microscopic changes of maternal vascular malperfusion. These include (1) placental hypoplasia, (2) placental **infarcts**, which can be large and numerous, (3) exaggerated ischemic changes in the chorionic villi and trophoblast, consisting of increased **syncytial knots** and small shrunken villi with increased maternal vascular space (**distal villous hypoplasia**), (4) frequent **retroplacental hematomas** due to bleeding and instability of uteroplacental vessels, and (5) **abnormal decidual vessels**, which may show thrombi, lack of normal remodeling (described earlier), fibrinoid necrosis, or intimal lipid deposition (acute atherosclerosis) (Fig. 22.52).

When present, liver lesions take the form of irregular, focal, subcapsular, and intraparenchymal hemorrhages. On histologic examination, there are fibrin thrombi in the portal capillaries and foci of hemorrhagic necrosis.

Renal lesions are variable. The glomeruli show marked swelling of endothelial cells, amorphous dense deposits on the endothelial side of the basement membrane, and mesangial cell hyperplasia. Immunofluorescent studies show abundant fibrin deposition in glomeruli. In advanced cases, fibrin thrombi are present in the glomeruli and capillaries of the cortex. If widespread and severe, these thrombi may produce **bilateral renal cortical necrosis** (Chapter 20), leading to complete destruction of the cortices. The brain may have gross or microscopic foci of hemorrhage along with small vessel thromboses. Similar changes are often found in the heart and the anterior pituitary gland.

Clinical Features

Preeclampsia/eclampsia is an important cause of maternal and fetal mortality worldwide due to hypertensive crises, strokes, and abnormal bleeding due to disseminated intravascular coagulation. Preeclampsia most commonly occurs after 34 weeks of gestation but begins earlier in females with hydatidiform mole (discussed later) or preexisting kidney disease, hypertension, or coagulopathies. The onset is typically marked by hypertension and edema, with proteinuria appearing several days later. Headaches and visual disturbances may follow and are indicative of preeclampsia with severe features, often requiring early delivery. Eclampsia is heralded by more severe central nervous system involvement, including seizures, eventual coma, and even death.

Management of preeclampsia differs depending on the gestational age and severity of disease. For term pregnancies, delivery is the treatment of choice regardless of disease severity. For preterm pregnancies in which delivery place the fetus at risk, mild disease can be managed expectantly by controlling maternal hypertension and by close monitoring of the mother and fetus. However, eclampsia, preeclampsia with severe features, fetal compromise, or the HELLP syndrome are indications for delivery regardless of gestational age.

Following delivery, proteinuria and hypertension usually disappear within 1 to 2 weeks later. Although in most instances preeclampsia has no lasting sequelae, about 20% of affected females develop hypertension and microalbuminuria within 7 years of a pregnancy complicated by preeclampsia. There is also a two-fold increase in the long-term risk of cardiovascular disease.

GESTATIONAL TROPHOBLASTIC DISEASE

Gestational trophoblastic disease encompasses a spectrum of tumors and tumorlike conditions characterized by proliferation of placental

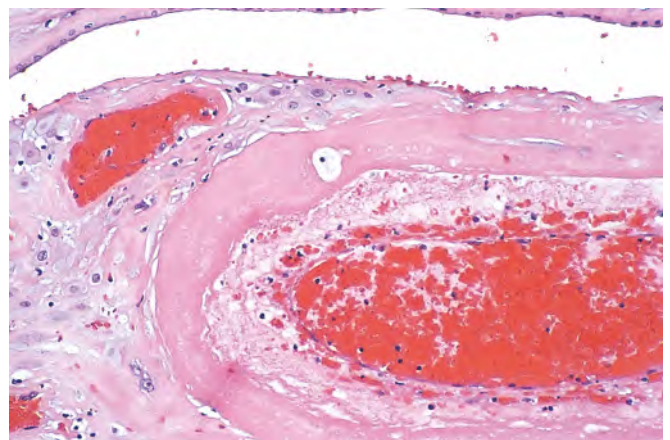


Fig. 22.52 Decidual arteriopathy. Decidual arteriopathy, classic type, showing acute atherosclerosis of uterine vessels in a placenta complicated with preeclampsia. Note fibrinoid necrosis of the vessel wall and sub-endothelial foamy macrophages.

tissue. The major disorders of this type are hydatidiform mole (complete and partial), invasive and metastatic mole, choriocarcinoma, placental site trophoblastic tumor, and epithelioid trophoblastic tumor.

Hydatidiform Mole

Hydatidiform moles are important to recognize because they are associated with an increased risk of persistent trophoblastic disease (invasive mole) or choriocarcinoma. Moles are characterized histologically by cystic swelling of the chorionic villi and circumferential biphasic trophoblastic hyperplasia. They are usually diagnosed during early pregnancy (average 9 weeks) by pelvic sonography. Molar pregnancy can develop at any age, but the risk is higher at the two ends of reproductive life, during puberty and in perimenopausal females. For unknown reasons, the incidence varies considerably in different parts of the world. Hydatidiform mole occurs in about 1 in 1000 to 2000 pregnancies in the United States, but it is twice as common in Southeast Asia. Two types of noninvasive moles—complete and partial—can be identified by cytogenetic and histologic studies.

Complete Mole

Complete mole results from fertilization of an egg that has lost (or subsequently loses) its female chromosomes, and as a result its nuclear genetic material is completely paternally derived. Ninety percent have a diploid 46,XX karyotype stemming from duplication of the genetic material of one sperm (a phenomenon called androgenesis that produces monospermic complete hydatidiform mole) (Fig. 22.53A). The remaining 10% result from the fertilization of an “empty” egg by two sperm (dispermic complete hydatidiform mole) (Fig. 22.53B); these may have a 46,XX or 46,XY karyotype. In complete moles, there is no embryonic development. Patients have 2.5% risk of subsequent choriocarcinoma and a 15% risk of persistent or invasive mole.

Partial Mole

Partial moles result from fertilization of an egg with two sperm (Fig. 22.53C). In these moles, the karyotype is triploid (e.g., 69,XXY) or occasionally tetraploid (92,XXXXY). Fetal development may be seen, but the fetus is characteristically growth restricted and has lethal anomalies of the central nervous system as well as skeletal abnormalities such as syndactyly. Partial moles have an increased risk of persistent molar disease but are not typically associated with postmolar choriocarcinoma.

MORPHOLOGY

Grossly, classic hydatidiform moles appear as a delicate, friable mass of thin-walled, translucent, cystic, grapelike structures consisting of swollen edematous (hydropic) villi (Fig. 22.54A). In complete mole, the microscopic abnormalities involve all or most of the villous tissue. The chorionic villi are enlarged, smooth and oval with central cavitation (cisterns) and are covered by a trophoblast proliferation that includes both cytotrophoblasts and syncytiotrophoblasts (Fig. 22.54B). In contrast, in partial moles, only a fraction of the villi is enlarged and edematous, the trophoblastic hyperplasia is focal and less marked, and only syncytiotrophoblasts are hyperplastic. Partial moles have irregular villous outlines, unlike the smooth round outlines of a complete mole, often with prominent trophoblastic inclusions (Fig. 22.55).

Clinical Features

Most females with partial or early complete mole present with spontaneous miscarriage or undergo curettage because of ultrasound findings of abnormal villous enlargement. In cases of complete mole, human chorionic gonadotropin (HCG) levels greatly exceed those of a normal preg-

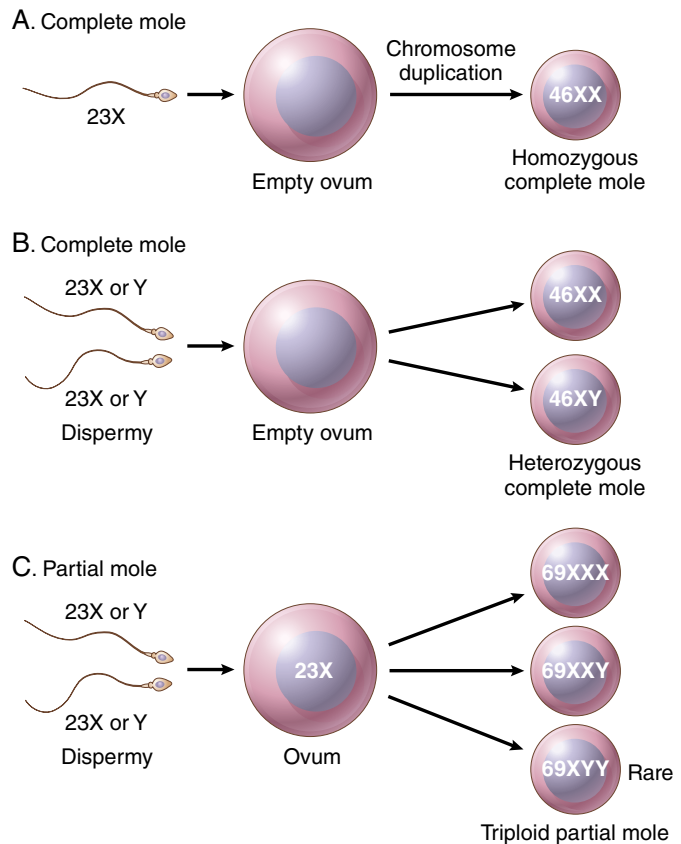


Fig. 22.53 Origin of complete and partial hydatidiform moles. (A) Complete moles most commonly arise from fertilization of an empty ovum by a single sperm that undergoes duplication of its chromosomes. (B) Less commonly, complete moles arise from dispermy in which two sperm fertilize an empty ovum. (C) Partial moles arise from two sperm fertilizing a single ovum.

nancy of similar gestational age. In addition, the rate at which HCG levels rise over time in molar pregnancy exceeds what is seen with normal single or even multiple pregnancies. Most moles are successfully removed by curettage. Patients are subsequently monitored for 6 months to 1 year to ensure that HCG levels decrease to nonpregnant levels. Continuous elevation of HCG may be indicative of persistent or invasive mole (described next) or choriocarcinoma and may require chemotherapy.

Invasive Mole

Invasive mole is an infiltrative lesion that penetrates and sometimes perforates the uterine wall. Invasive moles manifest clinically with vaginal bleeding, irregular uterine enlargement, and persistently elevated serum HCG. Bleeding is due to invasion of the myometrium by hydropic chorionic villi lined by proliferating cytotrophoblast and syncytiotrophoblast. The principal therapy is surgical evacuation of the uterus or hysterectomy. In some instances, however, the villi invade parametrial tissue and blood vessels and embolize to distant sites, such as the lungs and brain. Lung metastases occur in about 5% of complete hydatidiform moles. Small (<2cm) lung metastases usually spontaneously regress and do not require chemotherapy. By contrast, larger metastases require treatment to prevent progressive disease.

Choriocarcinoma

Gestational choriocarcinoma is a malignant neoplasm of trophoblastic cells originating from a previously normal or abnormal pregnancy,

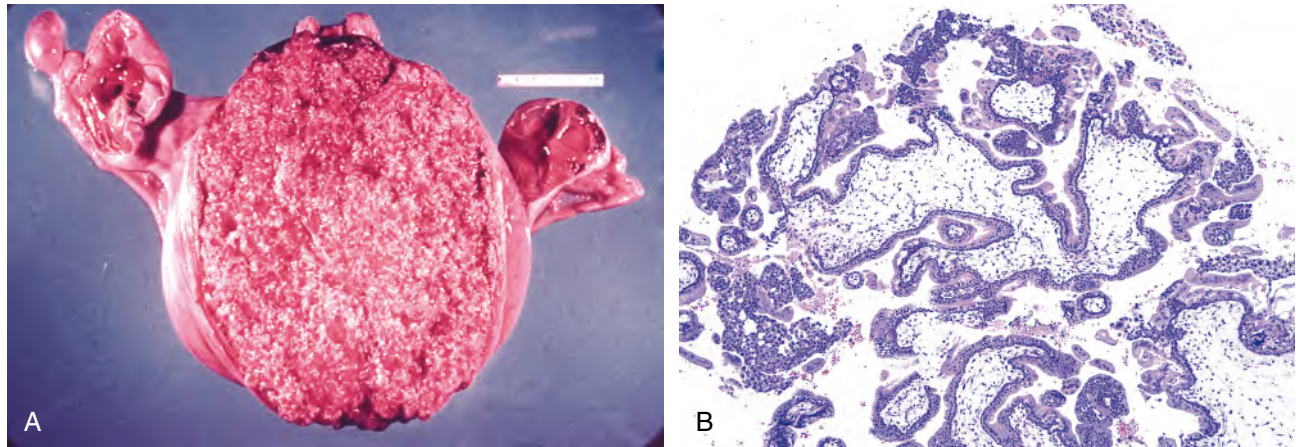


Fig. 22.54 Complete hydatidiform mole. (A) Gross of a well-developed complete hydatidiform mole. Note marked distention of the uterus by enlarged, vesicular chorionic villi. Adnexa (ovaries and fallopian tubes) are visible on the left and right side of the uterus. (B) Histology of a complete hydatidiform mole demonstrating marked villous enlargement, edema, cisternae, and circumferential trophoblast proliferation.

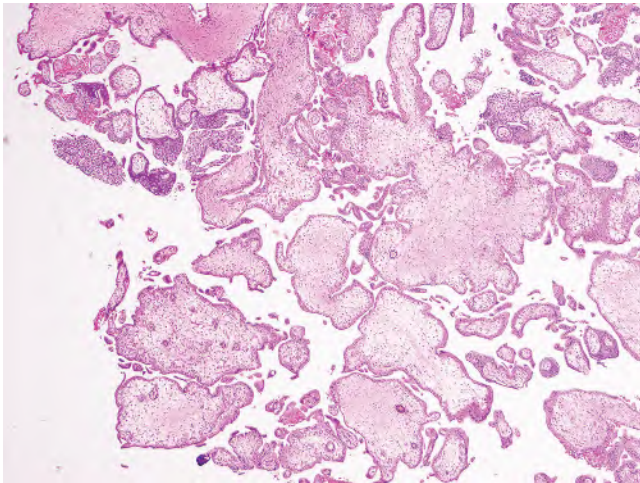


Fig. 22.55 Partial hydatidiform mole. Histology of a partial hydatidiform mole showing a subset of villi that are large and have irregular/scalloped outlines, syncytiotrophoblast inclusions, and focal syncytiotrophoblastic hyperplasia. This partial hydatidiform mole was triploid.

such as an extrauterine ectopic pregnancy or a complete hydatidiform mole. Gestational choriocarcinoma is an uncommon condition that arises in 1 in 20,000 to 30,000 pregnancies in the United States. It may be preceded by several conditions; 50% arise after a complete hydatidiform mole, 25% follow spontaneous abortions, approximately 22% follow normal pregnancies, and the remainder occur after ectopic pregnancies or arise in the placenta during pregnancy. Choriocarcinoma is characterized by destructive and hemorrhagic invasion and is often widely metastatic at presentation. Once identified, it responds well to chemotherapy and is considered a curable disease. Rarely, nongestational choriocarcinoma develops from germ cells in the ovaries or the mediastinum. These tumors are morphologically identical to gestational choriocarcinoma but can be distinguished by the absence of paternally derived DNA, which is present in all gestational choriocarcinomas. Nongestational choriocarcinomas do not have as good a prognosis as gestational choriocarcinomas.

MORPHOLOGY

Choriocarcinoma is a soft, fleshy, yellow-white tumor that usually has large pale areas of necrosis and extensive hemorrhage (Fig. 22.56A). Histologically, it consists of proliferating syncytiotrophoblasts and cytotrophoblasts (Fig. 22.56B); chorionic villi are absent. Mitoses are abundant and sometimes abnormal. The tumor invades the underlying myometrium, frequently penetrates blood vessels, and in some cases extends to the uterine serosa and adjacent structures.

Clinical Features

Gestational choriocarcinoma usually manifests as irregular vaginal spotting of a bloody, brown fluid. This discharge may appear during an apparently normal pregnancy, after a miscarriage, or after curettage. Sometimes the tumor does not appear until months after these events. This tumor has high propensity for hematogenous spread and by the time it is discovered imaging of the chest and bones may disclose the presence of metastatic lesions. The HCG levels are typically elevated to levels above those encountered in hydatidiform moles, but occasional tumors produce little hormone and some tumors are so necrotic that HCG levels are low. Widespread metastases are characteristic; the most common sites are the lungs (50%) and vagina (30% to 40%), followed by, in descending order of frequency, the brain, liver, bone, and kidney.

Intraplacental choriocarcinoma can present as a grossly identifiable mass or with widespread metastasis. It seems likely that all postgestational choriocarcinomas originate within the placenta but are not recognized at delivery, possibly due to small size. For this reason, all placental masses must be carefully examined histologically.

Treatment of gestational choriocarcinoma depends on the stage of the tumor and usually consists of evacuation of the uterine contents and chemotherapy. Chemotherapy is highly effective, with nearly 100% remission rate and high rates of cure. Many cured patients have normal subsequent pregnancies and deliveries. By contrast, rare nongestational choriocarcinomas arising outside of the uterus are more difficult to treat.

Tumors of Extravascular Trophoblasts

Two uncommon malignant tumors of extravascular trophoblasts have unique biologic and morphologic features and merit brief mention.

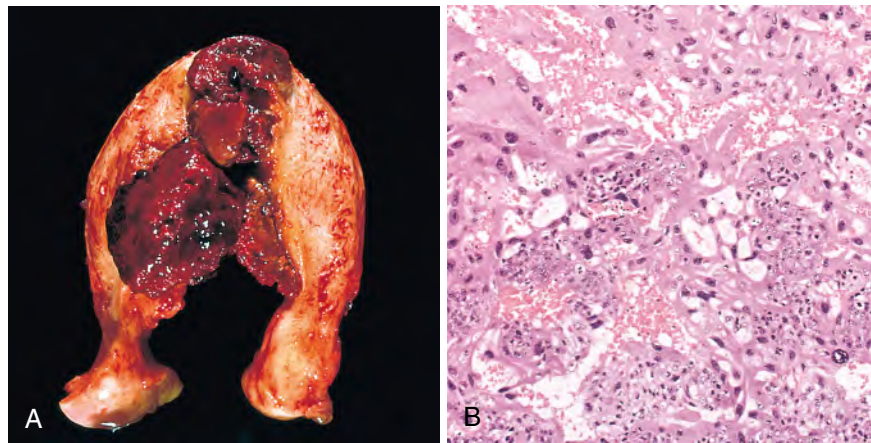


Fig. 22.56 Choriocarcinoma. (A) Choriocarcinoma presenting as a bulky hemorrhagic mass invading the uterine wall. (B) Photomicrograph illustrating the tumor composed of malignant cytotrophoblast and syncytiotrophoblast. (Courtesy Dr. David R. Genest, Brigham and Women's Hospital, Boston.)

- *Placenta site trophoblastic tumor (PSTT)* presents as a uterine mass, accompanied by either abnormal uterine bleeding or amenorrhea and moderately elevated HCG. It may follow a normal pregnancy (one-half of the cases), spontaneous abortion, or hydatidiform mole. The tumor is comprised of polygonal mononuclear or binucleated extravillous trophoblasts with abundant cytoplasm. Like normal extravillous trophoblasts, malignant extravillous trophoblasts are capable of invasion without local destruction or hemorrhage (Fig. 22.57) and show a tropism for maternal vessels. Patients with localized disease have a good prognosis with surgical removal alone, but 10% to 15% of females die of disseminated disease.
- *Epithelioid trophoblastic tumor (ETT)* most commonly presents following pregnancy with amenorrhea and sometimes produces a palpable mass; most are intrauterine or cervical, but extrauterine tumors also occur. It is a neoplasm of mononuclear trophoblast cells, often with abundant clear cytoplasm (Fig. 22.58), that lacks the overt invasive features of placental site trophoblastic tumor yet is capable of metastasis, most commonly to the lungs and to bone. About 10% to 15% of females with metastatic disease eventually succumb to the tumor.

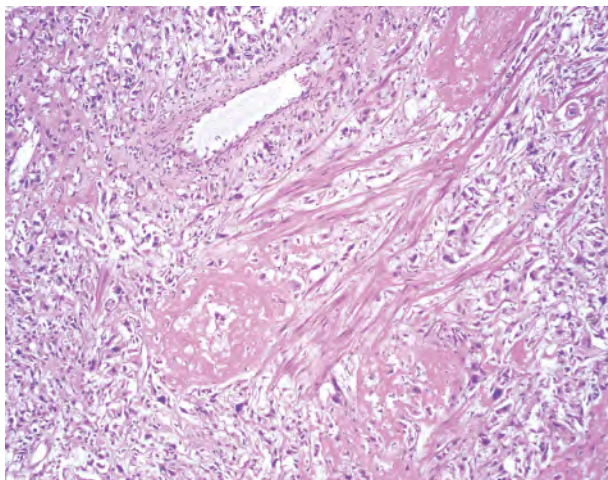


Fig. 22.57 Placental site trophoblastic tumor. Markedly atypical trophoblasts are seen splaying apart smooth muscle fibers as they invade the myometrium.

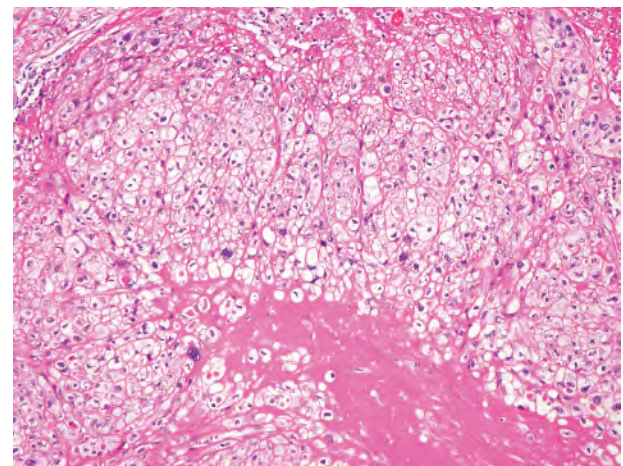


Fig. 22.58 Epithelioid trophoblastic tumor. Histology of an epithelioid trophoblastic tumor composed of sheets of mononuclear trophoblasts, many with clear cytoplasm, associated with characteristic pink hyaline material (lower right).

SELECTED READINGS

- Annual Capri Workshop Group. Early pregnancy loss: the default outcome for fertilized human oocytes, *J Assist Reprod Genet* 37:1057, 2020. [A review of early pregnancy failure.]
- Bell DW, Ellenson LH. Molecular genetics of endometrial carcinoma, *Annu Rev Pathol* 14:339, 2019. [Review of the molecular genetics of endometrial carcinoma.]
- Kraggerud SM, Høie-Hansen CE, Alagaratnam S, et al: Molecular characteristics of malignant ovarian germ cell tumors and comparison with testicular counterparts: implications for pathogenesis, *Endocr Rev* 34:339, 2013. [A detailed review of the molecular genetics of ovarian germ cell tumors.]
- Mathias-Guiu X, Stewart CJR: Endometriosis-associated ovarian neoplasia, *Pathology* 17:30384, 2017. [An overview of ovarian tumors associated with endometriosis.]
- Munoz N, Bosch FX, de Sanjose S, et al: Epidemiologic classification of human papillomavirus types associated with cervical cancer, *N Engl J Med* 348:518, 2003. [A seminal study of HPV detection in cervical carcinoma.]
- Pinar MH, Gibbins K, He M, et al: Early pregnancy losses: review of nomenclature, histopathology, and possible etiologies, *Fetal Pediatr Pathol* 37:191, 2018. [A review of early pregnancy loss.]

- Rana S, Lemoine E, Granger JB, Karumanchi SA: Preeclampsia: pathophysiology, challenges, and perspectives, *Circ Res* 124:1094, 2019. [A review of preeclampsia pathogenesis.]
- Redline RW, Roberts DJ, Parast MM, et al: Placental pathology is necessary to understand common pregnancy complications and achieve an improved taxonomy of obstetrical disease, *Am J Obstet Gynecol* 228:187, 2023. [A review highlighting the relevance of placental pathology.]
- Schiffman M, Castle PE, Jeronimo J, et al: Human papillomavirus and cervical cancer, *Lancet* 370:890, 2007. [A review of HPV-related cervical carcinogenesis.]
- Strickland AL, Gwin K: Gestational trophoblastic disease- rare, sometimes dramatic, and what we know so far, *Semin Diagn Pathol* 39:228, 2022. [A review of the pathology of gestational trophoblastic disease.]
- The Cancer Genome Atlas Research Network: Integrated genomic characterization of endometrial carcinoma, *Nature* 500:242, 2013. [A comprehensive genomic study of endometrioid and serous endometrial carcinoma.]
- Wang Y, Nicholes K, Shih IM. The origin and pathogenesis of endometriosis. *Annu Rev Pathol* 15:71, 2020. [Review of the pathogenesis of endometriosis.]
- Yoon JY, Chapel DB, Goebel E, et al: Genomic catastrophe, the peritoneal cavity and ovarian cancer prevention. *J Pathol* 257:255, 2022. [Review of the pathogenesis of high-grade serous ovarian carcinoma.]